

From cybernetics to brain theory, and more: A memoir

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Abstract

While structured as an autobiography, this memoir exemplifies ways in which classic contributions to cybernetics (e.g., by Wiener, McCulloch & Pitts, and von Neumann) have fed into a diversity of current research areas, including the mathematical theory of systems and computation, artificial intelligence and robotics, computational neuroscience, linguistics, and cognitive science. The challenges of brain theory receive special emphasis. Action-oriented perception and schema theory complement neural network modeling in analyzing cerebral cortex, cerebellum, hippocampus, and basal ganglia. Comparative studies of frog, rat, monkey, ape and human not only deepen insights into the human brain but also ground an EvoDevoSocio view of “how the brain got language.” The rapprochement between neuroscience and architecture provides a recent challenge. The essay also assesses some of the social and theological implications of this broad perspective.

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Chapter 1. Preamble

In the early 21st century, the wide use of the prefix *cyber-* has become the marker of the penetration of computer and information science, broadly conceived, into myriad facets of our lives. This use of the term can be traced back to a single book, Norbert Wiener’s *Cybernetics* (Wiener, 1948, 1961). I first read that book in early 1959, and my subsequent research and teaching career may be seen as the working out of key ideas not only of that book but of the related intellectual ferment that ushered in “the information age.” This article outlines that career as the basis for a history of some key ideas in artificial

intelligence, brain research and cognitive science, together with some other areas with stronger or weaker ties to cybernetics.

If citation of my own work seems at times excessive, the intention is not to claim undue influence so much as to enrich the reader’s understanding of present day science by recalling a broad range of scientific and philosophical perspectives that may be obscured by too narrow a focus on one specialty. This document is both personal and partial. The interviews in *Talking Nets: An Oral History of Neural Networks* (Anderson & Rosenfeld, 1998) demonstrate the diversity of perspectives on the development of this crucial component of cybernetics, and I have learned much from the researchers interviewed there, and many others, whether in person or through their writings, even if they are explicitly mentioned in what follows. Maggie

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Boden's *Mind as Machine* (Boden, 2006) provides a comprehensive history of cognitive science in two volumes and 1712 pages (I make a cameo appearance as "A Wizard from Oz"). It is also salutary to read *Mind, Brain and Adaptation in the Nineteenth Century: Cerebral Localization and its Biological Context from Gall to Ferrier* (Young, 1970) to see how many of our present questions were already being explored in the flush of new insights into evolution and brain localization decades before Santiago Ramon y Cajal (1899, 1911) and Charles Sherrington (1906) ushered the investigation of the anatomy and physiology of neural circuitry into the 20th century.

2. Chapter 2. The formative years (1940–1965)

2.1. Before University

I was born in England on May 28, 1940, the very day that Churchill succeeded in quashing those in Parliament who sought to appease Hitler (Lukacs, 1999). My father was in the Tank Corps of the British Army and, having subsequently been captured in North Africa, spent 2 years as a prisoner of war in Italy and then 2 more in Germany. I have always thought it a great achievement of my mother's that she made him a presence in our life through those difficult years so that there was a sense of completeness when he returned – he was not a stranger. He had decided during his time "in the bag" (as he called POW camp) that England's economy would be in poor state after the war, and having made friends there from various parts of the Empire decided that we should emigrate to New Zealand. Alas, this proved too provincial for my mother, a London girl – she said she cried herself to sleep each night there. Dad got the message, and after 2 years moved us to Sydney, where the family (except for me) has remained ever since – I had become a citizen of the (English-speaking) world.

At Scots College in Sydney, I spent 2 years at the Preparatory School where my Mathematics teacher was Mr. Tomlinson. He lost my respect when, in response to his teaching us that $.3333\dots$ was $1/3$ and $.6666\dots$ was $2/3$, I asked him whether $.9999\dots$ must then be 1. He appeared brush this off as a foolish question rather than a mathematical insight. I started in sixth class at Scots (the highest class of the prep school) and came top of the class. However, because I was about 2 years younger than most in the class, my parents and teachers agreed that I should repeat the final year at the prep school so that my high school years (a further 5 years at Scots College) would not be marred by the emotional dissonance of studying with boys who were passing through adolescence so far ahead of me. I do not know whether I was better adjusted for this year's delay or not, but I am struck by the fact that I had few friendships with my classmates from the first pass through sixth class when I reached the high school.

One of the great privileges of starting at the high school the year I did was that I had the same, superb, mathematics

teacher for all five years there. Fred Pollock usually taught only the fourth and fifth years, but by lucky chance he chose to take the class I was in right through from first year. Fred may not have been the best teacher for the poorer students in the class, but for the better students he was unbeatable, with at least 4 of his students over a 20-year period topping the final mathematics exams for students completing high school in New South Wales. It was not until the 40th Year Reunion of the class of 1956 that I learned from Fred that Mr. Tomlinson had informed him that my mathematical abilities were unusually promising (I had misjudged the man!) and it was because of this that Fred decided to take on my mathematics class from first year onward.

Soon after my arrival at the high school (when I was 11), Fred lent me his copy of *Mathematics and the Imagination* (Kasner & Newman, 1940) which introduced me to the mathematics of infinity, probability, and topology. The authors coined the terms *googol* for 10^{100} , and *googolplex* for 10^{googol} to make the point that incredibly large numbers are nowhere near infinity. Much later, this led to the naming of the company Google and the choice of Googleplex for the name of their complex of buildings. But for me, back in the 1950s, this book set the pattern of reading mathematical books for pleasure, expanding my horizons far beyond the classroom materials.

And one more debt to Fred: He also was very fond of puns, and encouraged his students to contribute puns, as well as mathematical insights, during class. Later, when I moved on to university, I found this "talent" was far less appreciated.

Where Fred Pollock developed my appreciation for the beauty of mathematics and the challenge and pleasure of seeking elegant proofs, Rhys Jones helped me appreciate the richness of language and the history of English. I also honed my debating skills and edited the student newspaper and magazine at Scots. The debates were nothing like the American model where students would be given one topic to debate again and again over the year and would thus conduct heavy research on the topic and prepare copious notes to bring to each debate. We were told the topic of each debate just 15 min before it began. This fostered an ability to think on our feet to generate persuasive arguments. This has usually served me well, but at times has been counterproductive, offending people who mistakenly thought I had not taken time to consider their arguments when formulating my reply.

2.2. Sydney University (1957–1960): from pure mathematics to cybernetics

When I entered Sydney University in early 1957 (at the end of the Southern Hemisphere Summer), it was the elegance of pure mathematics that fascinated me; the potential for scientific application was secondary at best. Nonetheless, my summers were spent learning to program the primitive computers of the day (the IBM 650 in Sydney

at that time had just 2000 words of accessible memory, as distinct from storage on magnetic tape) and it was the second and third of those summers that transformed my view of mathematics. Bill Skyvington, a fellow student from Sydney University working at IBM, got me to read Norbert Wiener's *Cybernetics*. Browsing in the IBM Journal of Research and Development led me to the classic paper by Rabin and Scott (1959) on finite automata and their decision problems. Indeed, whatever the limitations of computer technology at that time, I discovered as I read a range of books and papers of continuing importance that the theoretical foundations of computer science and cybernetics had been well and truly laid from the 1930s onwards. Wiener emphasized that advances in the study of feedback control and electronics as well as insights linking classic ideas on entropy in statistical mechanics to a (semantics-free) notion of information could inform the study of animals, including humans. His mention of McCulloch and Pitts led me to their work on modeling neural networks as *networks of logical devices* (McCulloch & Pitts, 1943) and on the use of group theory to analyze layered networks to achieve *invariant pattern recognition* (Pitts & McCulloch, 1947). Martin Davis's book *Computability and Unsolvability* (Davis, 1958) – which I cherish because it has only one figure, and that figure is labeled “a square of Turing machine tape” – led me to read Alan Turing's classic paper introducing what became known as universal Turing machines (Turing, 1936), and go on to understand Gödel's incompleteness theorem (Gödel, 1931) after translating it with a friend (Nick Whetton) who was studying German. Turing was not yet known to the general public (his role at Bletchley Park and the Enigma Machine was not declassified till much later) but his 1936 paper had by then been complemented by the papers in which he posed the “Turing test” (Turing, 1950) for machine intelligence and (although I became aware of this only later) introduced his theory of morphogenesis (Turing, 1952). The last paper, on pattern formation in arrays of biological cells, has stimulated a whole area of applied mathematics of the continuous variety, complementing the importance of the 1936 paper for the discrete mathematics of computation. (Turing was asked if his 1952 theory could explain the stripes of the zebra. He was said to have replied “The stripes are easy. It's the horse part I have trouble with.”)

Claude Shannon and John McCarthy had edited a book called *Automata Studies* (Shannon & McCarthy, 1956) that included papers by John von Neumann on constructing reliable networks from unreliable components, E.F. Moore on states of finite automata, S.C. Kleene on regular events and Marvin Minsky on neural networks. I also read Shannon's theory of reliable communication in the presence of noise (Shannon, 1948), which led me to two highly mathematical treatments of information theory and statistical mechanics by the Russian mathematician A.I. Khinchin, and these in turn led me to study Lebesgue integration and ergodic theory. Another avenue of exploration was provided by the *Journal of Symbolic Logic* – and I had

the delight of reading George Boole's *The Laws of Thought* (Boole, 1854) in a first edition which I found in the stacks of the original Fisher Library at Sydney University. This was before Xeroxing had become available at the University, so to get a copy of a book or article, I had to either buy the book, order a copy made by having photos printed for each page of interest, or copy out papers by hand as I did for, at least, the Turing and Gödel papers, correcting errors or looking for alternate proofs as I did so.

The *Sunday Sun*, a Sydney Sunday paper, reproduced an article from *Newsweek* on von Neumann's theory of self-reproducing automata, and said that this work had been incomplete at the time of his death but that the papers were being analyzed at the University of Michigan. Wanting to learn the technical details, I wrote a letter addressed simply to “The group working on von Neumann's theory of Automata” at the University of Michigan – and a long time went by. I finally got a reply from John Holland, whom many know as the father of evolutionary algorithms. He apologized for the delay because, he said, the letter had been sent to the School of Automotive Engineering. How impressive that professors in a school of automotive engineering would figure out where to send the letter. Holland was a member of the Logic of Computers Group led by the philosopher Arthur W. Burks, who had co-authored the paper with von Neumann that defined what became known as the von Neumann architecture for storing programs along with the data in the memories of digital computers (Burks, Goldstine, & von Neumann, 1946).

Another important encounter was with a Lecturer (essentially an assistant professor) in Physiology at Sydney University, Bill Levick. Bill was working on neurophysiology of the cat visual cortex. Our deal was that he would let me see what he and Professor Peter Bishop were doing to the cats and I would tutor him in mathematics. He introduced me to the then just published paper “What the Frog's Eye tells the Frog's brain” by Lettvin, Maturana, McCulloch, and Pitts (1959). The work was inspired by the 1947 Pitts-McCulloch paper. Indeed, Lettvin et al. found 4 types of feature detectors in the retina of the frog and showed that they projected to four separate layers in the tectum (the visual midbrain, homologous to the mammalian superior colliculus), but there was no evidence linking those layers to group theory. What was of immediate personal relevance was that the papers of McCulloch and Pitts from the 1940s came from the University of Chicago, but “What the Frog's Eye tells the Frog's Brain” had been written at MIT.

By the Honours year at Sydney Uni, the fourth year of study, I was no longer dedicated purely to pure mathematics but had become fascinated by cybernetics, theories of automata and computation, and the brain. This fascination – and some measure of expertise – was based entirely on my reading, not on any course work. However, one course in pure mathematics did open up an important dimension for my later career – it was a course in algebraic topology taught by Max Kelly that led me to the work of Samuel

Eilenberg and Saunders MacLane in defining the field of category theory.

At that time, all mathematics students from Sydney who wanted a PhD would go to Cambridge in England. However, among all the people whose work I had been reading, Wiener, McCulloch and Pitts, Shannon, Minsky and McCarthy, at least, were all at MIT so here was a real motivation to go instead to Cambridge in Massachusetts. Eventually the matter was settled because I never got a reply from Cambridge in England – until 10 years after I finished my PhD at MIT, when a professor of physics at Sydney University died and they emptied out his mailbox and found at the back the aerogramme accepting me.

By the end of 1960, Cybernetics had expanded beyond Control and Communication in the Animal and the Machine to, at least, Computation, Control and Communication in the Animal and the Machine. But in the years that followed, the very success of this spectrum of studies led to its fragmentation. People who became expert in automata theory or computability were unlikely to master control theory and its application to biological systems; many workers in artificial intelligence came to declare “airplanes don’t flap their wings” and turned their backs on studies of the brain. Many of these specialists rejected the term Cybernetics, but there always remained a core of people who responded to the broader challenges even if some lost their moorings in the process.

2.3. MIT (1961–1963) at the end of the “golden age of cybernetics”

On my first day at MIT, I met Warren McCulloch, who was then in his early 60s. Jack Cowan (see below) claims that I came in and asked, “Tell me how the brain works,” which he thought arrogant; but I recall my statement as “I want to understand how the brain works,” an aim that Jerry Lettvin dismissed as far too broad (and so it is, but I persist). In any case, Warren immediately welcomed me into the group, and it proved to be the most important (but not the only) environment for the growth of my ideas while at MIT. Those were days of less restrictive funding and, after one semester grading linear algebra (I’m unsure as to whether I was more shocked or relieved to discover that my mathematical abilities exceeded those of most MIT undergraduates), McCulloch gave me a no-strings-attached research assistantship that supported me for the rest of my time at MIT.

I spent two and a half years there. I started a thesis with Norbert Wiener but, unfortunately, his interests had switched from cybernetics to statistical mechanics. So, when he disappeared on sabbatical, I moved to Henry McKean Jr., a brilliant young probability theorist, and in due course wrote a thesis on stochastic processes. The thesis was rather a small part of my time at MIT and, reacting to my broad interests, McKean quoted Goethe to the effect that “He who would master the infinite should take the finite and master it from all sides.” I met this quote again

(in German) in the epigraph to von Békésy’s book on *Sensory Inhibition* (von Békésy, 1967) – he had illuminated our broad understanding of visual and tactile perception through a determinedly narrow focus.

I took advantage of a whole academic community rather than confining myself to one topic defined by one professor. The key switch from my Sydney style was from exploration based only on reading to talking to people who were doing the actual research, both those working at MIT and Harvard, and those visiting McCulloch in his office, Room 26-027. Giving me the opportunity to talk to these visitors was one of the ways in which McCulloch contributed more to my graduate education than any other professor at MIT. When I thanked him, he told me that the best thanks would be to offer similar kindness to the young scientists I would have a chance to mentor in future years, a commitment I have done my best to honor.

2.4. The McCulloch group

Warren McCulloch had a very interesting group at that time (for a biographical sketch of McCulloch, his motivations and contributions, see Arbib, 2000). He was particularly intrigued by von Neumann’s problem of reliable neural computation (von Neumann, 1956) and looked for networks of McCulloch-Pitts neurons that would retain their function despite fluctuations in threshold (McCulloch, 1959). By contrast, Jack Cowan, a visiting scholar, was applying a variant of Shannon’s theory to make the transition from communication to computation, recoding layers of neurons into economically larger layers to provide redundancy (Winograd & Cowan, 1963), rather than using inefficient multiplexing as was done by von Neumann (1956).

Manuel Blum was working on recursive function theory and came up with an amazing result that helped establish complexity theory for such functions (Blum, 1967): He showed the existence of one function that requires an enormous number of steps to be computed but has a “nearly quickest” program, and of another function such that no matter how fast a program may be for computing this function, another program exists for computing the function very much faster – a classic speed-up theorem.

The conceptually most important idea for me came from the work of Bill Kilmer, another visiting scholar, in developing a computational model, RETIC, to address McCulloch’s ideas on distributed computation, which were inspired in turn by Magoun’s (1952) work on the role of the reticular formation in the neural management of wakefulness and the findings of Madge and Arnold Scheibel describing its anatomy as a stack of “poker chips” or modules arrayed along the neuraxis (Scheibel & Scheibel, 1958). RETIC had to commit to one of several modes of behavior (e.g., sleep, eat, drink, fight, flee or mate) using modules that received different samples of sensory input and thus formed different *initial* estimates for the desirability of the various modes. The challenge was to connect the modules

in such a way that *modes* would compete and *modules* would cooperate to yield a consensus that could commit the organism to action. This model – developed in the early 1960s though not published in full till 1969 (Kilmer, McCulloch, & Blum, 1969) – seems to me to exemplify a notion crucial for understanding the brain: a general methodology for *cooperative computation* whereby competition and cooperation between elements of a network can yield an overall decision without the necessary involvement of a centralized controller. In the 1970s I generalized this to yield a variant of *schema theory* (Arbib, 1975) in which the interacting units could be functional (schemas that were possibly implemented across multiple brain regions) and not necessarily structural (e.g., specific neurons or neural circuits).

Meanwhile, experimental work was being carried out in McCulloch's group by Jerry Lettvin and Humberto Maturana (who was then visiting from Chile) as well as Pat Wall (whose work would soon thereafter yield a classic theory of pain mechanisms, Melzack & Wall, 1965), among others. As mentioned earlier, I learned that McCulloch and Pitts were at MIT because they were co-authors of "What the Frog's Eye Tells the Frog's Brain" (Lettvin et al., 1959). Lettvin did the neurophysiology and Humberto Maturana did the neuroanatomy, inspired by the theorizing of Pitts and McCulloch (1947) on "How we know universals" – in their case, recognition of a visual pattern when the shape, size or position of the pattern may vary – to study how the nervous system of a frog would respond to small moving stimuli like a fly or large moving stimuli like a predator. The distinction is crucial to the animal's behavior (snap, or escape) – *they were looking at vision in terms of action*.

By contrast, David Hubel and Thorsten Wiesel – I visited Hubel several times at Harvard Medical School – had reshaped the foundations of visual neurophysiology by discovering generic feature extraction in the visual cortex of the cat (and later, in monkey), demonstrating, e.g., how local edge information might be extracted (Hubel & Wiesel, 1959, 1962, 1977). Horace Barlow, when visiting McCulloch from the other Cambridge, explained to me that the Hubel-Wiesel edge detectors can extract the contours of an object and that this is why one can recognize a person from a caricature – to which I replied, "But then, Horace, how can I tell you are not a caricature?" The important point implicit here is that visual processing cannot be purely hierarchical. To the extent that higher levels of processing may extract key properties of an object, person or scene, our awareness is still enriched by the scene's lower-level aspects (e.g., shape, motion, color and texture).

In any case, two views of vision were in competition: a visual system that extracts features that are "general purpose" for pattern recognition (Hubel and Wiesel), and a visual system geared to recognizing features of immediate relevance to interacting with the world (Lettvin et al.). The latter view, of *Action-Oriented Perception*, has shaped an important aspect of my career. I will return to this

theme when I discuss *Rana computatrix*, the frog that computes.

2.5. Automata theory and control theory

On my way to MIT in January of 1961, I stopped in New York to spend time with relatives and make several scientific contacts, including a visit to Martin Davis at Yeshiva University, not only because he had written *Computability and Unsolvability* but also because he was an editor of the *Journal of the Association for Computing Machinery (JACM)* to which I had submitted a paper called "Turing machines, finite automata and neural nets," my Honours thesis at Sydney University. Indeed, *JACM* published the paper that October (Arbib, 1961). The review in *Mathematical Reviews* by R.M. Baer noted that "Notation is introduced to avoid the net diagram approach" – whose drollery was only apparent when one saw that his next review noted of another paper that "Net diagrams are used to avoid the notational approach."

When I arrived at MIT, I was innocent of control theory beyond the basic notions of feedback and oscillations developed in Wiener's book (the term *cybernetics* derived from the Greek word for the helmsman of a boat, a key link in a man-machine servo loop). However, a high school friend, Ron Acher, had read a textbook by three MIT Professors, Newton, Gould, and Kaiser (1957), and wrote to ask me to check with them about their current research. When I met with two of them and they learned that I was strongly mathematical, they referred me to Michael Athanasiades (now Athans) and Peter Falb of the nearby Lincoln Laboratories who were writing a book on mathematical control theory (Athans & Falb, 1966). Their draft contained an exposition of Rudolf Kalman's work on mathematical system theory. I read this with great excitement. At MIT at that time, the study of linear systems was conducted in the frequency domain, via the Laplace Transform. Kalman, on the other hand, transformed linear system theory by focusing on the concept of state. I had learned to think of finite automata as characterized by three finite sets X , Y and Q of inputs, outputs and states, along with two functions: $\delta: Q \times X \rightarrow Q$ to define the next state after q to be $\delta(q, x)$ if the input was x ; and $\beta: Q \rightarrow Y$ to define the output as $\beta(q)$ when the state was q . I immediately found Kalman's approach congenial – with X , Y and Q now vector spaces, and δ and β linear: $\delta(q, x) = Fq + Gx$, while $\beta(q) = Hx$. My challenge was to assess what these two categories of systems shared when one came to think about

- *reachability* – what states in Q could be reached from an initial state by applying sequences of inputs?
- *observability* – given the sequences of outputs generated by a system in response to sequences of inputs, what could one infer about a possible state space Q and related maps δ and β that might have generated the observed behaviors?

My understanding of these issues was enriched by many conversations with a wild young man named John Rhodes whom I got to know when we were both graduate students in mathematics (he went on to become a successful professor of mathematics at Berkeley). In particular, working with Ken Krohn, he showed how algebraic sense could be made of finite automata through the theory of semigroups (Krohn & Rhodes, 1965a, 1965b). They submitted identical theses, Rhodes to MIT and Krohn to Harvard. This introduction to algebraic approaches to automata was enriched with a specific link to the mathematics of Marco Schützenberger whom I got to know while he was on sabbatical, jointly at MIT and Harvard Medical School. Marco was a crucial influence on Noam Chomsky during the mathematical phase of Chomsky’s work, and characterized languages as formal power series (Chomsky & Schützenberger, 1963). Explaining semigroups and formal power series is beyond the scope of this paper; the point here is simply to suggest the rich mathematical content of automata theory as already established in the early 1960s.

2.6. Brains, machines & mathematics

A year and a half into my PhD studies (in the Northern summer/Southern winter of 1962), at the invitation of John Blatt, a friend of my parents and professor of applied mathematics at the University of New South Wales (UNSW) in Sydney, I gave a series of lectures at UNSW that I entitled “Brains, Machines and Mathematics” (BMM). One member of the audience, Harry Guss, introduced me to the book *The Thinking Machine* by C. Judson Herrick (1929) which provided another example of consideration of cybernetic issues well before the computer age – and introduced me to the work of a neuroanatomist who thought carefully about the function of the networks he revealed.

In addition to presenting the lectures to a live audience, I wrote up notes as the basis for a version of the lectures broadcast to subscribers on the university radio station. The book based on these lectures came out two years later (Arbib, 1964). Just a few comments on the book will give a sense of how I perceived the “state of the art” in 1962.

Chapter 1 basically combined my fourth-year work in Sydney on “Turing Machines, Finite Automata, and Neural Nets” with an exposition of the necessary background. Here, neural nets were simply computational elements with no connections to neurobiology beyond the original inspiration of McCulloch and Pitts to define threshold logic units on a discrete time scale.

Chapter 2 summarized the Lettvin et al. work on the visual system of the frog, and noted the challenge of linking the distinctive anatomy of ganglion cells seen by Maturana to the distinctive responses to visual features observed by Lettvin. This linkage cannot be addressed using Pitts-McCulloch neurons or the leaky integrator neurons that have featured in many of the brain models later developed

by my group. A major shortcoming of BMM, then, is that there is no mention of the Hodgkin-Huxley equations (Hodgkin & Huxley, 1952) which describe the propagation of spikes down the axon, but their inclusion would have been of little value pending Wilfrid Rall’s extension of this to compartmental modeling, dividing a neuron into various coupled components along with synapses, each amenable to Hodgkin-Huxley-inspired modeling (Rall, 1964; for a perspective written by Rall when he retired, see Rall, 1995).

Chapter 2 continued with an exposition of the training “with a teacher” of a simple Perceptron (Rosenblatt, 1958), a McCulloch Pitts feedforward network with synapses of only the output layer being adjustable. The gaping hole in this Chapter is that no mention was made of one of the most fundamental principles of neural learning, namely Hebb’s rule that synapses will be strengthened if the pre- and post-synaptic neurons fire “at the same time” – but I cannot recall now whether this was because I had not yet read Hebb’s *The Organization of Behavior* (Hebb, 1949) or had not found a mathematical treatment. (Chapter 4 does mention the study of machine learning by Samuel, 1959; which many years later could be seen as the precursor of temporal difference learning; Sutton, 1988; Sutton & Barto, 1998). In later work, I have paid much attention to Hebb’s rule and other models of synaptic plasticity, but – seeking to understand large scale neural activity serving cognitive functions – have generally used leaky integrator neurons rather than models of spiking neurons, let alone neurons based on the Hodgkin-Huxley equations.

Chapter 3 explained von Neumann’s multiplexing theory, Shannon’s basic result on reliable communication in the presence of noise, and the Winograd-Cowan theory of reliable computation in feedforward layered networks in the presence of noise.

Chapter 4 then offered diverse topics in cybernetics broadly construed: the basic theory of feedback and oscillations and Wiener’s linkage of that to the human motor dysfunction of ataxia; Peter Greene’s ideas on resonant frequencies in neural networks (Greene, 1962a, 1962b); notes on prosthesis and homeostasis; and the basic ideas on gestalt and universals from Pitts and McCulloch (1947). A notable omission here was any discussion of electroencephalography (EEG) – probably because my focus at that time was on relating brains to the neurons in neural networks (as in the work of Lettvin et al. and Hubel & Wiesel) rather than to global measures of brain activity whose role in ongoing computation was not at all obvious. Perhaps of most importance for the later emergence of cognitive science from cybernetic roots, all-too-briefly mentioned, was *Plans and the Structure of behavior* by George Miller, Eugene Galanter and Karl Pribram (1960) which combined ideas from early AI and psychology with hierarchies of cybernetic feedback units they called Test-Operate-Test-Exit (TOTE) units.

The centerpiece of Chapter 5 was a new and much simplified proof of Gödel’s Incompleteness Theorem – basically, that a formal logic in which one can prove

theorems about arithmetic must be *incomplete* (there are true facts about arithmetic that cannot be proved as theorems of that logic) if it is *consistent* (that is, one cannot derive both a statement and its negation within the logic). This was followed by arguments opposing the claim that Gödel's Incompleteness Theorem places limits on machines that the human mind transcends.

Looking decades ahead, one may say that the choice of the title “Brains, Machines and Mathematics” in 1962 has in some sense defined the core of my career, though I stopped proving theorems after 1986, and ceased working on robotics around 1990, with my studies in artificial intelligence thereafter being occasional corollaries to efforts to build a computational and cognitive neuroscience. Looking at this another way, we can see neural nets both as (i) challenges for mathematics and theory of computation, and (ii) as models of the brain. The two are not necessarily inconsistent, but the latter predominated in my work after 1980 or so. A second edition of BMM published by Springer in 1987 marked a finale to the automaton-theoretic phase, while other publications had long since pushed hard into seeking to understand “how the brain works.” The second edition was much harder to write than the first – for each chapter in the original, the intervening quarter century had yielded many expansions of old topics while developing new topics as well. How could one select what seemed most relevant to the spirit of the original and still keep the length of the new edition in bounds? In the end, I dropped the discussion of information theory (whereas I probably should have kept it but shifted the focus), while adding chapters on history, realization (i.e., going from observed behavior to a machine that could yield that behavior), learning networks, and automata that construct as well as compute. I also expanded the Gödel chapter to include speed-up theorems (if a logic is incomplete, one can effectively find new axioms that shorten proofs of old theorems and support the inference of new theorems) and greatly enriched the discussion of the brain-machine controversy, stressing that a brain cannot learn unless it makes mistakes and thus is not to be modeled as implementing a consistent logic.

2.7. Artificial intelligence

Soon after my arrival at MIT, I met Marvin Minsky just as he was opening the box of reprints of “Steps Towards Artificial Intelligence” (Minsky, 1961), and so I was the first person to receive a copy. Minsky and his wife Gloria were very kind to me during my time at MIT, while intellectually I learned about the growth of Artificial Intelligence (AI) as fostered in the range of studies underway in Minsky's group, many of which were later collected in *Semantic information processing* (Minsky, 1968). A crucial cybernetic idea was that of an *internal model*: to interact with an object or person, a human, animal or AI system needs to have some representation of that system to ground perception and interaction. This idea was first developed

for cybernetics by Kenneth Craik (1943) and figured in the writings of Richard Gregory (1961), Donald MacKay (1966) and Minsky himself (Minsky, 1965).

“Steps Towards Artificial Intelligence” is still worth reading because it offers a view of AI as part of cybernetics before the divergence between NNs (neural networks) and GOF AI (the label for what, perhaps misleadingly, was called good old-fashioned AI). It is ironic, then – jumping ahead a few years – that Minsky coauthored *Perceptrons: an introduction to computational geometry* (Minsky & Papert, 1969) which is often cited as the key text downplaying the computational power of NNs, and doubly ironic that it was Warren McCulloch who had arranged for Seymour Papert – at that time stateless – to get the paperwork that made it possible for him to come to the United States in 1963. However, I believe the rejection of NNs which for many greeted the reception of the Minsky-Papert book was based on too superficial a reading – all they showed (elegantly) was that the computational power of a single layer of McCulloch-Pitts neurons without loops was limited. Multi-layered networks could do much more, which is hardly surprising (Spira & Arbib, 1967). Which leads to the triple irony – that learning in NNs has become the “in thing” in current AI. The abstract of a 2015 overview (LeCun, Bengio, & Hinton, 2015) gives some sense of this turnabout:

Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction. ... Deep learning discovers intricate structure in large data sets by using the backpropagation algorithm to indicate how a machine should change its internal parameters that are used to compute the representation in each layer from the representation in the previous layer. Deep convolutional nets have brought about breakthroughs in processing images, video, speech and audio, whereas recurrent nets have shone light on sequential data such as text and speech.

But the insights that grounded this turnabout did not emerge till the 1980s, and their breakthrough into today's widespread applications was only made possible by post-2010 levels of speed and capacity of computers and the aggregation of enormous databases.

The day before my thesis defense, Henry McKean told me “We agreed you could get a PhD for solving that problem, but you found a much easier solution than we expected, and you really deserve to have a deeper thesis than that...” This was in April or May of 1963. I was already booked to go to a Summer School run by Allen Newell and Herbert Simon, co-developers of a then key approach to AI, the General Problem Solver (Newell, Shaw, & Simon, 1959; in those days, GPS had another meaning). If McKean had accepted the then near-completed thesis, I might today be an AI person without much work on the brain, having devoted much effort to programming computers myself rather than “programming students to program computers.” Instead, I spent the

summer of 1963 writing the next thesis (Arbib, 1965) at Dartmouth College, which was the nearest place to Henry McKean's summer home.

2.8. Linguistics and psychology

I would visit Noam Chomsky once a semester and ask what was new in linguistics. At that time, Chomsky was in a mathematical phase, with a specific link to the mathematics of Marco Schützenberger (mentioned above), but already Chomsky had drawn a line between *competence* – the abstract knowledge that underlies one's ability to judge the grammaticality of sentences – and *performance* – where, for example, limits in working memory may affect one's comprehension or production of utterances. Chomsky championed competence to the exclusion of performance; another MIT linguist, Victor Yngve, championed performance: “The precise significance of the rules will become clear with the description of the mechanism that applies them” (Yngve, 1960). In later years, as I sought to understand the brain mechanisms underlying language, the charms of a competence theory faded, save perhaps as an abstract boundary condition on performance models. An automaton that offers a yes-no decision of grammaticality is very different from one that can, e.g., produce a sentence that expresses a desired meaning.

I have already mentioned Miller, Galanter and Pribram's *Plans and the Structure of behavior*. George Miller worked with “the early Chomsky” to carry out psycholinguistic studies suggesting that finite state automata were inadequate models for language (Miller & Chomsky, 1963) and was well known for his classic paper “The magical number seven plus or minus two: some limits on our capacity for processing information” (Miller, 1956). His further importance to me was that, while I was at MIT, the neuropsychologist Hans-Lukas Teuber was hired to found a Ph.D. program in Psychology with an emphasis on brain mechanisms. I immediately applied to Teuber for permission to take part in the Proseminar for new graduate students. He refused because I was enrolled in Mathematics! But then Miller gave me permission to audit the Proseminar for Psychology graduate students at Harvard, which I did, greatly deepening my understanding of Psychology. Later, Karl Pribram was an importance influence on me during my years at Stanford.

2.9. Brain theory

All this helped lay the foundations for my work in brain theory. Cybernetics explores the processes of computation in machines, animals (including humans) and societies, and thus has spawned – but can continue to develop a framework for – many more focused disciplines. Artificial intelligence seeks to understand how programs may address, or help humans address, problems that humans categorize as requiring intelligence for their solution. Psychology seeks to understand mental processes, but without a neces-

sary concern for underlying mechanisms. Cognitive psychology and cognitive science have sought to work within the cybernetic and computer revolutions to explain action, perception, memory, language, and thought (singly or in tandem) in terms of computation/information processing. *Brain theory* carries this one step further to explore how interactions between brain regions and other structures down to neural circuits and plastic synapses *in the living brains of humans and other animals* may support these computations. As such, the aim – in my perspective – is not to reduce the brain or mind to the operation of present-day computers, but rather to contribute to the development of a much richer framework for *computer and information science* in which the operation of biological and social systems can enrich the study of technological systems, and vice versa (Arbib, 1972a, 1972b).

2.10. The Neurosciences Research Program (NRP)

Neuroscience seems like such an established field that it may surprise readers to learn that it (as distinct from neurophysiology and neuroanatomy as subdisciplines of physiology and anatomy) was established only in 1962, with the founding of the Neurosciences Research Program (NRP) at MIT by Francis O. Schmitt and a variety of scientists interested in the neural basis of behavior and mind. (The Society for Neuroscience was founded in 1969 with Ralph W. Gerard as Honorary President. I later discovered that I am the James Bond of neuroscience – my membership number in the Society is 000000007.) The NRP hosted four summer schools and multiple Work Sessions over the next 20 years to help establish the new field. Schmitt not only brought together a variety of brain-centered disciplines but also championed the application of molecular biology and genetics to the study of the brain. For a history of the NRP, see the essay by George Adelman (2010) who was for many years librarian – and more – for the NRP.

No doubt through Warren McCulloch's recommendation, I was fortunate enough to be invited to participate in a range of NRP events, starting with a Work Session on *Mathematical Concepts of Nervous System Function* (January 31 to February 1, 1964) which I took part in en route to starting my post-doc with Jack Cowan at Imperial College, London. More about the NRP below.

2.11. Post-doctoral adventures (September 1963–December 1965)

2.11.1. Making the grand tour 1: The USA

After completing my PhD, I set off for a tour of the United States in the (Northern) Fall of 1963, alternating between tourism and visits to universities to learn about current research in fields that interested me. This was the time when the Link computer had been introduced as the first commercially available computer for neurophysiological research. When I visited neurophysiology labs, there were ongoing debates as to whether one should use the

computers or whether doing so would remove the delicate touch on which experimenters prided themselves.

On the automata side, among those I visited were Juris Hartmanis and Bob Stearns at GE in Schenectady who were pioneers in the study of complexity of computation (Hartmanis & Stearns, 1965) and I presented Manuel Blum's results to them. I also visited (among many others) Lotfi Zadeh at UC Berkeley and Rudolf Kalman, then at the Research Institute for Advanced Studies (RIAS), an industry-funded think tank in Baltimore for research in control theory and nonlinear systems. Both were interested in my ideas on the relation between the state space approaches to automata theory and linear systems, and in due course each offered me a post-doc. I accepted the offer from Kalman who was about to move to Stanford. So – serendipity! – an inquiry from a high school friend about control theory at MIT is why I ended up at Stanford in mid-1965.

Among many other significant visits on the US tour was that to Heinz von Foerster at the University of Illinois in Champaign-Urbana. Frank Fremont Smith, as Medical Director of the Josiah Macy, Jr. Foundation, had introduced a range of conference series. One of these (chaired by Warren McCulloch) had, at its sixth conference and following the critical acclaim of Wiener's book, been renamed *Cybernetics: Circular, Causal, and Feedback Mechanisms in Biological and Social Systems*. Note the extension from Wiener's "animal and machine" to "social systems." The proceedings of the last five conferences (six through ten) were published by the Foundation under von Foerster's editorship. McCulloch, Shannon, von Neumann and Wiener were among those who represented "hard" cybernetics, linking mathematics, biology, and technology; von Foerster, Margaret Mead and Gregory Bateson were among those who represented "soft" cybernetics with the emphasis on persons as the units in cybernetic analysis of social systems. Not all would agree with my dichotomy – but it is simply a mnemonic for two very different levels of analysis, not a value judgement. Heims (1991) provides a history of the group's interactions, charting the broad range of agreements and disagreements whose gradations were much finer than my hard/soft dichotomy. Although I had met von Foerster before in McCulloch's company, this visit to Urbana gave me a chance – as someone who had focused on "hard" cybernetics at MIT – to get a better sense of how circular causality – A affects B, which then affects A, and so on, in a circle of events which modify each other – also operates at the interpersonal level, as when a child cries, the parent yells at the child, and the child cries more thus further stressing the parent, and so on.

von Foerster distinguished *first order* cybernetics as the cybernetics of observed systems from *second order cybernetics* as the cybernetics of observing systems – though in social systems each person is both observer and observed, a point which became important in my thinking only in the 1990s with the discovery of mirror neurons. von Foerster (1974) offers a broad range of papers on *Cybernet-*

ics of Cybernetics: or the Control of Control and the Communication of Communication. "Soft" cybernetics includes classic applications to anthropology (Bateson, 1987) and family therapy (Watzlawick, Beavin, & Jackson, 1967), including the basic pathology of the "double bind."

To break the timeline in following through on these ideas: Humberto Maturana was established back in Chile by the time of the presidency of Salvador Allende, and Allende's government had taken a great interest in the applicability of cybernetics to the restructuring of Chile's economy. But Allende was overthrown on September 11, 1973, by a military coup led by Augusto Pinochet in which Allende died. I became concerned for Maturana's safety but dared not write to ask him about it explicitly for fear the government would intercept my mail, to his detriment. Instead, I found funds to invite him to a conference on *Biologically Motivated Automata Theory* in MacLean Virginia, that I organized for June 1974. In the first day or two at my home in Amherst, Humberto was in a distressed state from the situation in Chile, but by the time we got to MacLean he had "thawed" enough to take an active part in the conference, with a presentation on "The Organization of the Living: A Theory of the Living Organization" (Maturana, 1975). This paper was my first indication of his transition from "hard" to "soft" cybernetics, introducing the theory of autopoiesis he came to develop with Francisco Varela (Maturana & Varela, 1991). Among the five proposals in his paper were (a) that autonomy in living systems is a feature of self-production (autopoiesis), and that a living system is properly characterized only as a network of processes of production of components that is continuously, and recursively, generated and realized as a concrete entity in the physical space, by the interactions of the same components that it produces as such a network; and (b) that language arises as a phenomenon proper to living systems from the reciprocal structural coupling of at least two organisms with nervous systems, and that self-consciousness arises as an individual phenomenon from the recursive structural coupling of an organism with language with its own structure through recursive self-description. I have some sympathy with (a) but later developed a view very different from that of (b).

Many years later, I spoke at a Gordon Research Conference on Cybernetics (New Hampton, NH, August 1984) where I had some trouble connecting with an audience which I found, to my dismay, to be almost entirely composed of people with almost no interest in, and perhaps even hostility to, "hard" cybernetics. Maturana gave a talk on autopoiesis, and in the discussion period I asked him how this theory related to the classic work on "What the Frog's Eye Tells the Frog's Brain." But instead of the informative response I had hoped to elicit, Humberto responded with anger – apparently seeing it as an attack for moving away from empirical neuroscience. The audience leapt to his defense. Ernst von Glasersfeld – with whom, on my tour of Europe and the Soviet Union in 1964 (discussed below), I had spent some pleasant hours

in Milano discussing a new theory of language – led the attack. “Michael. After you visited us in 1964, you wrote a trip report that showed you did not understand our theory.” A damning indictment? But I responded, “Ernst, that’s the greatest compliment I have ever received.” Stunned silence. “I visited you as a fresh Ph.D. twenty years ago, and you still believe I should have been able to fully understand your theory in just a few hours.” And thus the tension was defused.

But during the US tour, an event occurred that shattered my world view. Driving south from the Grand Canyon, I pulled into a gas station in Flagstaff, Arizona. The radio was on, “President Kennedy has been killed . . .” I assumed it was a spoof akin to Orson Welles’ “War of the Worlds.” The attendant saw the look on my face and asked, “Haven’t you heard?” I drove the next four hundred miles in a reckless state of disorientation.

2.11.2. *Post-doc 1: Imperial College, London*

After my tour of the US and my return to Boston for the NRP Work Session, I headed to London for a post-doc, in the first half of 1964, at Imperial College with Jack Cowan, as part of the group of Denis Gabor (who later received the Nobel Prize for his invention of holography). I had deep discussions with Jack about his application of statistical mechanics to the analysis of neural networks, and learned from him of continuing work on the genetic code, but we did not write a paper together.

Shortly after my arrival, I met Professor John Westcott who told me he would be hosting a conference on control theory at Imperial, and I flippantly commented that I should give a talk on the rapprochement between automata theory and control theory. Some weeks later, Westcott asked “How is your paper coming along?” I was nonplused. “What paper?” “The one on control and automata for the conference.” Hoisted by my own petard, I set to work on the paper (Arbib, 1966a), reading more of Kalman’s papers and a new book on the state space approach to linear systems (Zadeh & Desoer, 1963), plus various papers on automata theory, to affect that rapprochement in time for the conference, in April of 1964.

Among other adventures, I got to know people in London and Bristol working on Turing machines, gave a course on automata theory and a very poor lecture on the social implications of cybernetics at Imperial College, attended a weekly seminar on brain modeling run by J.Z. (Jay Zed) Young (I was intrigued by his work on the intelligence of octopuses and his just published book on modeling the octopus brain, Young 1964), and meeting various experts on cybernetics around Britain.

2.11.3. *Making the grand tour 2: Europe and the Soviet Union*

I spent the latter part of 1964 in Europe and the Soviet Union (funded by the Air Force Office of Scientific Research, AFOSR, thanks to Rowena Swanson, to whom I had been introduced by McCulloch); and subsequently

published an account of my experience of cybernetics in Eastern Europe and the Soviet Union (Arbib, 1966b). I prepared intensively for my stay in the Soviet Union, including very helpful discussions with a Hungarian émigré, George Paloczi Horvath, to whom I had been introduced by Gabor (Paloczi-Horvath, 1964). I met many researchers through the good graces of Warren McCulloch, and others through my own efforts, establishing a network of colleagues who have helped enrich my understanding of cybernetics and more. Among the memorable meetings was that with the great neuropsychologist Alexander Romanovich Luria at my hotel in Moscow – he apologized for being late, but there had been a party at his Institute because the Director’s daughter was one of the three cosmonauts who had just been launched into space. We talked about the brain and he told me about his work with Vygotsky (Luria & Vygotsky, 1992, for a much later English translation).

The trip not only taught me a great deal about cybernetics, but also taught me a great deal about society, as I came to understand something of the different reactions of people in Eastern Europe to their relation to the Soviet Union (the Hungarians wanted to return to capitalism; the Czechoslovaks wanted to build a better socialism), and of different currents within the Soviet Union itself. And a few days after I met Luria, I came down to breakfast in Tbilisi to see Pravda’s headline announcing the overthrow of Khrushchev by Brezhnev and Kosygin. In Leningrad, I had the strange experience of arguing about the early stages of the Vietnam war with an American couple on their honeymoon.

2.11.4. *Australia and post-doc 2: Stanford University, California*

I returned to Sydney as a visiting lecturer in mathematics at UNSW for a few months through May of 1965. On May 1, Keith Burrows, who was a mentor during my summers at IBM, introduced me to Prue Hassell, a girl from Perth in Western Australia who was then working as an Assistant TV Producer at the Australian Broadcasting Commission (ABC) in Sydney – her grandmother played mahjong with Keith’s mother.

My postdoc with Kalman at Stanford started in June of 1965. Kalman had organized a summer school whose faculty, in addition to Kalman and myself, included Paul Zeiger, another new PhD but there just for the summer, and Peter Falb who (if memory serves me correctly) had by then moved from Lincoln Labs to the faculty at Brown University. My own contribution was in great part based on the paper I had written for Westcott’s conference.

One of the highlights of the summer was a trip with Peter to Las Vegas where he taught me to play craps and where I converted \$100 (a great deal of money to me at that time) into \$500, a stake which enabled me to make a number of the then very costly trans-Pacific phone calls to Prue. When I returned to Sydney for Christmas, I proposed to Prue and we were married 12 days later. At that time,

women with Australian Government jobs were automatically fired when they were married – it was then up to them to be homemakers, while their husbands earned the money. Prue did not wear her wedding ring for her last few days at the ABC before she left Sydney to join me in California.

In response to the enthusiasm of my editor at the McGraw-Hill publishing company, it was agreed that the notes from the summer school should be converted into a book. The book, *Topics in Mathematical System Theory* (Kalman, Falb, & Arbib, 1969), remained more a collection of three pieces by Kalman, Falb and Arbib than an integrated whole but was delayed by Kalman's work on his new algebraic (module-theoretic) theory of linear systems. Kalman dedicated his book to Constantina (his wife), Fortuna, and Prudence. Falb was not married – Kalman invoked Fortuna to recall Peter's love of gambling.

Chapter 3. Stanford (1965–70)

Rather mysteriously, I became a faculty member just six months after arriving at Stanford. From January 1966 through August 1970, I was an assistant and then an associate professor of Electrical Engineering (EE). Not because I knew anything about electrical engineering but, presumably, because I had worked with Kalman and could thus lecture on mathematical systems theory in the EE curriculum. However, I also introduced two new courses. One, entitled *Brains, Machines and Mathematics*, allowed me to update material from my 1962 lectures in UNSW, while the other, on *Automata Theory*, was an outgrowth of the course I had given at Imperial College in 1964.

3.1. Mathematical theory of systems and computation

In addition to the book with Kalman and Falb, I published two other books while at Stanford, all three under the umbrella of mathematical theory of systems and computation.

Machines, Language, and Semigroups was an edited volume which was centered on the MIT/Harvard work of John Rhodes and Ken Krohn (they organized the 1966 conference at Asilomar, California, on which the book was in great part based) but also included a range of contributions from other approaches to automata theory and formal language theory. The serendipitous element here is that one of the people who was at the conference was Ed Blum, for many years a professor in mathematics and computer science at the University of Southern California (USC) and who thus provided my first link to that University, at which I would spend 30 years (1986–2016).

Theories of Abstract Automata built on my lectures at Imperial College and Stanford, synthesizing very broad reading of the literature in automata theory and computability theory, along with new proofs and new contributions by myself, my colleagues and my students. The last chapter followed through on my interest in self-reproducing automata, which had started with that Sunday

newspaper clipping. It introduced self-reproducing automata with huge state sets (contra von Neumann, 1966, and Thatcher, 1970), inspired by the richness of DNA. Alvy Ray Smith (my only student to win motion picture Oscars) wrote his thesis on self-reproducing automata (summarized as part of a review article written more than 20 years later: Smith, 1991). He came up with the cover of *Theories of Abstract Automata*, a fanciful conceptualization of a self-reproducing automaton.

Further books on brain theory had to wait, building on major conceptual advances in my understanding of computational and cognitive neuroscience at Stanford, where I supervised three Ph.D. students in automata theory and three in brain theory, the first of the 55 in all. I have been very fortunate in that most of those 55 students came to me with interesting new questions whose answers demanded that I learn from them and they learn from me (and Prue and I are fortunate to count many of them as good friends, some even after almost half a century). More generally, much of my work has been explicitly collaborative. For better and worse, I have cast a broad net in my research, and would have netted fewer and less interesting fish were it not for colleagues who were prepared to pool their knowledge with mine. But there have been failures in interdisciplinary conversation, too, such as a neurophysiologist who felt that as a modeler I would simply be stealing her data, or a psychomusicologist who walked out of a luncheon, offended that I wanted her to explain the basics of her discipline to me.

3.2. What the frog's eye tells the frog

The title “What the Frog's Eye tells the Frog's brain” left open the crucial question for action-oriented perception: “What does the frog's eye tell the frog?” Addressing this question was the real start of making the integration of action and perception a key focus for my work on modeling brains rather than artificial neural networks. This got going with Rich Didday's thesis on modeling of visuomotor coordination in the frog (Didday, 1970, 1976). The proximal stimulus for this work came from David Ingle (1968). Confronted with two fly-like stimuli, the frog will normally snap at just one of them (if at all), but when the stimuli are close, the animal will snap between them. Work on lateral inhibition (Ratliff, 1965; von Békésy, 1967) could indeed address the latter effect, but could not provide a distributed mechanism for snapping at the more potent of two widely separated stimuli. Rich Didday showed how, through distributed interaction – rather than serial computation – a network confronted with several different flies could determine which one to snap at. This was perhaps the first winner-take-all (WTA) circuit, though the need for such a circuit was implicit in Oliver Selfridge's Pandemonium model (Selfridge, 1959): individual demons would shout out their confidence that they had recognized a pattern, the demon-in-chief would then go with the choice of the one who screamed the loudest.

This emphasis on distributed interaction across layers of neurons (in this case with a retinotopic organization) was the basis for the manifesto “The brain is a layered somatotopic computer” (Arbib, 1971):

We present the notion of distributed computation in a layered somatotopically organized computer, present the Pitts-McCulloch scheme for obtaining standard forms, provide anarchic networks for ballistic and tracking modes of behavior, and relate this to the visuomotor activity of the frog. Much research in artificial intelligence seeks efficient ways to implement certain “intelligent” activities on computers, with little concern for the correspondence between the resultant mechanisms and those of the human brain. The present paper, on the contrary, belongs to that line of research which designs its artefacts as models to be used in increasing our understanding of brain mechanisms . . . but [we] may nonetheless hope that our studies will offer clues for the design of future, highly parallel, computers for use in the control systems of robots.

A visit to Japan in 1972 got me thinking about parallels between the frog tectum (posited to be the location of the frog’s WTA for prey selection) and the role of the homologous superior colliculus in primates, the controller of reflex eye movements and a target for cortical control of attention in directing eye movements. A crucial influence here was *Locating and identifying: Two modes of visual processing*, a symposium bringing together David Ingle, Gerald Schneider, Colwyn Trevarthen and Richard Held (1967). They indeed linked visual perception to action, and anticipated work of a decade later on “what” and “where” systems (see below), but emphasizing the critical role of subcortical mechanisms. Schneider’s study of hamsters distinguished a “where” system in the superior colliculus and a “what” system in cortex that allowed the hamster’s behavior to depend on visual patterns whose discrimination was beyond the capabilities of the frog brain. An intriguing follow-up was provided by Humphrey’s (1970) “What the frog’s eye tells the monkey’s brain,” demonstrating that a monkey without primary visual cortex could nonetheless navigate on simple visual cues like well-lit contours though having lost the capability of visual perception (a key precursor for understanding “blindsight” in humans, Stoerig & Cowey, 1997; Weiskrantz, 1986, 1996).

Such results led us (Didday & Arbib, 1975) to formulate a new, conceptual, model of visual scene recognition in which cortex could relay targets to tectum to there be selected by our WTA mechanism to direct visual attention (see also Itti, Koch, & Niebur, 1998; Koch & Ullman, 1985). But the real innovation was our *slide-box metaphor*. Inspired by techniques used to generate cartoons by the Walt Disney studio in which – rather than draw each frame *ab initio* – a frame could often be formed by simply moving and adjusting “cels” used in previous slides to provide a suitably updated image. I will return to this below, where “cels” become “perceptual schemas.”

3.3. Action and perception in the mammalian brain

Complementing the work on the frog were two efforts that got underway at Stanford but were completed at the University of Massachusetts. Both were modeling efforts addressed to systems for which monkey or cat neurophysiology could be directly applied to analysis of human brain mechanisms. Curt Boylls worked on the role of the cerebellum in locomotion, while Parvati Dev studied stereo vision.

3.3.1. Cerebellum

As a graduate student at MIT, I had (as noted earlier) got involved with the Neurosciences Research Program (NRP) founded by Frank Schmidt. After my post-PhD tour of the US in 1963, I took part in an NRP Work Session on mathematical models of the nervous system. In 1966, I was invited to spend the summer (my first with Prue) to discuss the cerebellum at the NRP’s home, the top floor of a mansion in Brookline, outside Boston. There I got to know three people: Curt Bell, an American; Masao Ito, a Japanese who went on to become the dean of the medical school in Tokyo University (the only neuroscientist I know who can successfully throw a boomerang, thanks to his postdoc in Canberra with Sir John Eccles), and Ray Kado, whose grandmother had died when Japanese-Americans were taken from their homes in California in WWII to internment camps.

The summer session in Brookline served as preparation for a conference on *Information Processing in the Cerebellum* held in Salishan, Oregon, May 15–18, 1967, as a collaboration between Curt Bell’s mentor Robert Dow and the NRP. At around the same time, Ito co-authored a book entitled *The Cerebellum as a Neuronal Machine* (Eccles, Ito, & Szentágothai, 1967) in which János (John) Szentágothai revealed the beautiful quasi-crystalline structure of the cerebellar cortex and Eccles and Ito revealed its neurophysiology. Despite the title, the book was not about modeling the cerebellum, but it proved to be an invaluable resource for modelers.

When one gets serious about study of the details of Fig. 1, one must make a life decision: Are you going to live your life at the rightmost level and focus on the neurochemistry and molecular biology, or are you going to live your life at the network level and try to embed the cerebellum in a larger system to try to understand its role in action. Masao Ito’s vote was eventually to focus on the former (Ito, 2002), while my focus was large scale modeling of the role of the cerebellum in motor control.

One of the first attempts to model the role of the cerebellum in the timing of movements was offered by Valentino Braitenberg and Nello Onesto (1962) who focused on the propagation of signals down the parallel fibers that spanned a whole set of Purkinje cells. (I got to know Nello while he was on sabbatical with McCulloch; diverse encounters with Valentino led in due course to my writing

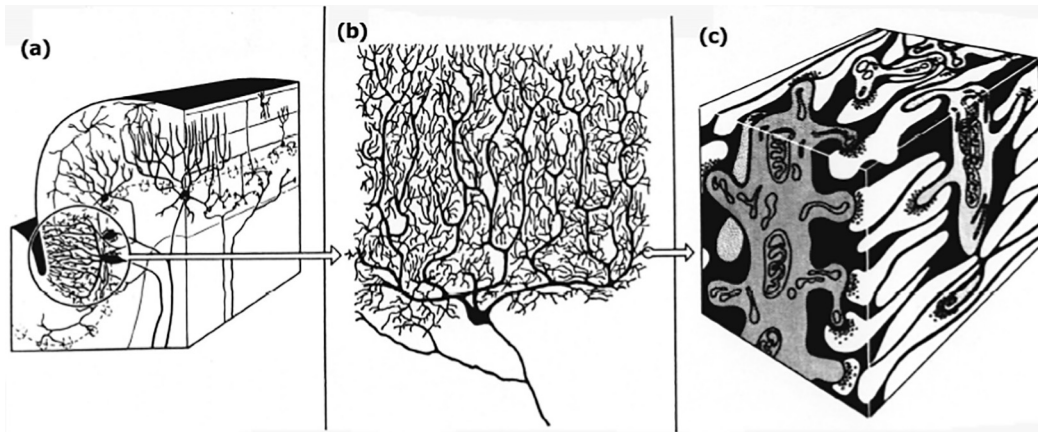


Fig. 1. Three levels of magnification of a fragment of cerebellar cortex. As the magnification increases, we go from a little slice through the cortex, the outer sheet of the cerebellum, to the Purkinje cell, which is the output cell of the cerebellar cortex, to a fragment of dendritic branches showing the synapses as revealed under the electron microscope.

the foreword to his classic book *Vehicles* (Braitenberg, 1984).) However, propagation speed of spikes along the parallel fibers were found to be too fast to support their timing hypothesis. Cerebellar theory was greatly advanced when David Marr (1969) and then Jim Albus (1971) came up with a theory of the cerebellar cortex as a learning machine. However, Curt Boylls and I decided not to look at the learning properties of the cerebellar cortex but instead look at the way this beautiful structure is embedded in much larger systems. We took particular note of work from the Soviet Union (Orlovsky, 1972a, 1972b) inspired by Nikolai Bernstein's (1967) analysis of motor control in terms of synergies, coordinating various muscle groups to act together in varied ways to achieve core motor tasks – if you lose the cerebellum, you can still act, but not gracefully (Holmes, 1939). We thus sought to understand the systems integration that allows graceful movement, rather than just getting shakily to your goal. In agreement with Ito, we viewed the cerebellar cortex and the associated cerebellar nuclei as divided into *microcomplexes*, united via loops with MPGs located elsewhere in the brain – seeing the cerebellum (cortex + nuclei) not as a set of motor controllers so much as the basis for parameter settings which would allow control systems to operate in a more coordinated fashion. In placing cerebellar cortex within a larger system, we were ahead of Marr and Albus, but we were mistaken not to assess the role of learning in our more integrative systems view, a mistake that was not corrected until after my move to USC.

Some insight into the then fraught relationship between theory and experiment in neuroscience can be gleaned from an exchange at a meeting on the cerebellum held in Portland, Oregon, August 1–4, 1972. Curt Boylls was a scrupulous reader of the literature, and studied several hundred papers on the cerebellum and motor control in preparing his thesis. He cited several of these in his talk on his model, but no sooner had he finished than a young German neurophysiologist jumped to his feet and

declared “You rely on the experiments by X and Y, but in my lab we have shown they are wrong. That is why I keep saying that it is premature to do modeling.” To this I replied that, if anything, his evidence showed that it was premature to do experiments! Of course, my true view is that it takes many of us, both modelers and experimentalists, to pursue an ongoing conversation in which we learn from each other's mistakes as well as each other's findings. One more observation about the meeting. Although there were talks by both neurophysiologists studying cerebellar circuits in animal brains and clinicians studying dysfunction related to damage to the human cerebellum, there was remarkably little interchange between the two groups. This situation was transformed by the subsequent development of PET and then fMRI studies in functional imaging of the human brain as a bridge between the two disciplines.

A Symposium on *Neural Modeling* was organized three years later by Ted Lewis (co-author of an early and very influential paper on neural modeling: Harmon & Lewis, 1966) as part of the Society for Neuroscience 5th Annual Meeting (New York City, November 2–6, 1975). There, it was notable that, although the attitude to modeling was supportive, the questions all concerned the experimental data linked to the models, not the methodology or processes of the models themselves. Even now, four decades later, many experimental papers in systems-level neuroscience neither discuss models in any depth nor show an understanding of the interacting processes that modeling seeks to reveal. A paper demonstrating great experimental expertise may conclude with a Discussion section that is simplistic and misleading. Nonetheless, through the efforts of many scholars, the field of computational neuroscience is now flourishing, with many fruitful interactions (often in the same person) between modeling and experimentation at all levels from the synaptic to the cognitive. However, it is outside the scope of this article to offer a roll call of these fine efforts.

3.3.2. Stereo vision

Parvati Dev built on the work of Bela Julesz, who had invented the random dot stereogram in which, amazingly, you can see depth even though the left eye input and right eye input are each totally random. The secret is that correlations between the two – how much one part of the left eye pattern has shifted relative to its position in the right eye pattern – provide disparity cues. Bela Julesz himself had come up with a sort of Rube Goldberg frame of magnets to suggest how he thought this could be done through distributed cooperative computation (Julesz, 1960, 1971). Parvati figured out how to use the available neurophysiology (Barlow, Blakemore, & Pettigrew, 1967) to develop a neural model. Her model (Arbib, Boylls, & Dev, 1974; Dev, 1975) employed a cooperative array of Didday's competitive maximum selectors, with one for each disparity to aid selection of just one perceived depth in each direction, with cooperation between nearby directions to encourage the emergence of surfaces rather than random depth patterns. We were truly upset when David Marr and Tommy Poggio published a minor variation on her model (Marr & Poggio, 1976) yet only mentioned it as an aside in footnote 23, a number which I remember to this day.

3.4. Interstellar communication

Strangely, I was invited to join the likes of Carl Sagan and Frank Drake at a conference held at the NASA-Ames Research Center near Stanford on *Interstellar Communication – Scientific Perspectives*. My contribution, “The Likelihood of the Evolution of Communicating Intelligences on Other Planets” (Arbib, 1974), combined a somewhat simplistic account of the evolution of early nervous systems on Earth with an exposition of von Neumann's theory of self-reproducing automata as the basis for the claim (made perhaps for the first time in the literature) that if spaceships from across the galaxy were to come to our planet, they were more likely to be such machines, rather than bringing living beings. A follow-up paper (Arbib, 1979), based on the estimate that the nearest technologically advanced civilization might be 100 light years away, asked “If we are sending a message to which we would not receive an answer for 200 years, what would it be?” When I asked this question during a talk at Bell Laboratories in Holmdel, New Jersey, a voice from the back of the room shouted out “Goodbye ...” My less pithy answer was to make the case for an *Encyclopedia Galactica* to distill knowledge in a form that could inform an alien intelligence – but with an even greater divergence in interpretations than that of the way people interpret the US Constitution since, no matter how much the culture has changed, we do share the biology of the Founders.

Thirty years later, I was invited to speak at a Symposium on *Interstellar Communication: Semiotic, Linguistic, and Cognitive Approaches*, to be held at Lund University,

Sweden, in May of 2011. I protested that I had nothing new to say, but was over-ruled. Fortunately, I had by then developed a theory of the evolution of human language (much more on this below) and so I set to work to sketch an account of the evolution of a truly alien language. Inspired by what I had learned from J.Z. Young about the octopus in London, and from others subsequently, I invented the *octoplus*. Cephalopods are the most intelligent invertebrates, but they die when they reproduce. My octopluses were an alien life form that had evolved culturally as well as biologically, not only living long after the birth of offspring to transmit social schemas, but also having developed language based on the display of iconic symbols and then more and more abstract ones through their chromatophores (Arbib, 2013a). But, fun though this was, it was a side show.

3.5. The metaphorical brain

Although published in 1972 and thus completed for publication at the University of Massachusetts at Amherst during my first year there, *The Metaphorical Brain: An Introduction to Cybernetics as Artificial Intelligence and Brain Theory* (Arbib, 1972b) may be seen as the capstone of my work at Stanford. It marked the transition in my study of neural networks from an emphasis on automata theory to a concern with their operation in the brains of humans and other animals, and its subtitle declared that both Artificial intelligence and Brain Theory were progeny of cybernetics. The book went very much against the grain of the time when most workers in AI had abandoned its roots in cybernetics and saw little relevance in the study of brain mechanisms. However, it cautioned that the view that “the brain is a computer” must not be read as reducing the brain to the level of the then current technology of serial computers, but rather must expand our concepts of computation to embrace the style of the brain. This pointed the way to a form of computation based on the constant interaction of a variety of concurrently active systems, many of which were expressed in the interplay of spatio-temporal patterns in layers of neurons.

The book's title was inspired by a comment by John McChesney (then studying at Stanford, later active on NPR). I had drawn a range of diagrams showing various functions and structures of the brain and taped them to the hallway wall in the house where Prue and I lived in Los Altos Hills, aiming to go back and forth between these sketches to better understand the diverse relationships that underlie our action and perception. John saw these, and said “Ah. Those are your metaphors.” Some critics derided the book's title, proclaiming that they study the real brain, whereas I only treat of metaphorical brains. They missed the point that it is detrimental to the progress of brain theory to confuse model with reality. A model of the brain is a metaphor which, by its own very special selections of data and modeling tools, enriches our understanding.

My later work on schema theory led me to understand that the title meant more than I had at first realized. A *schema* is a unit of perception, of action, of knowledge. It is acquired through experience, though it may be rooted in innate structures acquired through evolution. As such it is imperfect. We have no direct knowledge of reality, for it is our schemas which mediate it. Knowledge is thus inherently and fundamentally “metaphorical” rather than literal in nature. And so the book is both about metaphors *for* brains, models of how the brain works; and about metaphors *in* brains, about the schemas which mediate our knowledge of the world (as indeed was made fully explicit in *The Metaphorical Brain 2*, Arbib, 1989).

At a time when many people consider talk of “embodied cognition” as something new, and hotly debate whether all of cognition is captured at the sensorimotor level, it is salutary to see that, inspired by Ingle and others, I was already espousing “embodied cognition” (but I called it “action-oriented perception”) back in 1972, and had already placed it in perspective by stressing that human evolution had enriched fundamental sensorimotor mechanisms with new capabilities:

“The animal perceives its environment to the extent that it is *prepared to interact* with it. . . . Perception of an object generally involves the gaining of access to “programs” [schemas] for controlling interaction with the object, rather than simply generating a “name” for the object . . . [L]anguage can best be understood . . . as a ‘recently’ evolved refinement of an underlying ability to interact with the environment.”

This discussion opposed the notion that verbal mediation is *necessary* to link perception and action – there is much intelligent behavior in which it does not intervene. Language is rooted in embodiment and may modulate or be secondary to ongoing embodied behavior – but the argument still holds that language also supports inferences that are abstract rather than embodied. One might know that President Nixon was a male by summoning a visual image with his five o’clock shadow, but most of us cannot summon an image of President Polk, and instead know he is male by inference from the generalization “All presidents of the United States (up to this time of writing) have been male.”

The final section of the book was entitled “Possible Social Implications.” It grew out of concerns that had emerged from interacting with students and faculty at Stanford during the Vietnam war, out of wide reading that included Kurt Vonnegut’s *The Sirens of Titan*, and – especially – from long conversations with Curt Bell (mentioned in the cerebellum section) whose strong concerns about the ethics of conducting invasive neuroscience experiments on animals raised further implications of science and technology for the human condition.

3.6. Mathematics “versus” experience as a social being

When I was a mathematician, I relished the way in which a small set of axioms, coupled with some judicious definitions, could support proofs that yielded results that were totally unexpected. I enjoyed not only the challenge of discovering new theorems, but also that of mastering old proofs to the extent that I could “rediscover” them rather than memorize them, especially when, in the process, I could in turn discover a shorter or more elegant proof, expanding my toolkit of proof techniques in the process. I was impressed at the way in which new theorems might require long and complicated proofs and yet become in time part of the intuition with which I could further explore a mathematical terrain. We ascend “towers of abstraction” which take us further and further from an immediate grounding in everyday experience, and yet – astonishingly (Wigner, 1960) – time and again the purest of pure mathematics give us tools to analyze the physical world, as when Levi-Civita’s tensor calculus for non-Euclidean geometry aided Einstein’s quest for general relativity, or infinite-dimensional Hilbert spaces proved essential for developing quantum mechanics. There are reaches of fundamental theory in which there is no hard line between pure and applied mathematics, but when a scientist turns to the analysis of real systems, the mathematical purity evaporates and expeditious approximations may predominate. What is it about the system one wants to understand? Is the Earth a point on a trajectory around the Sun, or a sphere covered with water subject to tidal forces of the moon, or a complex geological system, or . . . ? Each choice draws one’s attention to certain features, yet leaves the concern that others may be crucial “hidden” variables, and scientific revolutions may turn on what data a scientific community agrees to focus on (Hesse, 1980). The mathematical mindset becomes more problematic a tool as the design of new experiments and the revelation of new features of the systems under study comes to hold sway. And, of course, when it comes to engineered systems, the human needs and desires of the posited future users may become as crucial a factor as any other – all these factors competing and cooperating in a process of “constraint satisfaction.”

Expertise in one field of science, adapting to a changing conception of that field as the years pass, does not guarantee that one can speak with authority on other sciences – and this holds a fortiori when it comes to affairs of the heart or notions of social justice, emotion, or self-interest, or the particular confines of one’s social upbringing. We operate by what I call (just in this section) a post-Gödelian pseudologic. We do not derive theorems from a single consistent set of axioms. Rather, the present circumstances bring forth (consciously or unconsciously) associations based on past experience, some of which may have little relevance to each other, and may even be contradictory. But time is short, and so we converge (whether

analogically or by somewhat logical deduction) on an opinion to utter or a course of action. Of course, one may offer a more or less reasoned account for a decision, but the choice of what to appeal to in the “proof” may depend on a desired conclusion – as in judicial arguments where anything can be “proved,” but some results require a very high-priced lawyer to “prove” them (Arbib, 1969). In short, not only does expertise in one area of science not guarantee expertise in others (e.g., knowing about cosmology confers no expertise in biophysics beyond the underlying facility in mathematics) but scientists with similar expertise may hold widely divergent beliefs about social justice, politics and religion, to mention just a few topics of conversation. One colleague using computational methods to understand gene structure was a creationist – but he told me that he sought to understand God’s design of His creatures.

Thus, when I write articles or books on society such as *Computers and the Cybernetic Society* (Arbib, 1977) or even on theology as in *The Construction of Reality* (Arbib & Hesse, 1986), my expertise in cybernetics conveys no particular authority to my statements on social issues or religion. However, I seek some sort of constraint satisfaction between the different themes covered in each volume, bringing to bear a respect for the available facts as assessed by some form of non-Gödelian pseudologic. In other words, I attempt to think clearly and logically, but there are no well-defined axioms. Instead, the “facts” are gleaned both from authorities (often in disagreement) in the various domains and from personal experience. My views on religion are shaped in part by my experience as a non-observant Jew attending a Presbyterian High School. My views on society are shaped by my experiences in leading various social groupings, by having had the opportunity to reflect on the Cold War while contrasting Western Europe with Eastern Europe and the Soviet Union. My years at Stanford included not only anti-Vietnam War sit-ins but also debate about the place (if any) of classified research on campus, and service on the University Senate. Stanford thus provided a major phase of my education in the complexity of social systems. I do not claim that my expertise in cybernetics makes me an expert on such systems, but I do claim that my knowledge of cybernetics and the brain have combined with my life experiences to allow me to say interesting things about both society and religion.

Chapter 4. The University of Massachusetts at Amherst (1970–1986)

In 1970, I moved to the University of Massachusetts at Amherst (UMass) to create a Ph.D. program and convert the Master’s Level Computer Science Program into a fully fledged department of Computer and Information Science (COINS), serving as its founding chairman. When I was a faculty member at Stanford, Conrad (Connie) Wogrin

came there to recruit new Ph.D. students to join the EE faculty at Yale, and we had a detailed conversation about students and research directions. However, several years later, Connie had moved to UMass to develop their Computer Center and had been given the charge of recruiting someone to head up the new effort in Computer Science. His memory of our conversation convinced him that I was the right person for the job, and he phoned to invite me to Amherst for an interview. Being only 29, this was certainly a great ego boost, but being only 29 perhaps I should have said “No.” However (serendipity), because Henry McKean had asked me to write a second thesis and I had thus spent the summer of 1963 in Dartmouth, I had become a friend of Bill Marsh, who was completing his Ph.D. there. Bill was one of the faculty spending the 1969–70 academic year preparing for the opening of Hampshire College in Amherst, and one effort was directed to getting people in the Four (soon to be Five) Colleges of UMass, Amherst College, Smith College and Mt. Holyoke College to consider Hampshire College as part of their intellectual circle. To this end, they hosted a series of monthly lectures. Bill had invited me to give one of these, I had accepted, and so I said to Connie “Well, I’m going to be in Amherst anyway ...” In September of 1970 I was installed as a Professor of Computer Science at UMass, and during that first year I worked with my new colleagues to develop the Ph.D. proposal and then got it approved by the Trustees. Two of my senior colleagues were unhappy at the appointment of a brash youngster to lead the new Department, but Connie proved a pillar of strength, as well as a good friend, during the following years as the Department made the arduous transition to becoming, as I phrased it, “one of the twenty computer science departments in the top ten.”

This PhD program had the distinction, I believe, of being the first in computer science with cybernetics, inclusive of AI and brain theory, as a strand of the required Ph.D. curriculum. As part of getting approval for the new Ph.D., we hosted a group of reviewers headed by Juris Hartmanis who by then had moved to the Computer Science Department at Cornell (where there was still debate as to whether AI was a legitimate part of CS). Prue and I hosted a party at our house on the evening of the Review – and I had the pleasure of making the introduction: “Juris ... Prudence.”

4.1. Category theory, automata and the semantics of programs

En route to MIT in January 1961, I had visited Columbia University to spend an hour with Samuel Eilenberg. In my final year at Sydney, Max Kelly had taught a course based on *Foundations of Algebraic Topology* by Eilenberg and Steenrod (1952), a book much influenced by the work of Eilenberg and Saunders MacLane (1945) introducing category theory. When my JACM paper was published, I

sent Eilenberg a copy with the jocular dedication “Towards a categorical theory of automata”. Surprisingly, within 15 years both Eilenberg and I were writing on that subject.

The background for my effort was in great part the rapprochement of automata theory and control theory discussed above. To continue this story, I must go back to my undergraduate reading of *The Bulletin of Mathematical Biophysics*. Turning from the McCulloch and Pitts papers of the 40 s, I came upon the then (late 50 s) contemporary papers of Robert Rosen, who was describing cells in terms of interacting automata: a metabolic machine interacting with a reproduction machine to form an (M, R) system (Rosen, 1958, 1959). What captured my attention was that Rosen was describing families of these machines making (very elementary) use of the language of the category theory of Eilenberg and MacLane. My next taste of using category theory to describe automata came with the Ph.D. thesis of Shafee Give'on, whom I met in 1963 when I visited the University of Michigan on my post-Ph.D. tour. Some years later, Shafee came to Stanford as a research associate of Kalman's, and he and I collaborated, building on a paper by Eilenberg and Wright (1967), to use category theory as a tool for transferring insights from the study of “ordinary” automata which process strings of symbols to tree automata (Arbib & Give'on, 1968; Give'on & Arbib, 1968). These are devices which accept as input tree-shaped arrays of symbols (such as a derivation tree to be checked for grammatical correctness; or a tree representing an arithmetic expression to be evaluated by the machine).

By the late 60s, then, it was clear to me that the way to carry through the rapprochement of automata theory and linear system theory was to develop a general theory of “machines in a category” with ordinary automata resulting when the category was chosen to be **Set**, while linear systems resulted from the choice of **Vect** as category — with some further magic being required to show how tree automata could also live in **Set**. I sketched out the theory, but simply did not know enough category theory to carry the sketch through to completion. I discussed my sketch with Joe Goguen, then of the University of Chicago, who had been thinking along somewhat similar lines. He told me that I was missing the concept of adjoint functor. I expected that through correspondence we would develop a joint paper on this, and was thus somewhat taken aback when Goguen published on his own, presumably judging that my intuitions had been too ill-formalized to merit coauthorship.

I had reached an impasse, because I modeled finite automata and linear systems in terms of a next-state map $Q \times X \rightarrow Q$ in the categories **Set** and **Vect** respectively, and did not see how to link this with tree automata. But a year after I moved to UMass, I was joined by Ernie Manes, whose thesis used category theory to describe universal algebras (a topic somewhat related to tree automata). Our conversations soon blossomed into a rich collaboration. Ernie had what I had been missing in my search for a theory of machines in a category, and it was given by his thesis work

on a category-theoretic (triples, monads, algebraic theories) approach to abstract algebra (Manes, 1975). Goguen had described the dynamics of a machine by a morphism

$$Q \otimes X \rightarrow Q$$

where Q is the state object and X is the input object in a suitable category, and Q and X are joined by the tensor product to provide the information required for determining the next state (Goguen, 1972a, 1972b).

Unfortunately, this general formalism could handle neither linear systems nor tree automata. Ernie realized that the key was to consider morphisms of the form $QX \rightarrow Q$ where X is no longer an object but is now a functor. If X was of the form $- \times X_0$ in **Set**, we captured conventional automata; with $- \otimes X_0$ in **Vect**, we got linear systems with input vector space X_0 . Different choices of X in **Set** could yield a desired choice of operators for tree automata. By imposing suitable axioms on X , we gave a general theory of reachability, observability, system identification and duality for machines in a category (Arbib & Manes, 1974). In particular, it was more than sufficient to assume that X had a right adjoint (Arbib & Manes, 1975a). This basic theory of “adjoint machines” included Goguen's theory and the related theory of Hartmut Ehrig of West Berlin as special cases. Of course, once one has an axiomatization the mathematician wants to know how far it can be relaxed while still yielding interesting results. This program was carried through masterfully by Vera Trnkova, Jiri Adamek and their colleagues in Prague (Trnková, Adámek, Koubek, & Reiterman, 1975).

Having seen how to capture the input structure of a machine with the action of a suitable functor on the state object Q , we turned to the issue of nondeterminism. In automata theory, a *nondeterministic* automaton is one in which the dynamics is of the form $Q \times X \rightarrow 2^Q$, i.e., the current state and input determine a *set* of possible next states; while a *probabilistic* automaton is one in which the dynamics is of the form $Q \times X \rightarrow \mathbf{P}(Q)$, i.e., the current state and input determines a probability distribution of next states. Zadeh had by then (1974) achieved great popularity with his concept of fuzzy set (Zadeh, 1965), e.g., characterizing the set of tall men not by an all-or-none criterion but by providing a “membership function” $f: M \rightarrow [0,1]$ from the set of men to the set of all real numbers from 0 to 1 with, say, $f(m)$ being zero for a man m of height less than 5', 1 for men of height more than 6'6", and increasing from 0 to 1 for heights in between. We thus rounded out our gallery by considering dynamics $Q \times X \rightarrow \mathbf{Z}(Q)$ for $\mathbf{Z}(Q)$ the fuzzy sets on Q .

The right general theory should have a dynamics of the form $QX \rightarrow QT$ for functors T with suitable properties, and with 2^Q , $\mathbf{P}(Q)$, and $\mathbf{Z}(Q)$ as examples of QT . It turned out that every example satisfied the requirement that T be part of a larger structure (T, e, m) which was an algebraic theory, the very structure central to Ernie's thesis. To make the point that Zadeh's formalization was only one way of many for axiomatizing the intuitions behind colloquial

use of the word “fuzzy,” we called the resultant paper “Fuzzy Machines in a Category” (Arbib & Manes, 1975c). Unfortunately, it seems that many have read the title but few the paper, for the paper has been cited as an endorsement of Zadeh’s formalization.

During this period (1971–76) we also wrote a book, *Arrows, Structures and Functors: The Categorical Imperative*, to make the category theory used in our work accessible to computer scientists and control theorists. Our research continued. Our first Ph.D. student, Suad Alagić from Sarajevo in the then-existing country of Yugoslavia (little did we know of the tragedies lurking in the wings of history), was in fact the first person to receive a Ph.D. from COINS at UMass. It was a strange feeling for me. When my students had received a Ph.D. at Stanford, I could feel that it was the institution that was conferring the title. But here I was all too aware that it was an instrument that I had created (with a little help from my friends) that allowed Suad to call himself Dr. Alagić for the rest of his life. Fortunately, the ensuing growth of the Department meant that his Ph.D. is indeed a worthwhile credential.

While I was on sabbatical in Edinburgh in 1976–77 (more on this later) and after he had returned to Sarajevo, Suad arrived with the Serbo-Croatian draft of a book on the Hoare-Floyd approach to program semantics. I started by helping him with the English, ended up writing much of the exposition, even doing some original research by showing that the Hoare approach to proof rules for **gotos** was unsatisfactory and introducing and developing a rigorous and practical alternative. (A Dutch computer scientist named Edsger Dijkstra had argued that the goto statement was liable to be misused by programmers. His slogan was “Eliminate the go to.” There was a computer scientist named Eichi Goto in Tokyo. I wondered if his slogan was “Eliminate the Dijkstra.”) The result was our 1978 text, *The Design of Well-Structured and Correct Programs* (Alagic & Arbib, 2013).

Also in Edinburgh, I attended the Burstall-Plotkin-Milner seminar on theory of computation, and found that the tools that Ernie and I had developed were akin to those being developed in algebraic semantics. This led us to work on program semantics after I returned to Amherst. The most influential approach to the semantics of programs was then that of Dana Scott (the Scott of Rabin & Scott, 1959; our careers had overlapped at Stanford), based on viewing the semantics of a program as a continuous map between suitably axiomatized ordered sets. We felt strongly that the semantics of deterministic programs lived in **Pfn**, the category of sets and partial functions (partial because not all computations terminate; thus the importance of Turing’s halting problem). Initially, however, we could not get a handle on this and, influenced by work of Dijkstra, Ernie suggested we focus on nondeterministic programs whose semantics lived in **Mfn**, the category of sets and multi-valued functions (relations) – noting that one could get some genuine insights by viewing this as a special case of what was known as an additive category. However,

while lecturing on this work during a visit to McGill University, I discovered how to axiomatize the “partial addition” of partial functions, and this led to the notion of a partially additive category and thence to our theory of partially-additive semantics (Manes & Arbib, 1980). This confirmed my anti-Platonist take on mathematics. By the time one gets to partially additive categories, it seems to me that we are far from the discovery of pre-existing Platonic forms and firmly in the realm of invention. In *Conversations on mind, matter, and mathematics* (Changeux & Connes, 1999), the neuroscientist Jean-Pierre Changeux and the mathematician Alain Connes revisit this age-old debate from a new perspective.

We made one attempt to bring our ideas to a non-mathematical audience, namely that of philosophers of science (Arbib & Manes, 1975b, republished 1983) at a Seminar at Boston University, with discussants Dan Dennett, the philosopher of cognitive science, and Joe Weizenbaum, programmer of the primitive but somewhat persuasive virtual psychiatrist, ELIZA. The most memorable part of the evening was that Weizenbaum, having failed to understand the mathematics, chose to use his time to claim that the whole effort was a fraud. It was the only time I have used an obscenity in responding to a discussant. He had had a draft of the talk for over a month, but rather than using his incomprehension as a basis for contacting us for clarification, he instead chose a cheap stunt.

In our last collaboration, started in Amherst and completed after I had moved to Los Angeles, Ernie and I wrote the book *Algebraic Approaches to Program Semantics* (Manes & Arbib, 1986). It provides an exposition of all our work on algebraic semantics as well as of Scott’s approach and work on equational specification. Ernie has continued to work in the areas that we had defined and in related areas of pure mathematics, but this proved to be my farewell appearance as a mathematician. Since then, my students and I have used mathematics to establish the formal framework for computer simulation of diverse functions and structures of the brains of humans and other animals, but I have proved no more theorems. One of the penalties of a full intellectual life is that one may become engaged in too many areas to be able to meet all the challenges that each provides. Old avenues of research must be closed to leave time to explore new avenues as they open up before one.

4.2. My “frog period”: *Rana computatrix*

Modeling of action-oriented perception in the frog was reinvigorated by Rolando Lara, the first of 11 post-Diddy students to work with me on this topic at UMass and then USC. I called our series of frog models *Rana Computatrix*, and I forgot (or was not conscious of) why I did this until Dan Dennett suggested that it must be based on Grey Walter’s use of the term *Machina speculatrix* for what was perhaps the first interesting biological robot of the cybernetics era (Walter, 1953). This was a mechanical

“tortoise” which would roam around at random, but whenever its battery was low it would return to its hutch to recharge. Damper (2003) presents papers from a workshop on the legacy of Grey Walter’s work. The same Damper had revealed that ARBIB was an acronym for *Autonomous Robot Based on Inspirations from Biology* (Damper, French, & Scutt, 2000).

4.2.1. From ethology to neuroethology

The field of *ethology*, the study of animal behavior, received a great boost when the Nobel Prize in Physiology or Medicine for 1973 was awarded jointly to Karl von Frisch, Konrad Lorenz and Nikolaas Tinbergen “for their discoveries concerning organization and elicitation of individual and social behavior patterns”. Karl von Frisch was recognized for his research on the “language” of bees (which is not a language in the same sense as human languages); and Konrad Lorenz for his studies of “fixed action patterns” in birds that were elicited only by specific “key stimuli”, and his discovery of “imprinting”; while Nikolaas Tinbergen used dummies to precisely define the features of key stimuli eliciting specific behaviors, and also studied the organization of instinctive behavior, e.g., the complicated structuring of actions that constitutes the stickleback’s courtship and reproductive behavior. But there had also been many studies in *neuroethology*, seeking to tease apart various neural networks linking perception and action in diverse creatures. This work in turn was complemented by computational studies, such as J.Z. Young’s interest in models of learning in the octopus and the Ewert-von Seelen study of toad approach and avoidance described below. One particularly influential set of studies came from the laboratory of Werner Reichardt in Tübingen who also advanced the field by founding the journal *Kybernetik* (later renamed *Biological Cybernetics* as English became the lingua franca of science) to publish cybernetic models of animal behavior.

Much of the recent research on “embodied cognition” reduces to the observation that some cognitive behavior is correlated with overt actions of the hand or other part of the human body and seeks to reduce cognition to sensorimotor mechanisms without paying heed to the complementary mechanisms that must have evolved in the human brain to support, e.g., language and abstract thought (Arbib, Gasser, & Barrès, 2014). Moreover, much of this work pays no heed to the earlier insights on “action-oriented perception” or neuroethology (see Ewert, 1980, for a classic exposition; and von Uexküll, 1957, for an early perspective on the different perceptual worlds of different species). The ignorance of neuroethology is a dramatic weakness because most of the work pays no heed to what sort of body the cognition is embodied in – something to which neuroethology pays particular attention when, e.g., comparing the role of the tectum in directing whole body movements in the frog in relation to snapping movements of the tongue, and the role of its homologue the superior colliculus in monkeys and humans

directing eye movements in relation to reaching movements of the arm.

4.2.2. Schema theory for *Rana computatrix*

Much of the work on *Rana computatrix* involved modeling biologically plausible networks of neurons, preferably with testable hypotheses on the neuroanatomical localization of these networks, but the work also introduced a higher-level approach based on interacting functional units rather than interacting neurons. It was Richard Reiss who suggested that I call these units “schemas” with these being assessed within a version of *schema theory* akin in some way to that of Piaget, but with major differences as well (more on that below). As noted earlier, my approach to schema theory was informed in part by the competition and cooperation of modes and modules in the Kilmer-McCulloch RETIC model of the reticular formation. The classic example of schemas for *Rana computatrix* is the comparison of two models of pattern-recognition in the toad based on the work of two German colleagues, Peter Ewert and Werner von Seelen (Ewert & von Seelen, 1974). I distinguish perceptual schemas from motor schemas. A *perceptual schema* corresponds to a particular class of objects, and becomes more active to the extent that an observed object appears to belong to that class. Conversely, a *motor schema* corresponds to a course of action, and increasing activity there signals increasing readiness to perform that action. A crucial tenet of my version of schema theory is that perceptual schemas may pass parameters to motor schemas, such as the position of a prey or predator. The first model is based on the behavioral observation that if a toad sees a small moving object it will snap at it, but if the toad recognizes a large moving object, then the motor schema for avoidance (jumping away) would be activated.

To make this into a brain model, even at this coarse level of schemas, we need to hypothesize where the schemas are implemented. Ewert and von Seelen first hypothesized that the perceptual schema for small moving objects is mediated by the tectum while that for large moving objects is mediated by the pretectum. The first mini-model (Fig. 2, left) then predicts that if its pretectum were removed, the toad would snap at small moving objects but ignore large moving objects. However, Peter Ewert did the experiment – he lesioned the pretectum of a toad – and found that, surprisingly, the creature snapped at *all* moving objects.

This led to a slightly modified model (Fig. 2, right) in which the tectum has a perceptual schema for “all moving objects” (not just small ones) whereas the pretectum is still the recognizer for large moving objects. Crucially, when the pretectum recognizes a large moving object, it not only triggers the motor schema for avoidance but also inhibits the motor schema for snapping. The model then not only explains the behavior of the normal toad but also explains why, when the pretectum is lesioned, any moving object can trigger a snap. Note how the *apparently* unitary perceptual schema for small moving objects of the first model is

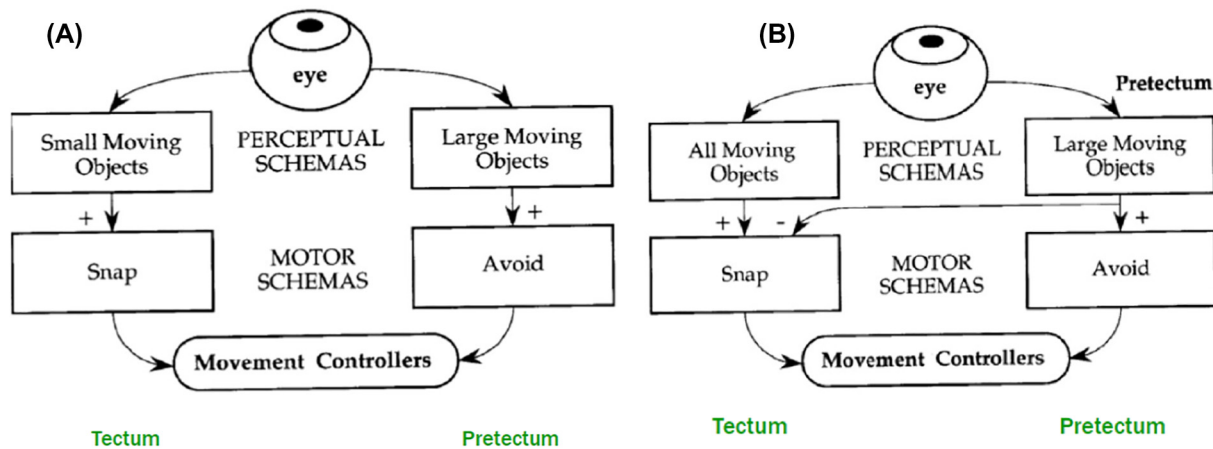


Fig. 2. Two schema-level models of approach and avoidance in the toad. Model B is more robust in that it can handle data on lesion of the pretectum as well as observations on normal behavior.

realized in the revised model by a network of subschemas distributed across multiple regions.

The moral is that even gross lesion studies can distinguish between alternative top-down schema-based analyses relating brain and behavior. In any case, we may distinguish between two variants of schema theory: “Basic” schema theory seeks to assess action, perception and cognition in terms of interacting schemas but does not concern itself with whether or how schemas relate to brain regions or neural networks. As such, it has proved useful not only in (non-neuro) psychology but also in robotics. “Neural” schema theory then brings neuroscience data into the picture. It is to neural network models for *Rana Computatrix* that we now turn.

4.3. *Rana Computatrix* at UMass and USC

Rather than burden this article with details of further work on *Rana Computatrix*, I will simply list the eleven post-Diddy theses (1982–1997, with a late follow-up in 2009), and then offer a few comments.

Rana computatrix at the University of Massachusetts

Rolando Lara (1982): Neural Models of the Visuomotor System of Amphibia.

Donald House (1984): Models of Anuran Depth Perception.

Francisco Cervantes-Pérez (1985): Modeling and Analysis of Neural Networks in the Visuomotor System of Anuran Amphibia.

Yillbyung Lee (1986): A Neural Network Model of Frog Retina.

Jeffrey Teeters (1989): A Simulation System and Model for the Anuran Retina.

Rana computatrix at the University of Southern California

DeLiang Wang (1991): Neural Networks for Temporal Order Learning and Stimulus Specific Habituation

Jim-Shih Liaw (1993): Visuomotor Coordination in Anurans, Mammals, and Robots

Hyun-bong Lee (1994): Visuomotor Coordination in Anurans – Detour Behavior

Fernando Corbacho (1997): Schema Based Learning: Towards a Theory of Organization for Adaptive Autonomous Agents

Mathew Lamb (1997): Modeling Behavior Based Depth Vision in Frog and Salamander

Jeffrey Begley (2009) Modeling the Integration of Salamander Vision and Behavior.

The work on *Rana computatrix* benefited from close interaction between modelers and experimentalists, and included Workshops at the University of Massachusetts, the Universidad Autonoma de Mexico (organized by Roland Lara), Kassel in (then) West Germany (organized by Peter Ewert), and Sedona, Arizona (organized by Kiisa Nishikawa), and two related symposium volumes (Arbib & Ewert, 1991; Ewert & Arbib, 1989). See also a pair of overviews, with commentaries, by Peter Ewert and myself (Arbib, 1987; Ewert, 1987).

Ewert’s work (Ewert, 1984, 1987) on how changing parameters in the shape of a moving rectangle (from “worm” to “antiworm”) can affect whether or not it is responded to as prey, and with what frequency, motivated the work of Lara; Cervantes and, to some extent, Wang. Interactions with both Ingle and Ewert were greatly stimulating yet also revealed tensions in the modeler-experimentalist relation. Ingle was distressed that we modeled diverse aspects of frog behavior rather than focusing on just those problems of current interest to him – though Ingle and Hoff (1990) certainly inspired Liaw. With Ewert the problem was that parametric variation in a model by Cervantes-Perez that could explain some of his data could – with different parameter settings – also explain

conflicting results from another group that he strongly contested. It took some effort to persuade him that it was indeed a power of modeling to define parameters that could be subject to new experiments as the basis for possible resolution of such conflicts.

Another source of inspiration came from the work of Tom Collett (1982) on depth and detours. This inspired the neural models of Donald House which in turn inspired the work on mobile robots that Ron Arkin conducted while at UMass (Arkin, 1989). Work of Paul Grobstein (Grobstein, 1991; Masino & Grobstein, 1989) inspired new schema-based models of prey-catching and predator-avoidance which decomposed the overall behaviors into subactions, and took spatial relations between frog and stimulus into account (Cobas & Arbib, 1992). And many other experimentalists enriched the set of data which challenged our diverse efforts. Working with data from Kiisa Nishikawa and Ananda Weerasuriya, Fernando Corbacho introduced learning into the schema analyses of *Rana computatrix* (Corbacho, Nishikawa, Weerasuriya, Liaw, & Arbib, 2005a, 2005b).

Our collaborations did have one tragic outcome, however.

On 19 January 1985, brain theory and neuroethology suffered a grievous loss. A car bearing Rolando Lara, Elena Sandoval, and Willi Borchers on a road near Cuernavaca, Mexico, was hit by a truck. All three were killed. Rolando Lara and Elena Sandoval were colleagues at the Centro de Investigaciones en Fisiología Celular of the Universidad Autónoma de México, he a brain theorist, she a neurochemist; while Willi Borchers was from the Arbeitsgruppe Neuroethologie und Biokybernetik, Universität des Landes Hessen, Kassel, where he was a neuroethologist with a background in biological control theory. Borchers was in México for a month to help Lara and Sandoval establish a laboratory in which both theoretical and experimental studies of brain mechanisms of visuomotor coordination could go hand in hand.

[Arbib, 1985b]

4.4. Conceptual models of neural organization

Back to the NRP. Szentágothai, the Hungarian whose beautiful work on the quasi-crystalline anatomy of the cerebellar cortex has already been mentioned, was invited to organize a Work Session on stereology, the development of methods to chart the three-dimensional structure of neural circuitry. I was invited to join the small group helping Szentágothai develop the guest list and the program for the meeting. For better or worse, I persuaded the group to include computational models of the function of brain regions, and not just their neuroanatomy, in our remit, switching from a focus on stereology to a more computationally informed perspective.

The Work Session was held October 1–3, 1972, and was published in due course in the *NRP Bulletin* (Szentágothai & Arbib, 1974). It was later issued as a free-standing book (Szentágothai & Arbib, 1975) which was published in a Russian translation by Olga Vinogradova (an expert in neural mechanisms of habituation) in 1976 with a foreword by Luria. The book was both co-authored and co-edited in that it combined extended material based on talks at the Workshop with a great deal of material written by Szentágothai and myself. It included a functional overview, a structural overview, and in-depth treatments of stereopsis and cerebellum. Both Dev and Boylls were participants and their work is clearly presented in the book along with data from neurophysiology, anatomy and modeling from the other participants.

A sequel, *Neural Organization: Structure, Function, and Dynamics* (Arbib, Érdi, & Szentágothai, 1998), was published 26 years after the Workshop. The idea of writing this book was formed when the three authors took part in the first week of a School organized by Francesco Ventriglia on “Neural Modeling and Neural Networks” on the Isle of Capri in October of 1992, a week which included the celebration of John Szentágothai’s eightieth birthday. We discussed the idea of revisiting *Conceptual Models of Neural Organization* but John felt that the task of developing a genuinely new volume might be too onerous without the aid of a Hungarian colleague. The choice naturally fell to Peter Érdi with whom John had written papers on the self-organization of the nervous system. The work proceeded through the exchange of numerous drafts between Budapest and California (I had by then moved to USC) and two visits to Budapest to review what each of us had written and to better coordinate our contributions, including places where each of us could contribute to sections that the others had drafted. A memorable event during the first visit was having lunch in a crowded restaurant when John noticed two men waiting for a table, one of whom was George Soros, the billionaire who had by then devoted so much of his fortune to easing the transition of Eastern Europe to democracy. John invited them to share our table while we finished our meal. I had recently read an article in the *New York Review of Books* by Soros about the plight of Sarajevo in the war that destroyed Yugoslavia. He talked of the millions he had spent for wells to be drilled to keep Sarajevo supplied with water, and of the millions he was setting aside to build a multi-ethnic university to help heal the wounds of “ethnic cleansing” once the war was over. How strange to have a meal at which, discussing the events of the day, one is no longer saying “They should do X and Y,” whoever they may be, but is talking to someone who was actually doing it. (By contrast, lunch with “my other billionaire,” Bill Gates – at the conference where I spelled out my ideas on Sixth Generation Computing (Arbib, 1988) – was boring because he was then still running Microsoft so that as soon as we began to touch

on any interesting topic on the future of computing, the blinds of proprietary information would quickly close off the topic. By now, though, with his passion for the work of the Bill and Melinda Gates Foundation, I imagine that he would be a fascinating luncheon companion indeed.)

John died in September of 1994, between the planning of the second visit and the visit itself. A lengthy draft of the book had been completed by the time and, indeed, John was working on the book the very day he died. (The Preface of the book concludes with a short account of John's seminal career in neuroscience.) Nonetheless, Prue and I went ahead with our visit to Budapest so I could work with Peter. We visited Alice, John's widow (to whom we dedicated the book), while Peter and I developed our strategy for integrating the material John had left us into the final version. When Prue and I went to check out of the hotel, we learned that "Professor Szentágothai had already paid the bill. He arranged for payment by the Academy of Sciences." A spooky moment, with John as a living presence in that hotel.

Where *Conceptual Models of Neural Organization* started with an overview of the Structural and Functional Approaches to neural organization; Peter Érdi added a third dimension to the new book, the Dynamic Approach. Before reviewing the form these three approaches took in *Neural Organization: Structure, Function, and Dynamics*, it is worth a short trip back to 1964. I had the sad good fortune that Norbert Wiener died on March 18, 1964, so that the subject of cybernetics was very much under discussion when *Brains, Machines and Mathematics* was published a couple of months later. A side effect of this was that the book received the lead review in the June 1964 issue of *Scientific American* (my career peaked when I was 24!) by the scientist/philosopher Jacob Bronowski. Overall, he seemed to approve of the book, but he ended on a prophetic note:

Arbib's book traverses all the cybernetic subjects (rather too many) in a compact manner and with more mathematical backing than is usual. He cannot be blamed for the sense of disappointment we feel at the end of the book: the sense that the cybernetic models already belong to the past, and begin to have as old-fashioned an air as the model that used to describe the brain as a telephone exchange. Perhaps we can go no further in elucidating the machinery of life by analogies drawn from engineering. If so, the time has come to learn a more radical approach from more modern sciences. At the beginning of this century physics was at a standstill until Max Planck and Albert Einstein broke away from the explanations of natural phenomena that would satisfy engineers. It is my belief that the mechanism of the brain will not be explained until a new generation of biologists invents new concepts as unexpected and as audacious as Planck's and Einstein's.

[Bronowski, 1964, p.134]

Looking back more than 50 years later, we can certainly agree that the study of finite automata and Turing machines is at best a philosophical constraint for brain theory (the limits of the computable). Perhaps we are still waiting for a Planck or an Einstein, but for me the comparison with physics is mistaken. Some scholars do argue that the brain's secrets can only be unraveled at the most basic of physical levels. For example, [Hameroff and Penrose \(2014\)](#) propose that consciousness depends on biologically orchestrated coherent quantum processes in collections of microtubules within neurons and that their theory "establishes a connection between the brain's biomolecular processes and the basic structure of the universe" and that "consciousness plays an intrinsic role in the universe." However, I agree with the current consensus in neuroscience that this is irrelevant to "how the brain works." We must approach each brain as a system of systems, with different classes of subsystems each demanding their own mathematical treatment, and with different patterns of subsystem integration linking structure to behavior posing their own challenges. The brain is a highly evolved subsystem which is best analyzed at multiple levels from the basic biophysics and molecular biology of membranes and synapses to the properties of neurons and small neural circuits on up through layers and columns of neurons to interacting schemas distributed across multiple brain regions – an analysis which must even extend to "multi-agent" analyses of social interaction. Just as statistical mechanics supports thermodynamics, so may we examine processes in the brain at a single level, or make forays down the levels to seek more detailed mechanisms or up the levels to understand a system's contribution to overall functioning. In short, I doubt that we will see a Planck-Einstein breakthrough that reshapes the foundations, but I agree with Bronowski that the cybernetics of 1964 left much to be learned, even had I supplemented my book with an account of the Hodgkin-Huxley equations and Hebbian learning.

Feedback and feedforward remain vital to the study of motor control and behavior, while much has changed in the last half century. Schema theory, constraint satisfaction, notions of cooperative computation (competition and cooperation) have all made their mark in high-level analyses of brain function. Analysis of dynamic fields (described later) provides tools for assessing the interaction of neural layers, complemented by models of pattern formation as "nature" and "nurture" interact in the individual "wiring up" of each brain. Powerful tools probe the dynamics of interacting neurons, the propagation of membrane potentials within neurons, and the functioning of synapses. Finally, the brain is a dynamic system not only within the ongoing action-perception cycle linking perception, decision-making and action ([Fuster, 2004](#); [Neisser, 1976](#)) but also on the longer scale of learning and memory (and, alas, aging) and here new theories of learning (e.g.,

spike timing dependent plasticity, reinforcement learning) have emerged, and have all been subject to both mathematical analysis and computer simulation. In short and in my opinion, the analysis of brains requires a diversity of breakthroughs – but the purpose of this riff on Bronowski’s review was to make clear the importance of adding Peter’s Dynamical Approach to John’s Structural Approach and my Functional Approach in our 1998 book. Here then is an outline of the three approaches (Arbib et al., 1998, pp ix-x):

Structural Approach: Studies of brain function and dynamics build on, and contribute to, an understanding of many brain regions and of the neural circuits which constitute them. We thus review anatomical data that integrates the overall spatial relations between a variety of brain regions with a selection of critical details of neural morphology and synaptic connectivity. This analysis of neural structure is guided by a developmental view which approaches the complexity of the adult nervous system through an understanding of the way in which that complexity emerges during embryogenesis, thus linking the structural approach to dynamical models of self-organization. The developing nervous system can generate movement before it becomes responsive to sensory stimuli, consonant with the emphasis on action-oriented perception in our functional studies, analyzing the ways in which sensory systems are specialized to serve a variety of behaviors. As a basis for our functional and dynamical analysis of a variety of systems, later chapters progress through regions of the brain which, singly or in combination, underlie these systems: the segmented part of the neuraxis, the olfactory system, the hippocampus, the thalamus, the cerebral cortex, the cerebellum, and, finally, the basal ganglia.

Functional Approach: We first approach complex functions such as the control of eye movements, reaching and grasping, the use of a cognitive map for navigation, and the roles of vision in these behaviors, by the use of *schemas* in the sense of units which provide a *functional* decomposition of the overall skill or behavior. A schema account becomes a brain model when we offer hypotheses as to how each schema is implemented through the interaction of specific brain regions. A brain-based schema model may be tested by analysis of the behavior of animals with localized lesions or reversible inactivation of specific brain regions or by human brain imaging; such a model provides the basis for modeling the overall function by neural networks which plausibly implement (usually in a distributed fashion) the schemas in the brain. Further analysis may then proceed bottom up (as the neural data drive further research) as well as top down (as we refine our schema-theoretic formulations).

Dynamical Approach: Dynamic system theory offers a conceptual and mathematical framework to analyze spa-

tiotemporal neural phenomena occurring at different levels of organization, such as oscillatory and chaotic activity both in single neurons and in (often synchronized) neural networks, the self-organizing development and plasticity of ordered neural structures, and learning and memory phenomena associated with synaptic modification. We discuss a variety of rhythms (and arrhythmia) found in the olfactory bulb and olfactory cortex, in the hippocampus, and in the thalamocortical system. In most cases we relate these rhythms to memory functions. We also study learning rules in both developmental processes (self-organization) and in the acquisition of a variety of behaviors. In this way, we ground our functional analysis of neural organization in a dynamic systems analysis of the neural networks which implement the basic schemas.

We not only offered a tripartite framework but also showed how the pattern of application of its various components (whether mathematical, through computer simulation, or conceptual) must change as we shift attention from one neural system to another or the larger systems they form. I believe that the book remains worthy of careful study even after twenty years, as researchers realize that their increasingly specialized studies need to be calibrated against, and help calibrate, an overarching framework for understanding the brain.

4.5. Schema theory in perspective

Let’s briefly assess schema theory within a historical perspective. The concept of schema in neurology goes back to [Head and Holmes \(1911\)](#) who introduced the notion of *body schema* to make sense of those people who had parietal lesions and “lost” half their body, so that in getting dressed they would put clothing on one side of the body and leave it off the other. Common sense would deny that one could ignore half of one’s body but, in fact, with a parietal lobe lesion on one side of the brain, the representation of the other side of the body may be lost. Building on this, Frederic Bartlett, who was a student of Head’s, published his book *Remembering* ([Bartlett, 1932](#)) on the role of schemas in memory. He observed that people do not generally remember in a photographic way. If you tell somebody something and ask them to repeat it somewhat later, they do not repeat the exact words, but offer a reconstruction, recalling certain aspects and then providing words to re-express them. Here, then, is the notion of remembering not as a passive reactivation but as a constructive process in terms of the schemas that each person brings to bear.

Thereafter come various papers within cybernetics, starting perhaps with [Kenneth Craik’s \(1943\)](#) little book on *The Nature of Explanation*, which says that the job of the brain is to model the world, so that when you act it is because you have been able to simulate the effects of your action before you do it. In our terms, we might say that perceptual schemas activate motor schemas that can be

run off-line to predict the outcome of various actions before deciding to proceed with one of them. Here we see, far in advance, the ideas on forward and inverse models brought into play by Mitsuo Kawato, Daniel Wolpert and others (Haruno, Wolpert, & Kawato, 2001; Wolpert & Kawato, 1998).

Piaget belongs to another stream in schema theory that perhaps begins with Kant rather than Head & Holmes. One of his books, *The Construction of Reality in the Child* (Piaget, 1954), offers a constructivist view of the mental development of the child as starting with a few sensorimotor schemas. He offers a rich descriptive theory of the types of schemas the child develops as she (or he) grows from infant to adult – going from very simple motor skills to more abstract skills in which the child begins to have a notion of an object in general rather than just interacting with particular things; to a situation in which she begins to use formal arguments; and finally to the stage where, when appropriate, she can use logical arguments to solve a problem by using abstract knowledge (Arbib, 1990). A key notion is *assimilation*: if you are in a new situation, you try to make sense of it in terms of the schemas you have, assimilating the situation to what you already know. But from time to time you will find yourself in a situation where these schemas are inadequate; what you know will not suffice and your stock of schemas must change. Piaget calls this *accommodation*; the schemas accommodate to the data they cannot assimilate; learning modifies old schemas and creates new ones.

When I talk about schema theory, I am usually talking about my own brand (further detailed in later sections), but owing a debt to earlier contributions. I find many of Piaget's insights very helpful. The key difference is that I think in terms of assemblages of schemas whereas Piaget talks more in terms of the influence of one type of schema at a time. Another difference is that Piaget seems to talk as if the stages are predetermined steps in maturation whereas I want to see the interaction and accommodation of schemas as underlying a process whereby eventually the schemas will develop a “common style” that we can describe as a new stage. Piaget does not give a system theory or computational model of how schemas change, although he had a very interesting collaboration with the mathematical logician Evert Beth (Beth & Piaget, 1966) and co-authored a little known work with Seymour Papert and others (Apostel, Grize, Papert, & Piaget, 1963).

Why many schemas rather than one at a time? When you behave in a particular situation, you have to recognize many things – the people around the table, the table itself, where the margarita is – and this means that you have different perceptual schemas for recognizing the drink, the table, and the people. You also have ways for combining those schemas so that you can represent a novel situation and yet call upon prior knowledge to make sense of that situation. I thus introduced the notion of a *schema assemblage*, a network of interacting schemas pulled together at any time that represents the situation. Thus, there must

be mechanisms whereby schemas can be activated, whereby schemas can compete and cooperate with each other, and whereby this can lead to activation of *motor schemas*. Moreover, in most of my work I have the extra dimension of concern for brain mechanisms so that, as in the case of Fig. 2, the initially hypothesized schemas may need to be changed. Piaget is biological, but his biology is not that of the brain. He makes analogies between embryology and mental development, his *genetic epistemology*, and that's a different game. (McCulloch talked of *experimental epistemology*.)

4.6. Vision

4.6.1. High-level: schemas for human-style understanding of visual scenes

An effort at UMass that has had a great influence on my subsequent thinking was the development of the VISIONS system for schema-based interpretation of visual scenes by my colleagues Ed Riseman and Allen Hanson (Draper, Collins, Brolio, Hanson, & Riseman, 1989; Hanson & Riseman, 1978; Riseman & Hanson, 1987). This was inspired in part by the work on schema theory for *Rana* computatrix, and in part by the HEARSAY-II system (Lesser, Fennel, Erman, & Reddy, 1975), perhaps the first AI system to develop a blackboard architecture.

In HEARSAY, data are processed at different blackboard levels, separating data such as phonemes, words, phrases and the overall interpretation. There may at any time be alternative hypotheses at each level for any one stretch of the speech stream, each with different confidence values. Processes called *knowledge sources* link hypotheses at different levels, supporting the competition and cooperation which will limit ambiguities. In addition to data-driven processing which works upward from the speech data, HEARSAY also uses hypothesis-driven processing so that when a hypothesis is formed on the basis of partial data, a search may be initiated to find supporting data at lower levels. A hypothesis activated with sufficient confidence will provide context for determination of other hypotheses. However, such an *island of reliability* need not survive into the final interpretation of the sentence. All we can ask is that it forwards the process which eventually converges on this interpretation – e.g., by raising only the most relevant hypotheses above some threshold. The key point for brain theory is that, although HEARSAY was implemented on a serial computer and thus required a scheduler to specify the order in which knowledge sources were activated, the underlying logic is not serial – rather it involves competition and cooperation between units with assigned confidence levels which vary until a state is achieved in which the units with highest confidence levels can be read out to provide the overall interpretation.

The VISIONS system integrates low-level processes (akin to those well studied in the neurophysiology of visual cortex from Hubel & Wiesel onwards) with high-level pro-

cesses for recognizing objects and relations. (Neurophysiology, after all these years, still has relatively little to tell us about these higher-order relations, so schema models still seem most appropriate.) The low-level processes take an image of an outdoor visual scene and extract an *intermediate representation* – including contours and regions tagged with features such as color, texture, shape, size and location. Adapting terminology from the work on frogs, the processes that then analyze different features of the intermediate representation to form confidence values for the presence of objects like houses, walls and trees are called *perceptual schemas*. The knowledge required for interpretation is stored in LTM (long-term memory) as a network of schemas, while the state of interpretation of the particular scene unfolds in Working Memory (WM – though Hanson and Riseman used the less suitable term short-term memory, STM) as a network of schema instances defined in terms of continuing relevance.

The innovation for schema theory here was in the notion of *schema instance*. In the frog models, the data supported the implementation of a perceptual schema for, e.g., small moving objects as a retinotopic array of neurons whose local activity would signal the confidence that the sort of object that could elicit approach behavior was present in the corresponding locations in the external world. However, it is implausible that we have retinotopic arrays of house detectors or other classes of objects that were not part of our deep evolutionary history. Rather (recall the slide-box metaphor of Didday & Arbib, 1975 – though VISIONS treated static images, rather than considering the updating of representations of a dynamically changing scene), humans exploit eye movements – driven by peripheral cues as well as top-down influences from working memory – to attend to different parts of the scene and assess, with varying confidence levels, what each part of the scene may hold. We are thus led to distinguish the perceptual schema for an object (the general process that can assess its presence, primarily from foveal data) from the *schema instance* that combines each application of the schema with the region for which it was invoked, and which then lays down a trace in working memory of the confidence level associated with that *instantiation* of the schema. In a computer program, we could posit that each region is associated with pointers to the schemas for which it has served to base an instantiation, along with the confidence values associated with each schema instance. However, the neural mechanism for building such an array of schema instances in the mammalian brain (“populating and updating the slide box”) remains, as far as I know, an open and under-discussed challenge (see Arbib & Liaw, 1995, for further discussion).

Interpretation of a novel scene in VISIONS starts from the intermediate database with the data-driven instantiation of several schemas (e.g., a certain range of color and texture might cue an instance of the foliage schema for a certain region of the image). When a schema instance is activated, it is linked with an associated area of the image

as segmented in the intermediate database, and an associated set of local variables. Each schema instance in WM has an associated confidence level which changes on the basis of interactions with other units in WM – each object represents a context for further processing. Thus, once several schema instances are active, they may instantiate others in a “hypothesis-driven” way (e.g., recognizing what appears to be a roof will activate an instance of the house schema to seek confirming evidence in the region below that of the putative roof). Ensuing computation is based on the competition and cooperation of concurrently active schema instances until a coherent scene interpretation of (part of) the scene is obtained. Cooperation yields a pattern of “strengthened alliances” between mutually consistent schema instances that allows them to achieve high activity levels to constitute the overall solution of a problem. As a result of competition, instances which do not meet the evolving consensus lose activity, and thus are not part of this solution (though their continuing subthreshold activity may possibly affect later behavior). Successful instances of perceptual schemas become part of the current understanding of the environment. On occasion, the interaction of schema instances may trigger further processes that update the intermediate data base, as when irresolvable competition between two schemas for a region calls for its resegmentation.

Moving beyond purely visual WM for objects in retinotopic location, sensory stimulation is always interpreted in action-oriented perception current within the ongoing state of the organism. Thus, in general, WM is dynamic and task-oriented and must include a representation of goals and needs, linking instances of perceptual schemas to motor schemas, providing parameters and changing activity (confidence) levels. As their activity levels reach threshold, certain motor schemas create patterns of overt behavior but with parameters set by the relevant instances of perceptual schemas. For example, consider a driver instructed to “Turn right at the red barn”. At first the person drives along looking for something large and red, after which the perceptual schema for barns is brought to bear. Once a barn is identified, the emphasis shifts to recognition of spatial relations appropriate to executing a right turn “at” the barn, but determined rather by the placement and angle of the roadway, and so on. All this is “planning” in a flexible representation strongly conditioned by current goals. Arbib and Liaw (1995) thus suggested extending VISIONS by the inclusion of motor as well as perceptual schemas and the dynamic interaction of working memory with changing sensory input. Activity in IT (inferotemporal cortex) accentuates perceptual schemas for the current focus of attention. In this case, only a few intermediate schemas may be active, with WM being updated as new results come in from this focal processing. This leads us to reverse the view of activity/passivity of schemas and instances in the VISIONS system. There, the schema in LTM is the passive code for processes (the program for deciding if a region is a roof, for example), while the

schema instance is an active copy of that process (the execution of that program to test a region for “roofness”). By contrast, it may be that in the brain, the active circuitry is the *schema*, so that only one or a few instances can apply that circuitry at a time to update WM.

Such considerations offer a different perspective on the neuropsychological view of working memory offered by Baddeley (2003). The initial three-component model of working memory proposed by Baddeley and Hitch (1974) posits a *central executive* (an attentional controller) coordinating two subsidiary systems, the *phonological loop*, capable of holding speech-based information, and the *visuospatial sketchpad*. However, the latter is passive, since Baddeley focused on the role of working memory in sentence processing. Baddeley (2003) added an episodic LTM to the Baddeley-Hitch model, with the ability to hold language information complementing the phonological loop and (the idea is less well developed) an LTM for visual semantics complementing the visuospatial sketchpad. He further added an *episodic buffer*, controlled by the central executive, which is assumed to provide a temporary interface between the phonological loop and the visuospatial sketchpad and LTM. The Arbib-Liaw scheme seems far more general, because it integrates dynamic visual analysis with the ongoing control of action (and see “Modeling How the Brain May Support Language” in Chapter 7).

4.6.2. Low-level: affordances and optic flow

J.J. Gibson (1966, 1979) stressed that there is much more to vision than the conscious classification of objects. His key notion is that of *affordances* – invitations to/indications for action. For example, optic flow – as you move, patterns of visual features flow across the retinas – can provide key information for navigation including possible collisions and “time until contact” unless you change your trajectory (Lee & Kalmus, 1980; Lee & Lishman, 1977). In stereopsis, you have two views of the world from a slightly different place and can use it to estimate the depth of objects in the world. In optic flow, separation is in terms not of the distance between the eyes, but of the times between the snapshots, as it were – so that time rather than space provides the measure of separation. (Poggio, Torre, & Koch, 1985, provide a shared mathematical framework for stereo and optic flow.)

Gibson not only pioneered affordances, he saw them as examples of direct, unmediated perception. We had a debate at UMass in which I agreed with Gibson on the importance of affordances but stressed that, while their extraction may be “unconscious,” it nonetheless depends on the operation of suitable neural networks. John Prager and I developed a cooperative computation algorithm for optic flow (Prager & Arbib, 1982).

4.6.3. From theoretical biology to linguistics

A thread that started at MIT that has not yet been mentioned is an interest in theoretical biology beyond the confines of brain theory to include, for example, models of

embryological development. I came to this via the work of Brian Goodwin (Goodwin, 1963) who spent several months with the McCulloch lab, and through him I came to know his supervisor from his days at the University of Edinburgh, C.H. Waddington, whose publications include the highly influential *The Strategy of the Genes* (Waddington, 1957) with its stress on epigenesis. And this in turn led to my invitation to participate in two of the four meetings that Waddington arranged at Bellagio on Lake Como (Waddington, 1968–1972). Here is how they are recalled by Brian Goodwin (2008, p.286)

When I was at MIT in Cambridge, MA, Wad had visited and we discussed with various friends I had made there the possibility of a series of meetings exploring theoretical biology. Wad then organized these, the result being a remarkable series of conferences involving physicists, mathematicians, computer scientists, geneticists, developmental biologists, philosophers, linguists, and others on the conceptual foundations of biology. I remember animated discussions between David Bohm, John Maynard Smith, Waddington, Howard Pattee, Richard Gregory, and other participants on the nature of biological organization in relation to physical principles. In subsequent conferences with Richard Lewontin, Lewis Wolpert, Stuart Kauffman, and others, the role of information, genetic programs, and the relationships between development and evolution were discussed, anticipating the contemporary emergence of EvoDevo as an integrative movement in biology of the type that Waddington always promoted.

At one of these meetings, I became friendly with Jimmy Thorne, a delightful Welshman and a Reader in English Literature at the University of Edinburgh. Through Jimmy, I got to know Samuel Jay Keyser, who was then the head of linguistics at UMass. I was invited to speak at the Linguistic Society of America Golden Anniversary Linguistic Institute which was held at UMass in June of 1974. The Program is still available online (LSA, 1974) and records the remarkable roster of linguistics stars who spoke there, as well as the abstract for my somewhat off-beat take on the subject:

Highlights from “The Metaphorical Brain: An Introduction to Cybernetics as Artificial Intelligence and Brain Theory.” Relation of this material to problems of Interest to linguistics. Topics Include the secondary role of language in human perception; the evolution of language; comparative studies of human language with the song of birds and the communication systems of other organisms; analogies between human language and the techniques currently used for communicating with robots.

More importantly, Jay opened his house to a faculty seminar (possibly starting in 1974 or 1975) which brought together people in the linguistics department with people at the cybernetics/AI end of COINS. What was most fasci-

nating was how very differently we thought about language. It took us about two months to understand that we were using the term “semantics” in completely different ways. For the linguists, it meant logical form, so that “Every girl loves a sailor” could mean.

$$\forall x(\text{Girl}(x) \Rightarrow \exists y(\text{Sailor}(y) \wedge (\text{Loves}(x, y)) \vee \exists y((\text{Sailor}(y) \wedge \forall x(\text{Girl}(x) \Rightarrow (\text{Loves}(x, y))).$$

By contrast, the COINS view of semantics involved understanding how computers might recognize a girl or a sailor and thus what features characterize girls and sailors, while also building on ideas from the HEARSAY system for speech understanding. However, the discussion of our initial misunderstandings provided a firm basis for developing shared interests without ignoring the different perspectives on offer.

In 1975–76, Jimmy became my host for a sabbatical in Edinburgh at the School of Epistemics in Edinburgh, which has since mutated into the School of Cognitive Science. When asked to define epistemics, I explain that epistemics is to epistemology as physics is to physiology, which is about as clear as it gets. At that time, the School’s main function was to put on a public lecture about twice a term, after which the group would retire to the Staff Club (which had an excellent collection of single malts) and drink the night away. I had the privilege of giving three of those talks during my year there. By then Waddington had died, but I nonetheless had a stimulating year learning more about theoretical biology (including discussions with Donald Ede, who also spent time at UMass), and discussing philosophy with Jimmy and Barry Richards (who later spent a sabbatical at UMass, where we co-taught a seminar “Beyond Chomsky and Piaget,” which scans in ironically similar fashion to B.F. Skinner’s *Beyond Freedom and Dignity*). While in Edinburgh, I also worked on some of my LSA ideas on looking at neurolinguistics from a computational perspective with the help of John Marshall (more on this later), and interacted with people versed in theory of computation (as mentioned above), and even (see Chapter 5) paved the way to add neurotheology to neuroethology in my list of interests.

4.7. The Sloan years

Grants from the Sloan Foundation proved important in the latter half of the 1970s. One supported the short life of the Center for Systems Neuroscience at UMass, the other provided funds to inaugurate the Five-College Cognitive Science Program, which is still going strong today.

4.7.1. The Center for Systems Neuroscience (CNE)

George Moore, a long-time professor of biomedical engineering at USC, introduced me to Ken Klivington, who was program manager for neuroscience at the Sloan Foundation, and this in turn led to Sloan funding for the Center for Systems Neuroscience (CNE, 1975–1978), which I co-directed with Bill Kilmer and Nico Spinelli, both of

whom I had hired as professors in COINS. After his time with McCulloch at MIT, Bill had joined the faculty at Montana State University and then Michigan State University before joining me to develop brain theory at UMass. Nico Spinelli had been a neurophysiologist working with Karl Pribram at Stanford and was best known for his work on the effects of early experience on the neural populations of visual cortex (Hirsch & Spinelli, 1970), which he built upon at UMass (Spinelli & Jensen, 1979).

Sloan Funding allowed us to bring in several visiting scientists for a year each, and to support two researchers as staff people even as they worked on their Ph.D.s. Art Karshmer handled our (very primitive by today’s standards) computing facility and Fred Lenherr acted as general scientific facilitator for the visitors while also editing *The Brain Theory Newsletter* (Efraim Katzir, then President of Israel, was one of our subscribers). The Newsletter became part of a new journal, *Cognition and Brain Theory*, which in due course became folded into *Cognitive Science* in its early days as the journal of the Cognitive Science Society. Alas, the latter journal is currently rather sparse in its offerings in brain theory/systems neuroscience. One of our visitors, Jacqueline Metzler – well-known for foundational work on mental rotation (Shepard & Metzler, 1971) – edited a book, *Systems Neuroscience* (Metzler, 1977), which assembled a range of contributions from the Center. Karshmer and Lenherr wrote on design criteria for CORETEX (magnetic cores had been the primary medium for fast computer memory), a simulation language for brain models. This got nowhere, but survived as the inspiration for development of the Neural Simulation Language (Weitzenfeld, Arbib, & Alexander, 2002) at USC. For me, the most significant benefits of the Center were the collaborations with Israel Liebliich and Shun-Ichi Amari.

4.7.2. Beyond the hippocampus to the world graph

Israel Liebliich was a Professor at the Hebrew University in Jerusalem, and he changed the focus from frogs to rats, assessing the large literature on maze running by rats such as the observations on detour behavior that grounded Tolman’s (1948) “Cognitive maps in rats and men” and lesser known papers such as those of Strain (1953) on “Establishment of an avoidance gradient under latent-learning conditions” and N. E. Miller (1959) on “Extensions of liberalized S-R theory.” The result was “Motivational learning of spatial behavior” (Arbib & Liebliich, 1977) which was later elaborated as a target article for commentary in *Behavioral and Brain Sciences* (Liebliich & Arbib, 1982). Our challenge was to go beyond Tolman in explaining the data in a way that escaped the stimulus-response confines of behaviorism. Instead, we exploited an internal state together with plausible processes for linking input, output and state to explain the animal’s behavior. We introduced the notion of the *World Graph* as a collection of nodes that represented landmarks in the animal environment, with the edges representing the actions for getting from one node to

a nearby node. Crucially, the nodes were not only associated with recognition criteria (what makes me think I am here?) but also with data related to motivation, such as the availability of food or water, or the likelihood of receiving an electric shock. Algorithms were then supplied for the computation of need-dependent paths to get to a node which satisfied an appetitive drive or to avoid a node which was aversive. We also included an account of exploration which could add new nodes or merge nodes or add new edges as well as updating motivational information. If the modeled animal knew that a certain path was obstructed, the available processes could still find a route to a desirable place, thus yielding detour behavior.

The work of Strain and Miller showed that the “navigating by landmarks” approach based on the World Graph alone was insufficient. In one study, a rat was placed in a linear maze with landmarks A-B-C-D-E-F. On some occasions, the rat received an electric shock at F, on other occasions the hungry rat received food. Starting at A, B, C or D, the animal would move swiftly to E and start toward F. But here’s the interesting thing. Once it got close enough to F, the fear of the shock would overwhelm its desire for the food and it would retreat – until its fear receded and it again advanced towards the food that might be in F. The animal oscillated back and forth. We thus had to complement the World Graph for confidently planning the overall behavior (feedforward, as it were) with the availability of local “locometric” maps which, when life got complicated, could support gradients which could underlie less confident behavior.

Note the date, 1977, for our initial publication. By that time, the basic discovery of place cells in the hippocampus (O’Keefe & Dostrovsky, 1971) was well known, but the seminal book *The Hippocampus as a Cognitive Map* by John O’Keefe and Lynn Nadel (1978) had not yet been published. Our hypothesis was that the hippocampus itself is *not*, by itself, a cognitive map – it is more like a chart table on which different maps for different locales can be displayed in that a cell may be active for different mazes, coding a place in each of them, but with no obvious relationship between the 2 places. This would seem to accord well with the representation of the animal’s local space, but we posited that this must be complemented by a World Graph style of representation, possibly located in prefrontal cortex. The “you are here, in the current neighborhood” of the place cells must be complemented by a representation of where you want to be in the larger environment, together with mechanisms for converting place representations for X and Y into a plan for getting from X to Y. Later work at USC built on these ideas.

4.7.3. From frog prey-catching & stereo vision to dynamic fields

We earlier discussed how Bronowski, in his review in *Scientific American*, complained that *Brains, Machines and Mathematics* did not contain a new mathematics

appropriate to the study of the brain. This raises the question “What is the Mathematics of the Brain?” and I suggested earlier that different questions about the brain will require different approaches. We also saw that one crucial form of mathematics, applicable to modeling the activity of neural networks as well as pattern formation and learning was that of nonlinear dynamics, as Érdi emphasized in helping develop *Neural Organization: Structure, Function, and Dynamics* (Arbib et al., 1998). Indeed, expertise in non-linear dynamics and stability theory for neural networks is what Shun-Ichi Amari, from Tokyo University, brought to his year at the Center for Systems Neuroscience. Our collaboration built on my work with Rich Didday and Parvati Dev and Amari’s prior theory of neural networks, to come up with a mathematical theory of computation and cooperation in neural nets (Amari & Arbib, 1977). The paper developed stability theory for a variant of Didday’s winner-take-all network, a purely competitive network, showing that the number of “winners” when the network achieved equilibrium for a given stimulus pattern could be controlled by varying the strengths of excitation and inhibition in the network in relation to the range of strengths of stimuli the network encountered. Consideration of the Dev model then yielded insights into stability conditions for a network which exploited cooperation between the elements in a spatial array of competitive modules. This paper was one of the founding documents for the theory of *dynamical fields*, which may be seen as the realization of the earlier notion of a layered somatotopic computer. Another vital, but quite separate, contribution to this field came from Jack Cowan and his colleagues, after he had established himself at the University of Chicago, with the earliest well-cited contribution dating back to 1972 (Wilson & Cowan, 1972, 1973). Ermentrout and Cowan (1979, 1980) provided further early examples of Cowan’s still ongoing research program. Meanwhile, Amari has developed *information geometry*, of great importance in neural network learning and in machine learning more generally (Amari & Nagaoka, 2007).

An open question: How can we best combine the non-linear mathematics that Amari and Érdi opened up for us with the competition and cooperation of schema instances in the version of schema theory I developed, building a bridge between the dynamics and the symbolic? And how indeed are schema instances implemented in the brain?

4.7.4. Reinforcement learning

The greatest contribution to brain theory developed at UMass was that of temporal difference (TD) learning. Here my role was more that of midwife than contributor. Harry Klopf, who was a research manager for AFOSR, had thought long and hard about the then current models of learning in neural networks (primarily, adjusting synaptic weights according to the Hebb or Perceptron rules) and asked the question “What’s in it for the neuron?” He offered the first pass on an answer in his account of “pleasure-seeking” neurons that “seek” excitation and

“avoid” inhibition, which was later developed into his book *The Hedonistic Neuron* (Klopf, 1982). Before I left for my Edinburgh sabbatical (1977–78), I put it to him that UMass would be the ideal place for research to develop his ideas more rigorously. He agreed, on one condition: that funding for a PhD student should be provided to Rich Sutton who had impressed him with his work as an undergraduate in Psychology at Stanford University, work that was to culminate in 1978 with a senior thesis on “A unified theory of expectation in classical and instrumental conditioning.” The funding was eventually approved and, since I was by then on sabbatical, Nico Spinelli and Bill Kilmer took on the task of hiring a post-doc on the grant, and their choice was Andy Barto (whom I had met while he was a graduate student at the University of Michigan; we had kept in touch thereafter). In due course, the partnership of Sutton and Barto yielded Rich’s Ph.D. thesis, “Temporal credit assignment in reinforcement learning,” with Barto as adviser. The thesis was to be followed in turn by several papers and the seminal book *Reinforcement Learning: An Introduction* (Sutton & Barto, 1998) and neurophysiological discoveries linking dopaminergic input to the basal ganglia to the prediction error in the temporal difference equations. All this set the course for long and productive careers for Sutton (now a Professor at the University of Alberta) and Barto (who stayed on at UMass and is now Professor Emeritus).

4.7.5. The diversity of cognitive science(s)

As Sloan Foundation funding for the Center came to an end, so did its funding for Neuroscience. Instead, Ken Klivington redirected the attention of his part of the Foundation to Cognitive Science – a field as ill-defined as it was exciting. George Miller, he of *Plans and the Structure of Behavior*, wrote his personal history of “The cognitive revolution” (Miller, 2003) for which he was indeed one of the key revolutionaries. He charts his growing disenchantment with behaviorism (recall the earlier discussion of the World Graph), and cites his Harvard colleague Jerome Bruner as well as Bartlett, Piaget and Luria as among those who contributed vigorously to non-behaviorist approaches to psychology during the dominance of behaviorism in the US. He also noted with approval Wiener’s cybernetics and the early work on AI by Minsky, McCarthy, Newell and Simon. But for him, a prime factor was that “Chomsky was single-handedly redefining linguistics. (p.142)” However, while Chomsky’s automaton-theoretic formulation intrigued me as a graduate student I hold that it eventually proved “anti-cognitive.” (I had seen that his formulation of classes of grammars defined new and insightful special cases of Post string rewriting systems (Post, 1936, 1944) which were computationally equivalent to Turing machines, but I also came to see that such computations were far removed from “the style of the brain.”)

Miller’s story again intersects my own in 1978

The Sloan Foundation had just completed a highly successful program of support for a new field called ‘neuroscience’ and ... [was] thinking that the next step would be to bridge the gap between brain and mind. I [Miller] learned of the Foundation’s interest in 1977 from Kenneth A. Klivington ... I lobbied for support of cognitive science, in which six disciplines were involved: psychology, linguistics, neuroscience, computer science, anthropology and philosophy. I saw psychology, linguistics and computer science as central, the other three as peripheral. ... Each, by historical accident, had inherited a way of looking at cognition and each had progressed far enough to recognize that the solution to some of its problems depended crucially on the solution of problems traditionally allocated to other disciplines. The Sloan Foundation accepted my argument and a committee of people from the several fields was assembled to summarize the state of cognitive science in 1978, and to write a report recommending appropriate action. The committee met once, in Kansas City. (p.143)

Barbara Hall Partee of Linguistics (an MIT graduate student contemporary) and I were the two UMass members of that committee; Samuel Jay Keyser who had, alas, by then left UMass for MIT was there as well. Although no consensus emerged in Kansas City as to the exact nature of cognitive science, Barbara and I were subsequently able to argue (aided in no small part by the interactions that Jay had previously helped us foster) that the linguists, philosophers, neuroscientists and computer scientists around Amherst could offer an exemplary approach to cognitive science. We succeeded, and Sloan funded the first few years of the 5-College Cognitive Science Program, which continues to this day. Unlike the CNE which was a group tightly organized around the activity of three professors, the Cognitive Science Program helped faculty from many different departments across the five colleges organize seminars on topics of shared interest, run workshops to bring scholars to campus to address cognitive approaches to some focused theme, bring in visitors for longer periods, and even hire new faculty (Lyn Frazier was the first).

A particular payoff of Sloan support for me was that it brought David Caplan, a leading aphasiologist, to UMass. We co-taught a graduate seminar in which I learned a great deal from him about aphasia and psycholinguistics, while I developed ideas on a schematheoretic approach to the Interaction of brain regions supporting language, embedding language in a framework of action-oriented perception. This collaboration led to a paper we published in *Behavioral and Brain Sciences*, “Neurolinguistics Must Be Computational” (Arbib & Caplan, 1979). The commentaries, while stimulating, revealed that my theoretical framework was hard for the psycholinguists and neurolinguists of that period to comprehend (and subsequent writings by David Caplan show no trace whatsoever of my theory!). Sloan funding also

supported a workshop on *Neural Models of Language Processes*. One of the participants was John Marshall, my guide in matters aphasiological back in Edinburgh, and he joined us in editing the conference volume (Arbib, Caplan, & Marshall, 1982).

The schema-theoretic approach to language was then developed in four PhD theses.

Helen Gigley (1982): “Neurolinguistically Constrained Simulation of Sentence Comprehension: Integrating Artificial Intelligence and Brain Theory” approached aphasia from a schema-theoretic point of view.

Jane Hill (1982): “A Computational Model of Language Acquisition in the Two-Year-Old” modeled processes underlying the week-by-week changes in language production Jane observed in a 2-year-old.

Jeff Conklin (1983; David McDonald was his primary advisor) “Data-driven indelible planning of discourse generation using salience.”

Bipin Indurkha (1985): “A Computational Theory of Metaphor Comprehension and Analogical Reasoning” This was inspired in part by Mary Hesse’s theory of metaphor which we had combined with schema theory in *The Construction of Reality* (Arbib & Hesse, 1986) (see Chapter 5).

The first three were presented in the book *From Schema Theory to Language* (Arbib, Conklin, & Hill, 1987). The most important of these theses for my own development was Jane Hill’s. Her model of language acquisition was very different from the then reigning view that Noam Chomsky had espoused – the idea of an innate Universal Grammar such that if you put a child in a language environment, her experience would trigger the settings within the universal grammar encoded in the brain for the child to yield the grammar for that language community. I sent a paper that Jane and I had written to Noam and wrote, “Here’s a very different approach to acquisition from yours and I’d really like your comments.” His reply was to the effect of “Well, Michael, we know your theory is wrong – so I’m not going to read your paper, but here is the truth.” This was in the days before word processing, so he kindly typed out a whole page and a half of what “the truth” was about language acquisition. Needless to say, I’m not convinced that his “truth” could address the week-by-week data that Jane had accumulated (Arbib, 2007).

Back to George Miller. He concluded his reminiscences by saying

Some veterans of those days question whether the program was successful, and whether there really is something now that we can call ‘cognitive science’. For myself, I prefer to speak of the cognitive sciences, in the plural. But the original dream of a unified science that would discover the rep-

resentational and computational capacities of the human mind and their structural and functional realization in the human brain still has an appeal that I cannot resist. (p.144)

I am happy to agree with George Miller. My own research program can be labeled cybernetics or cognitive science. Both my degrees are in mathematics, yet I can now claim to be a cognitive scientist, a computer scientist and a neuroscientist, and to the extent that I have succeeded it is in great part because my research has involved collaboration with many colleagues with many different departmental affiliations. Thus, whether a university has a department of cognitive science (as at UCSD) or a program to encourage interactions across departments (as at UMass), or simply enables people with different expertise to talk to each other, people can work together to “discover the representational and computational capacities of the human mind and their structural and functional realization in the human brain” using diverse methods linking researchers across a range of disciplines.

When I designed the core for the Ph.D. program for COINS at UMass, including cybernetics and biological models of computation was a daring innovation. Yet, at the same time, faculty at Cornell were still debating whether AI was properly part of computer science, let alone at its core. In short, for any challenging subject, the range of issues is so broad, and the data sets and methodologies so diverse, that one cannot expect a small set of courses or concepts to be accepted as a comprehensive “core” for all these efforts, and no single department can field all the talents needed even for any one university’s take on that subject. (Even physics spans so far, e.g., from cosmology to biophysics, that even with an agreement that all students must take basic courses in Newtonian dynamics, quantum mechanics and relativity, each specialization branches into realms in which other specialists have no expertise). The challenge is to create a network where, even though no one expert can comprehend the whole subject in detail, each one has a good enough understanding of aspects of one or two other specialties for paths of shared comprehension to be formed which allow breakthroughs in one area to in due course to diffuse to and inform even rather separate subfields. Breakdowns occur when (as is indeed more common than not) experts believe that their subsubfield holds the only key to unlocking the riddles of the subject as a whole.

4.8. Schemas linking eye and hand

David Ingle, who had helped launch my research on “what the frog’s eye tells the frog,” co-organized with Mel Goodale and Richard Mansfield a summer school at Brandeis University in 1979 on *Analysis of visual behavior* (the papers were published by the MIT Press in 1982). Two of the papers there added a crucial dimension to my

brain theory – the linkage of eye and hand in reaching and grasping, one by Jeannerod & Biguer on humans, the other by Ungerleider & Mishkin on monkeys.

Marc Jeannerod, who became a valued friend and colleague and was indeed a pioneer in action-oriented perception as approached from neuropsychology (Jeannerod, 1988, 1997), presented his analysis with Jean Biguer (Jeannerod & Biguer, 1982) of how, as the hand moves to grasp an object, the brain uses visual data to coordinate the movement of the arm to bring the hand to the object, with the preshaping of the hand and its orientation approximating (with a safety margin) the shape and orientation of the object (the affordances of the object for grasping). Seeking to understand how the linkage of perceptual and motor schemas as studied in the frog might apply to this example was catalyzed by an invitation from Vernon Brooks (whom I knew through my involvement in the cerebellum community following my time with Bell, Ito and Kado in the summer of 1966) to write the one theoretical article for the volumes on *Motor Control* he was editing for the *Handbook of Physiology – The Nervous System*. The resulting article was an extended meditation on “Perceptual structures and distributed motor control” (Arbib, 1981) which included a schema-theoretic analysis of the Jeannerod-Biguer observations. The dots at the bottom of their figure (Fig. 3, left) indicate where the thumb tip is in successive frames – showing a very fast movement and then a slow movement as the hand encloses the object. In the schema model (Fig. 3, right), perceptual schemas capture where the object is, what its size is, and what its orientation is. These can drive two different motor schemas: one telling the arm how to reach to position the hand (with sub-schemas for a fast feedforward phase and a slow feedback phase), and one telling the hand how to preshape and orient during the fast phase and then (with a motor sub-schema triggered by the transition within the reaching schema) how to enclose and grasp the object with tactile

and visual feedback. This model was widely adopted for instruction in kinesiology, complementing an earlier approach to schemas for motor skill learning (Schmidt, 1975). Although it has now been replaced by more subtle models, it remains a successful first approximation with continuing pedagogical value.

The other important Brandeis paper for my later work was that of Mishkin and Ungerleider (1982) who reported on monkey experiments on visual memory which distinguished a ventral path for “What,” supporting memory of what visual cue indicated the position of food for later retrieval, and a dorsal path for “Where,” supporting memory of the location of food for later retrieval. (Contrast the earlier what/where distinction posited by Schneider for cortical “what” versus superior colliculus “where.”) However, the crucial paradigm for my later work was based on a different task (and thus, presumably, involving somewhat different subpathways of the dorsal and ventral streams) which distinguished a ventral “What” pathway for being able to describe (either verbally or through pantomime) properties such as the size or orientation of an object and a dorsal “How” pathway (expanding upon “Where”) that could exploit the size and orientation of the object in pre-shaping accurately during the reach-to-grasp (Decety, Jeannerod, & Prablanc, 1989; Goodale, Milner, Jakobson, & Carey, 1991).

More serendipity: Ian Darian-Smith, an expert on the cerebral cortex and the control of the hand, read my *Handbook* article and mistakenly thought I was an expert. He invited me to speak at a 1983 IUPS Satellite Symposium on *Hand Function and the Neocortex* in Melbourne. But I didn’t have anything to say on the subject except for that one schema-diagram. Fortunately, two graduate students at UMass – a man with a charming Irish accent, Damian Lyons, and a very energetic woman, Thea Iberall – agreed to work with me to assemble a new paper on control of the hand in time for the conference (Arbib, Iberall, & Lyons,

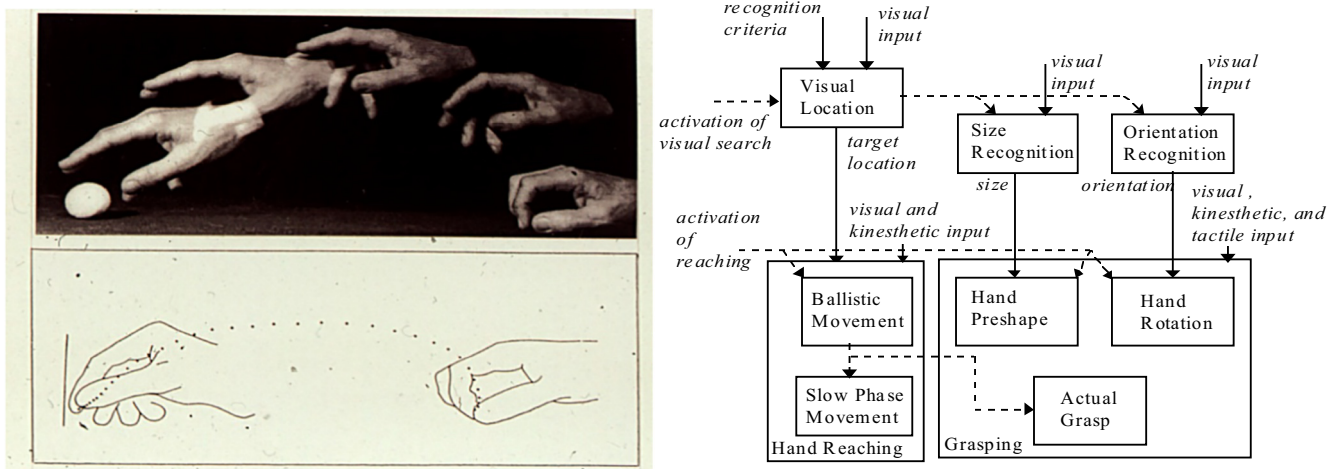


Fig. 3. (Left) Preshaping of the hand while reaching to grasp (top); the position of the thumb-tip traced from successive frames shows a fast initial movement followed by a slow completion of the grasp (bottom). (Adapted from Jeannerod & Biguer, 1982.) (Right) A coordinated control program linking perceptual and motor schemas to represent this behavior. Solid lines show transfer of data; dashed lines show transfer of control. The transition from ballistic to slow reaching provides the control signal to initiate the enclose phase of the actual grasp. (Adapted from Arbib, 1981.)

1985). This provided the notion of “opposition spaces,” linking affordances of the object to “effectivities” of the hand (Iberall, Bingham, & Arbib, 1986) and to “RS robot schemas” (Lyons & Arbib, 1989), so the collaboration not only led to new insight into what the brain has to do to control the hand at the schema level, but also gave us ideas for robot control. Indeed, Thea took a post-doctoral position with George Bekey when I moved to USC, and they had a fruitful collaboration centered on the control of the USC-Belgrade robot hand (Bekey & Goldberg, 1993; Bekey, Tomovic, & Zeljkovic, 1990).

At UMass, a Salisbury hand (a robotic hand developed by Ken Salisbury at MIT) coupled with a robot arm under the control of the control theorist Theodore E. Djaferis provided the core for the development of the Laboratory for Perceptual Robotics (LPR) which helped pioneer the transition from robots engaged in purely stereotypic actions to robots whose action-oriented perception could adapt their performance to current circumstances. LPR saw not only the application of perceptual and motor schemas to robot arms and hands (and recall Arkin’s frog-inspired mobile robots), but also supported the work of students engaged in what was then state of the art touch sensors (Arbib, Overton, & Lawton, 1984).

Chapter 5. Natural theology: Mary Hesse and *The Construction of Reality*

Perhaps the most surprising outcome of my Edinburgh sabbatical came from a conversation with Donald Michie, who was both the grand old man and enfant terrible of artificial intelligence, or machine intelligence as he called it, in Edinburgh. The University of Edinburgh and other Scottish Universities annually host the Gifford lectures in Natural Theology, the attempt to infer the nature of the Creator from the nature of His Creation. There was a precursor of the Gifford lectures in the early 1800s called the Bridgewater Treatises “on the power, wisdom, and goodness of God as manifested in the Creation.” One of these is by the great neuroanatomist Sir Charles Bell, *The Hand: Its Mechanism and Vital Endowments, as Evidencing Design* (Bell, 1834), Treatise IV of the series. He looked at how the fin of the whale, the hands of different animals, and the hand of the human share a basic skeletal ground plan. A few years later, Darwin would enable us to see this as evidence for natural selection, but Bell saw it in terms of the parsimony of God’s design. There were eight Bridgewater Treatises, and Charles Babbage, he of the universal computing machine of the early 1800s, wrote the unofficial ninth treatise (Babbage, 1837), where he countered the claim by the Rev. William Whewell in an earlier treatise that “We may ... deny to the mechanical philosophers and mathematicians ... any authority with regard to their view of the administration of the universe.”

Having read several books derived from the Gifford lectures (especially James, 1902; Sherrington, 1906; and Whitehead, 1929), I said to Donald Michie that it would

be fun to give the Gifford Lectures. I carefully said this in such a way that if he were to burst into helpless laughter at the suggestion, I would not have to appear offended because it could be treated as if I had been joking. But he said yes and put my name to the Gifford committee, but he also suggested that a theologian share the lectures with me. In due course, I was invited and in 1980 they found a partner to share the 1983 lectures, namely Mary Hesse, Professor of History and Philosophy of Science at Cambridge University. Mary was not a theologian, but did have a deep understanding of philosophy of religion. We had never met each other and did not know each other’s work, but it turned out to be a good choice. We had three years to get ready to give the lectures. I put to Mary the notion that we would write a book together – eventually published three years after the lectures (Arbib & Hesse, 1986) – and then each of us would base our lectures on the drafts of certain chapters that would have already been our joint effort. We met twice in Cambridge in her College and once at UMass, and slowly built this book. But first we set to educating each other. Mary had given the Stanton Lectures on the Philosophy of Religion in the Faculty of Divinity in Cambridge in 1978–1980 and shared the text of these lectures with me as we sought to establish common ground. I sent her *The Metaphorical Brain*.

5.1. The individual and social construction of reality

Our theme was “The Construction of Reality,” which is a Piagetian theme, looking at the way in which the schemas we already have shape the way we make sense of the world to create new schemas. I approached this as one who had done brain modeling and artificial intelligence and robotics in terms of the schemas in the head. Mary had been actively engaged in understanding *social* construction, namely how, given a plethora of data, a group of scientists could come to agree on what data were most important as well as the structure of a theory that could make sense of these data and lead on to novel predictions about the world (cf. Kuhn, 1962, on the structure of scientific revolutions). Thus, where I looked at how the brain constructs the individual’s reality (e.g., her understanding of the external world), Mary focused on how a community creates a social reality, a shared understanding. Despite the great gap between our initial perspectives, we eventually developed a coherent epistemology where we brought together schemas as social structures and schemas as functional entities “in the head.”

A further contrast was that Mary was a member of the Church of England, and both a supporter and critic of The World Council of Churches, whereas I was (and am) an atheist. But rather than being a cause for dissension between us, these differences served as challenges for us to address in our lectures. We took seriously the natural theology charge for the lectures, and agreed to disagree about two issues. One was the freedom of the will, and

the other was the existence of God! On the freedom of the will, I took the view that freedom is basically a social construct, whereas Mary wanted there to be some quantum indeterminacy which made room for it. As for God, she believed, and I did not. We formulated the issue as this: “Is God more like embarrassment or gravitation?” The former is not the vulgar claim that God is an embarrassment but rather that God might be a human construct, varying from one social group to another – just as what is embarrassing (consider nudity) may vary greatly from one culture to another – and yet be rooted in human nature. Similarly, the social construction of God might emerge from a human need for protection or to still fears of death or otherwise add a specifically human meaning to life (Arbib). Conversely, God, like gravitation, may be an external reality which we can only partially apprehend (Hesse). For the latter, consider that Newton and Einstein offer different theories of gravitation but, no matter what the theory, gravitation itself is not a social construct – under normal conditions (e.g., no jet pack) you will still fall to the ground if you jump from a building.

At UMass, with support from the Institute for Advanced Study in the Humanities (of which I was a founding board member), I organized a faculty seminar with diverse colleagues to better understand the relevance of their disciplines to the development of our Gifford Lectures. When Mary visited Amherst, she gave a talk on free will with which I strongly disagreed. Far from being offended, Mary was delighted that we had defined an important theme to debate in the lectures. In this way, the lectures became truly collaborative. When we presented the lectures in Edinburgh, Mary gave the lecture on free will, but she presented both sides of the debate.

Another example of how ideas from one of us informed a lecture by the other came from my reading of Northrop Frye’s *The Great Code* (Frye, 1982). He stressed the notion that much of Western literature is informed by the Bible, and his challenge was how to help students understand literature when their knowledge of the Bible was sketchy at best. This inspired Mary’s lecture on “The Great Schema,” the social schema rooted in the text of the Bible that has shaped so many lives. For Mary, it provided a glimpse of a greater reality. For me, it was a social construct rooted in European and Middle Eastern history.

5.2. Bridging the levels: extending schema theory from “inside the head” to social schemas

These 5 or 6 years working with Mary were important for me. Even more than 30 years later, the ideas we shared and that I learned from her continue to be important to my intellectual development. A crucial term we came up with was *Two-Way Reduction*, which has great relevance to the question of *downward causation* in neuroscience, asking whether mental activity can operate “downward” to change the activity of neurons (Arbib, 2013c; Sperry, 1980; Szentágothai, 1984). My answer was illustrated by

the example of a wooden wheel – while the wheel is intact, its “wheel-level” rolling determines the motion of all its particles, but if it hits a rock and breaks, only a lower-level analysis will explain the pattern of breakage. A classic challenge in science has been to explain chemistry by reducing it to physics, and biology by reducing it to chemistry. Our counter point is that one cannot simply turn the crank in one field and get the key results of a “higher level” field. It requires prior effort to assess what is important at one level that we need to explain, as well as what is relevant at another level that might supply an explanation. For example, we need thermodynamics to make precise the notions of pressure and temperature before we can develop statistical mechanics to explore their relationship at the level of large-scale molecular interactions. The challenge is to build a conversation between the two levels.

The primary sense of “schema” in my earlier work was as a unit of brain or mental function, which neural schema theory then seeks to relate to distributed patterns of neural interaction. A schema “in the head” can be looked at either from the inside (the mechanisms internal to the brain that support the schema), or from the outside (the external patterns of behavior evidenced by an animal or human perceiving and acting according to the designated schema). We can then bring in the social dimension by noting that related schemas in the heads of individuals within a community create patterns of behavior that help provide the environment in which a new member may perceive skills needed within that community. This led to a key distinction between the individual’s schemas about the society which she “holds” in her own head, schemas which embody her knowledge (possibly tacit) of her relations with and within society; and what Hesse and I call a *social schema*. This notion is an addition to my prior approach to schema theory. It addresses the fact that entities like “The Law” or “Presbyterianism” or “The English Language” are not exhausted by any one individual’s stock of schemas in the head, but are constituted by a “collective representation” (to adapt a term from Durkheim, 1915) which is experienced by each individual as an external reality constituted by patterns of behavior exhibited by many individuals as well as by various writings and artifacts. A related concept is that of a meme (Dawkins, 1976, from the Greek word μιμητισμός, mimetismos, for “something imitated”) as a unit of ideas, symbols or practices, which can be transmitted from one mind to another through social interaction rather than via the genome. However, a social schema may be a more overarching “style of thought and behavior” rather than a discrete “package”.

Hesse and I explored ways in which individuals respond to a social schema to acquire individual schemas which enable them to play a role in society, whether as conformists, or as rebels who reject and possibly change the social schemas which define society. Such change may involve a process of critique, whereby individual experience and social schemas are engaged in a process of accommodation in which either or both classes of schema may

change. Note that no individual's schemas may exhaust the social schema. In the case of "The English Language", we each know words and grammatical turns that others do not know. As the child comes to internalize the language, she too creates internal schemas that constitute an idiolect as she learns lexical and grammatical patterns from those around her to express her meanings, while picking up some idiosyncrasies but not others.

In summary, we may now distinguish three flavors of schema theory:

Basic schema theory studies schemas as dynamical, interacting systems which underlie mental and overt behavior (and not just conscious processes). Basic schema theory is defined at a functional level which associates schemas with specific perceptual, motor, and cognitive abilities and then stresses how our mental life (whether consciously or subconsciously) results from the dynamic interaction – the competition and cooperation – of many schema instances. It refines and extends an overly phenomenological account of the "mental level."

Neural schema theory: The "downward" extension of schema theory seeks to understand how schemas and their interactions may indeed be played out over neural circuitry – a move from psychology (viewing the mind "from the outside") to cognitive neuroscience. Neural schema theory analyzes data from neurophysiology, lesion studies and brain imaging to see how schemas may be restructured to relate to distributed neural mechanisms.

Social schema theory: The "upward" extension of schema theory seeks to understand how "social schemas" constituted by collective patterns of behavior in a society may provide an external reality for a person's acquisition of schemas "in the head." Conversely, it is the collective effect of behaviors which express schemas within the heads of many individuals that constitute, and change, this social reality. Social schemas represent the collective effect of behavior – whether related to everyday events, language, religion, ideology, myth, or scientific society – governed by related schemas in the individuals of a community.

5.3. Aftermath

Around 10 years after publication of the Gifford book, I received an intriguing invitation. A group of scholars based on the Center for Theology and the Natural Sciences in Berkeley, California, and cosmologists from the Vatican Observatory shared both a deep Christian faith and a thoroughgoing commitment to modern science. To this end they had held a series of three symposia, under the general rubric of *Scientific Perspectives on Divine Action*, addressing quantum cosmology and the laws of nature, chaos and complexity, and evolutionary and molecular biology. Would I join the committee organizing a fourth symposium, this time centered around neuroscience? (Had they noticed that not only had I co-authored the 1983 Gifford Lectures in Natural Theology – certainly a qualification –

but was also an atheist?) I accepted readily, and in the end joined two California theologians, Robert John Russell and Nancey Murphy, and the Dutch philosopher of science, Theo Meyering, in assembling the speakers for a meeting held in Paserbiec (near Kraków, in Poland) in June of 1988, and co-editing the volume *Neuroscience and the Person: Scientific Perspectives on Divine Action* (Russell, Murphy, Meyering, & Arbib, 1999).

I had assumed that the debate in the Workshop would pit monist neuroscientists against dualists who believed in the separation of soul and body. But when I met Nancey Murphy to discuss preparations I learned this was not so – that many of the Christians at the meeting believed in the resurrection of the body. I asked how old one would be when resurrected and, without missing a beat, Nancey replied "38 – when one is old enough to establish who one truly is, but young enough that the ravages of age have not set in." Perhaps she was pulling my leg, perhaps not.

My invitees were Joe LeDoux, an expert on emotion, Marc Jeannerod, and Leslie Brothers, a psychiatrist whose book on social neuroscience, *Friday's footprint: how society shapes the human mind* (Brothers, 1997) – published when the field was still in its very early stages – had certainly influenced my thinking on better linking the Gifford theme of social schemas to neuroscience data. The fifth neuroscientist, invited by Theo, was Peter Hagoort, a leading researcher in neurolinguistics. I contributed two papers to the symposium: the first, "Towards a Neuroscience of the Person" (Arbib, 1999b) sought to present schema theory as a bridge to neuroscience in a fashion attuned to the needs of the meeting, with discussion of neurology and the person and the relation between science, religion and the understanding of personhood. Theo proved an excellent critic in helping me refine my views on schema theory for a very different audience. Recalling the question "Is God more like embarrassment or gravitation?" the second paper, "Crusoe's Brain: Of Solitude and Society" (Arbib, 1999a), may be seen not only as a "tip of the hat" to Leslie Brothers but specifically as the case for the embarrassment view. As such it was vigorously countered by "Intimations of Transcendence: Relations of the mind and God" (Ellis, 1999) by George Ellis who is not only a staunch Christian but also a leading expert on gravitation in general and black holes in particular, a collaborator with Stephen Hawking (Hawking & Ellis, 1973). For the meeting, Leslie Brothers teamed up with a theologian, Wesley Wildman. Their paper, "A neuropsychological-semiotic model of religious experiences" (Wildman & Brothers, 1999), was no easy read, but I was struck by the fact that they extended the notion of varieties of religious experience, as charted by William James in his Gifford Lectures (James, 1902), to the notion of "experiences of ultimacy" – a sense of losing the boundaries between oneself and the world in which one is immersed; and which can be joyous and uplifting even to an atheist, but which a religious person may interpret in terms of the symbol system of their religion.

Briefly, two and a half further sequels. Marc Jeannerod and Giorgio Auletta, professor of the Pontifical Gregorian University in Rome, organized an international workshop on “Neuroscience Approaches to Top-Down Causation” held at Domus Sanctae Marthae (the House of Holy Martha) in the Vatican in 2009. Domus Sanctae Marthae was built on the order of Pope John Paul II to serve as the residence for cardinals during each enclave to select a new pope, but serves as a guest house for visitors to the Vatican in between – and is where Pope Francis now lives, eschewing palatial pomp. My own talk (Arbib, 2013c) has already been mentioned in the discussion of downward causation and two-way reduction. One of the other speakers was Juan José Sanguinetti, a philosopher of science at the Pontifical University of the Holy Cross (Santa Croce). Another was the German neurophysiologist Wolf Singer, and there, in the Vatican, he invited me to host a Strüngmann Forum in Frankfurt. More of this in the section “Music and the Brain.”

The conference in Paserbiec in 1998 was hosted by Michał Heller, a Roman Catholic priest in Kraków, a mathematical cosmologist and close friend of John Paul II. He was later awarded the 2008 Templeton Prize for Progress Toward Research or Discoveries about Spiritual Realities, and used the prize money to establish the Copernicus Institute in Kraków, devoted to promoting the interchange between science and religion. Some years later, I was invited to talk on the soul at the *Copernicus Festival: Revolutions* to be held in May of 2014, sponsored by this Institute. It later turned out “soul” was being used as a synonym for “mind” in the invitation, but by then it was too late, and my talk was titled “Your Soul is a Distributed Property of the Brains of Yourself and Others” (Arbib, 2016b). To my pleasure, the discussant was my former student Bipin Indurkha, who now teaches cognitive science at the Jagellonian University in Kraków. In Rome the following December, I gave a version of my talk at Santa Croce, at the invitation of Juan José Sanguinetti, to an audience primarily of Jesuit priests and students. They were not swayed by my atheistic conception of the soul, but we had a vigorous and interesting discussion and parted on good terms, with at least two members of the audience saying they would pray for me.

Chapter 6. Intermezzo

My strength is my weakness – namely, the very breadth of my interests. What if I had not abandoned mathematics after the fruitful collaboration with Ernie Manes but had, right from the start, made mathematical investigation a core part of my approach to brain theory? What if I had not made a major investigation of the relevance of AI, brain theory and cognitive science to natural theology? What if I had written and edited only 3 or 4 books instead of 40, and applied the time saved to focused research? Indeed, looking back there are books I wish I had not spent time writing and papers that now seem unimportant, and

yet I must confess that it has been a great adventure to reach at least some depth of research (beyond mere exposition) across a wide range of ideas.

In Chapter 7, I will report on models of detailed circuitry of cerebellum, hippocampus, basal ganglia, and parieto-frontal interactions. I believe them to be good anchors not only for understanding each system but also for gaining a deeper appreciation of the brain as a “system of systems.” Having such a diversity of models provided a solid complement to the anatomy of John Szentágothai and the dynamical systems of Peter Érdi for our integrated view of *Neural Organization: Structure, Function, and Dynamics* (Arbib et al., 1998). But many neurophysiologists devote a whole career to certain aspects of just one of these systems, and thousands of papers have been published on each one. If I had been similarly devoted to just one system, my group would certainly have developed more elaborate models, incorporating and explaining a wider range of data than that which we sampled. Of course, there are indeed computational neuroscientists who seek to get a general handle on neural systems and have gone deep as well as broad. As mentioned in the Preamble, this article does not claim to be comprehensive in its coverage, so let me just mention Stephen Grossberg as one example of a researcher devoted to this comprehensive approach.

A great deal of my energy has gone into editing, and this fits into my general need to understand the broader picture while still going moderately deep into the inner workings of whatever concerns me. Around 1971, I submitted a draft of *The Metaphorical Brain* to a book competition run by Phi Beta Kappa, publisher of *The American Scholar*, “a quarterly for the independent thinker.” I did not win an award, but got something perhaps even better, a letter from Hiram Haydn, a member of the jury who was then editor of *The American Scholar*. The book had made a positive impression on him and he invited me to write an article based on its concluding section. The article was eventually published as “Complex Systems: The Case for a Marriage of Science and Intuition” (Arbib, 1972a). As I wrote in the Preface to *The Handbook of Brain Theory and Neural Networks* (Arbib, 1995), which I edited with Prue’s invaluable assistance more than 20 years later:

What stays in my mind from the ensuing correspondence was the sympathetic way in which he helped me articulate the connections that were at best implicit in my draft, and find the right voice in which to “speak” with the readers of a publication so different from the usual scientific journal. I now realize that it is his example I have tried to follow as I have worked with these hundreds of authors in the quest to see the subject of *Brain Theory and Neural Networks* whole, and to share it with readers of diverse interests and backgrounds.

A key difference between myself and Haydn is that he would edit articles across a broad range of topics that might interest readers of *The American Scholar* whereas

most of my editing has been for books or workshops on topics of current interest to me. I offered authors two kinds – expert feedback (where I knew enough to provide it), and a simulation (or, in some cases, the real thing) of a reader new to the topic. There I would point out how the everyday parlance in one area might need exposition and background to make the article accessible to a wider audience. Something of this carries over to my research collaborations (whether with established researchers or my students) where I am not only engaged in contributing to our shared research, but am also endeavoring to learn from the other(s) while helping write a paper, book or thesis that will be both informative and readable to a somewhat wider readership than might otherwise have been the case. Having said this, I must confess that I am lucky if a book of mine sells more than a few thousand copies. I seem incapable of “gearing down” to produce a book that can gain a wide readership with the general public. Perhaps I am too keen to explain details where a leisurely exploration of the general feeling for a topic might have proved more seductive.

Chapter 7. The University of Southern California, USC (1986–2016)

I spent my sabbatical of 1985–86 in La Jolla with the Cognitive Science Program (not yet a Department) at UCSD. From David Rumelhart and others in Cognitive Science, I got a good feel for the learning-dominated approach to artificial neural networks, perhaps linked more to psychology and machine learning than to neuroscience, that became so influential with the publication of the two volumes on *Parallel Distributed Processing* (McClelland & Rumelhart, 1986; Rumelhart & McClelland, 1986). These volumes included the most widely cited paper on backpropagation (Rumelhart, Hinton, & Williams, 1986). But perhaps even more influential for my later career was the work of Ursula Bellugi, across the street from UCSD at the Salk Institute, on sign language and the extent to which it shared brain mechanisms with spoken language. The book based on this work was entitled *What the hands reveal about the brain* (Poizner, Klima, & Bellugi, 1987), thus raising questions about its relation to the work of Jeannerod and my work with Iberall and Lyons.

During this sabbatical year, Prue decided that 16 New England winters were enough for one lifetime. After I had interviewed at UCSD, UC Santa Barbara and UCLA, the USC Mathematics professor Ed Blum, an acquaintance from way back at the Asilomar (Krohn-Rhodes) meeting twenty years before who had since become interested in neural modeling, and George Bekey, at that time Chairman of Computer Science at USC and who was an expert on robotic hands (and much more), together got me appointed as a professor of Computer Science at USC. At that time, Bill Wagner (a physicist and dean of Natural Science and Mathematics) and Bill McClure (a neurobiologist with

whom Prue and I had become friends at one of the summer institutes hosted by the NRP in the 1960s) had established something called the Program in Neural Information and Behavioral Sciences (NIBS) at USC. NIBS was designed to link neuroscience to the study of cognition and behavior, and to computation. Wagner secured funding to provide startups so that different departments could be encouraged to hire new faculty members for a growing cross-campus community (compare the distributed model for Cognitive Science discussed earlier). I was the first professor hired for NIBS and Richard Thompson, who later became Director, was the second. NIBS was thereafter successful in continuing to strengthen neuroscience expertise in multiple departments. After some years, the research collaboration was complemented by a Ph.D. Program, the Neuroscience Graduate Program (NGP), so that students had the choice of enrolling in either the NGP or one of the NIBS-affiliated departments. Indeed, most of my USC students at USC enrolled in either Computer Science or NGP but soon shared the talents that my group fostered whatever their affiliation. In due course, though, the interdisciplinary spirit of NIBS was lost as faculty with diverse interests each thought their specialty was neuroscience enough, and NIBS was renamed simply as the Neuroscience Institute. Similarly, some years after I left UMass, the faculty renamed COINS as Computer Science (apparently thinking that a more conformist name would better attract funding) – though, ironically, when the Department became a School, they called it the School of Information and Computer Sciences, downplaying but vaguely echoing the initial history. But if the name change for COINS was recognized as a show of strength, this was not the case for NIBS. Soon after the name change, the Neuroscience Institute was killed by Max Nikias (then Provost and later President of USC) who felt that he knew better than any collection of neuroscience professors who should be “star” neuroscience hires from then on. Happily, though, NGP endures. Despite this and other disappointments, many good things happened during my thirty years at USC (from September 1986 till mid August 2016) and it is these that I will emphasize in this Chapter.

As already mentioned, with the completion of *Algebraic Approaches to Program Semantics* (Manes & Arbib, 1986), I abandoned study of automata theory and the mathematical theory of systems and computation. Norbert Wiener called the two parts of his autobiography *Ex-Prodigy* and *I am a Mathematician*; perhaps the rest of this essay should be called *Ex-Mathematician*. In any case, my main focus at USC was on efforts in brain theory which included an eventual return to a concern with language.

Various topics from my time at USC have already been covered where they rounded out earlier efforts, as in the discussion of *Rana computatrix*, or the work on perceptual robotics, or the aftermath of the Gifford lectures. In what follows, I will focus on other developments at USC, while seeking to show their relationship with what went before.

7.1. Mammalian brains: it's not just the cerebral cortex

Not only did work continue on *Rana computatrix*, but major effort went into the modeling of neural systems in rats, monkeys and humans. The major carry-over from the frog work was a respect for both action-oriented perception and the crucial role of subcortical systems (though not every model included them). In addition to modeling interaction between regions of the cerebral cortex, we brought in their interactions with basal ganglia, cerebellum, and hippocampus.

7.1.1. A major turning point: linking eye and hand

Shun-Ichi Amari invited me to talk at an IBM-Japan sponsored conference on *Neuro-Computers* in Oiso, Japan, in November of 1988. There I encountered Hideo Sakata, a neurophysiologist from Tokyo whom I had met at conferences on cortical control of movement. He told me that he and Marc Jeannerod, along with Giacomo Rizzolatti from Parma in Italy, were putting together a proposal to the Human Frontier Science Program (HFSP), a program to support multi-national research efforts initiated by Yasuhiro Nakasone when he was prime minister of Japan. Because of that chance meeting, I was invited to join the proposal, and this proved to be the basis for an exceptionally fruitful partnership (Fig. 4).

Where the collaboration with Marc focused on analysis of human behavior (and, later, brain imaging) in reaching to grasp, Hideo and Giacomo focused on neural correlates in the brains of macaque monkeys. Sakata's lab demonstrated that neurons in AIP (anterior intraparietal sulcus) responded to vision of an object with activity that correlated with "how" the object was to be grasped (which we viewed as an instance of affordances in the sense of

Gibson, 1966), whereas Rizzolatti's lab showed how neurons in the area of premotor cortex they labeled F5 coded something akin to motor schemas for grasping and manipulation. The insights from the first stage of our collaboration, integrating macaque neurophysiology, human behavior, schema theory and computational modeling, were set forth under the title "Grasping objects: the cortical mechanisms of visuomotor transformation" (Jeannerod, Arbib, Rizzolatti, & Sakata, 1995).

A recurrent theme in the analysis of sensorimotor coordination is that parietal affordances are linked to frontal motor schemas. We see this for saccades, locomotion, and (here) grasping (where AIP → F5). Andy Fagg came up with the FARS (Fagg-Arbib-Rizzolatti-Sakata) model of how AIP affordances for grasping could activate motor schemas in F5 (Fig. 5). Our key contribution was that a given object might have multiple affordances, with the one selected to guide current action being task-dependent. Our model posited two different pathways. The dorsal "how" can take the metrics of any affordance of an object and direct motor cortex (M1, aka F1), via F5, to reach out and pick up the object by grasping along this affordance (recall the notion of opposition spaces). The ventral "what" pathway could plan which affordance to use, leaving the details of execution to the dorsally specified metrics. For example, it could recognize a cup providing the basis for prefrontal cortex to make a decision (whether consciously or not) such as "I'm going to drink from it, so I need to grab it by the handle" or "I just want to move it, so I will grab it by the rim."

In the FARS model, the set of affordances and grasps is "built in." Erhan Oztop complemented FARS by developing a more detailed model of hand shape as a basis for assessing the stability of a grasp in holding an object and



Fig. 4. Marc and Jacqueline Jeannerod, Michael and Prue Arbib, Giacomo Rizzolatti, and Hideo Sakata at the HFSP Workshop on Cognitive Control of Movements and Actions, Lake Ashinoko, Hakone, Japan, on November 20, 1991.

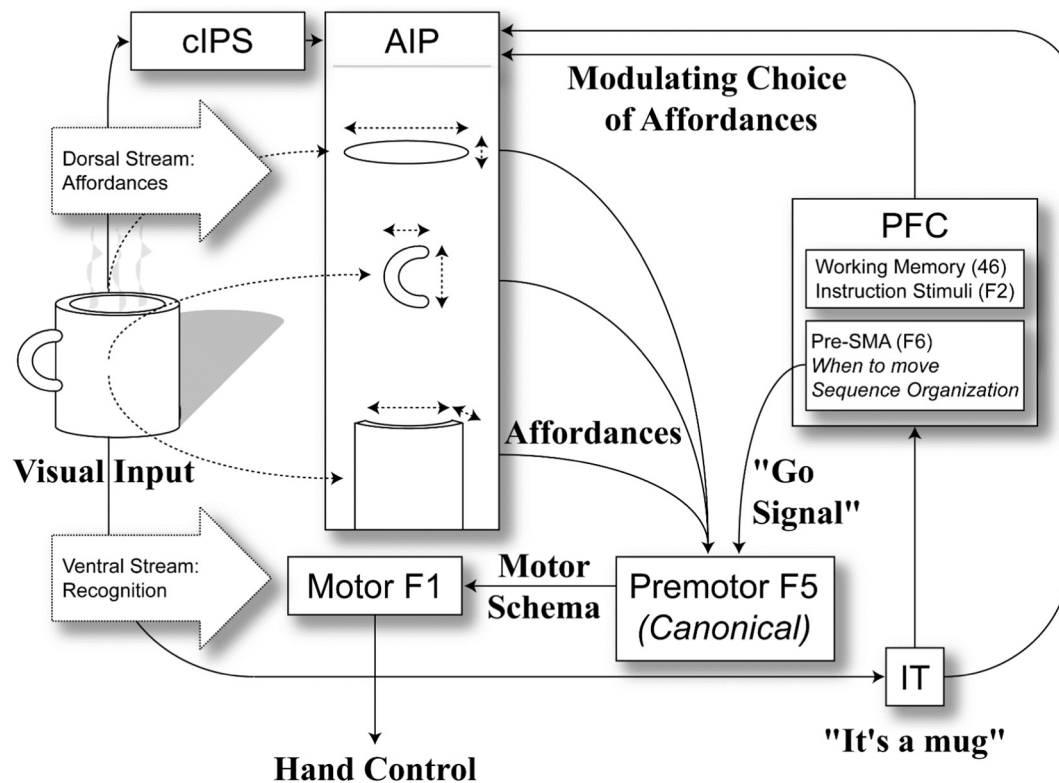


Fig. 5. The FARS Model of two visual systems in the control of grasping. The ventral pathway recognizes objects to provide prefrontal cortex (PFC) the data it needs, in concert with current task, motivation and working memory, to plan action and, as a result, to modulate the choice of affordances in AIP. The dorsal pathway passes the metrics of the selected affordance from AIP to premotor (F5) and thence motor (F1) cortex to control the metrics of the selected action. The full FARS model (Fagg & Arbib, 1998) also incorporates the role of basal ganglia in sequencing actions.

used this measure as the reinforcement signal (he called it “joy of grasping”) for learning what hand shape was appropriate for grasping a given object in a given context (Oztop, Bradley, & Arbib, 2004), addressing a range of data on infants learning to grasp. A separate model addressed the learning of affordances (Oztop, Imamizu, Cheng, & Kawato, 2006). Years later, Jimmy Bonaiuto provided an integrated model of the learning of grasps and affordances (Bonaiuto & Arbib, 2015).

A complementary line of research with Bruce Hoff refined the schema-level model of Fig. 3 by replacing the schemas with control systems and providing a timing mechanism to coordinate them. The new model addressed a range of kinematic data on human reach-to-grasp (it did not address dynamics). The key change was to include the possibility of feedback to modify (after some delay) even an apparently ballistic trajectory when conditions changed. This state-dependent approach (Hoff & Arbib, 1991, 1993) could thus give new insights into optimality criteria for reaching (e.g., Flash & Hogan, 1985). We added an optimality criterion for preshaping to address surprising data from the Jeannerod lab on changes in arm trajectory and preshaping when the size or position of the target object changed after a reach-to-grasp was initiated (Paulignan, Jeannerod, MacKenzie, & Marteniuk, 1991; Paulignan, MacKenzie, Marteniuk, & Jeannerod, 1991).

Such work provided a further demonstration that brain models can be falsified and, hopefully, revised even at the schema level. A complementary collaboration, with Marc Jeannerod’s colleague Jean-Paul Joseph in Lyon, led to the work on “Basal ganglia and the control of saccades” described in the next section.

The most famous discovery made during the HFSP collaboration was that of mirror neurons. In an influential paper, Jeannerod (1994) argued for a functional equivalence between motor imagery and motor preparation based on the positive effects of imagining movements on motor learning and the similarity of the neural structures involved. It is interesting to see how he talked about mirror neurons shortly after their discovery but before they got that name.

Rizzolatti and his group have described a class of neurons in the rostral part of the inferior premotor cortex, which fire prior to and during specific actions performed by the animal (e.g., picking up a food morsel with a precision grip). Neuron discharge is usually not conditional to the hand used, nor to the orientation of the grip, it relates to the fact that the monkey performs that particular action (Rizzolatti et al., 1988). Recently, these authors noticed that the same [correction: some] neurons also fire while the monkey observes the experimenter performing the same action (di

Pellegrino, Fadiga, Fogassi, Gallese, & Rizzolatti, 1992). According to the authors, “there is always a clear link between the effective observed movement and that executed by the monkey and, often, only movements of the experimenter identical to those controlled by a given neuron are able to activate it.” This very striking result supports the idea of representation neurons as a common substrate for motor preparation and imagery.

In short, *mirror neurons*, first discovered in macaque F5, are those neurons active both when the monkey executes a specific set of actions (e.g., precision pinches but not power grasps) and when the monkey observes similar manual actions performed by others. We may say that F5 is endowed with an observation/execution matching system. The modeling this inspired will be taken up in the section “Modeling Mirror Neurons in the Macaque.”

7.1.2. Basal ganglia and the control of saccades

Didday and I (1975) had related the role of tectum in directing frog body movements to catch prey to the role of superior colliculus in directing saccadic eye movements in cats and primates to attend to visual targets. In each case the retinotopic location of a target was translated into the direction of the movement. (Note the contrary situation for the frog escaping from a predator.) But what if A and B are flashed on briefly as saccade targets but turned off before the first saccade is initiated? Subjects saccade to A and then to B, but the problem is that when the saccade to A is completed, the retinotopic position of B (had it still been visible) would have shifted. Let's call that new position B-A. Satisfyingly, Mays and Sparks (1980) found activity at the B-A location following the first saccade. We say that *dynamic remapping* had occurred during a double saccade, and (inspired in part by Droulez & Berthoz, 1991), we developed a model of how dynamic remapping in area LIP (lateral intraparietal sulcus) might drive the shifting map in superior colliculus. Meanwhile, Hikosaka and Wurtz (1983a, 1983b) had demonstrated the critical role of the basal ganglia in the control of saccades, with a saccade only occurring when and where the basal ganglia released the inhibition of superior colliculus (SC) administered via the substantia nigra pars reticulata (SNr). Based on such work, Peter Dominey (Dominey & Arbib, 1992) modeled the control of simple saccades, memory saccades and double saccades, with essential roles for parietal cortex, frontal eye fields, thalamus, and superior colliculus. The key was that a reflex saccade could be elicited without cortical involvement but (addressing the data of Hikosaka and Wurtz) it required cortex to work with basal ganglia to release SNr inhibition of SC when and where a saccade primed in SC could be performed.

While on a Fellowship in Lyon (where he has stayed ever since), Peter worked with Jean-Paul Joseph, and extended the 1992 model to address Joseph's results on sequence learning (Barone & Joseph, 1989a, 1989b). A monkey faced with 3 lights, each with a button beneath

it, could – but only after extended training – learn that, having seen the lights flash in a given order, it should then press the buttons in the same order. It is worth stressing that the monkey did *not* “get” the general idea of “use the same order” – it required separate training to master each of the 6 possible sequences. Like the first model, the new model (Dominey, Arbib, & Joseph, 1995) considered only the direct path of the basal ganglia. It considered saccades rather than reaching for the response sequence. Observation of the visual sequence triggered a sequence of neutral states in PFC. The key to the model was a pathway from frontal cortex to the striatum, with “vanilla” (i.e., non-TD) reinforcement adjusting synapses from PFC to striatum to ensure that when the neutral sequence was cycled through in PFC, striatum would get excitation that would get it to sequentially inhibit activity in SNr so that the last three states would disinhibit SC to allow the appropriate saccades in the correct order. The model also predicted conditional learning, and gave insight into the firing patterns Barone and Joseph had seen in PFC. This model can be seen as providing the learning mechanism for the way the basal ganglia are posited to sequence action in the FARS model (Fagg & Arbib, 1998). The 1995 model is now seen as a precursor of reservoir computing (Enel, Procyk, Quilodran, & Dominey, 2016; Lukoševičius & Jaeger, 2009).

Two more theses followed on the basal ganglia. Michael Crowley updated Dominey's superior colliculus model and offered a new model of dynamic remapping. He distinguished direct and indirect pathways to study effects of lowering dopamine (Peter had modeled only the direct path). Amanda Bischoff developed a new model with roles for the direct and indirect paths in control of alternating, conditional, and sequential reaching movements. She employed a Hoff-Arbib motor pattern generator (MPG) for the arm, with key roles for pre-SMA and SMA-proper (SMA = supplementary motor area). Alas, neither published a detailed account of their work, though a joint summary is available (Bischoff-Grethe, Crowley, & Arbib, 2003). A post-doc, Roland Suri, did make a detailed model of striatum – up-states, down-states, striosomes and matrixes, and dopamine, with links to temporal difference learning – and, thankfully, published the results (Suri, Bargas, & Arbib, 2001).

7.1.3. The cerebellum, again

The second round of modeling of the cerebellum was conducted with Nicolas Schweighofer and Jacob Spoelstra. They both spent extended periods of time with Mitsuo Kawato's group in Japan (as did Erhan Oztop), to our mutual benefit. As with Boylls, we differed from Marr and Albus by regarding cerebellar cortex as a tuner and coordinator of motor schemas, *not* as a storehouse of elemental movements, but at last we followed them in bringing in learning. The only outputs of cerebellar cortex are the axons of Purkinje cells. These outputs are all inhibitory. Each Purkinje cell receives two types of input – many many

parallel fibers arising from a large number of granule cells, and one climbing fiber from a single cell in the inferior olive. The Marr-Albus theory is that the parallel fibers provide sensory patterns while the climbing fiber to a Purkinje cell provides a training signal. Coincidence of climbing fiber activity with that of any parallel fiber will strengthen (Marr) or weaken (Albus) the strength of the parallel fiber's synapse with the Purkinje cell. Ito established that Albus was closer to the mark, using neurophysiology to detect long-term depression (LTD) of those synapses (Ito, Sakurai, & Tongroach, 1982). By embedding such a cerebellar cortex in a system-level view of reaching (incorporating cerebellar nuclei, cortical areas, visual feedback and more), we came to understand that the model of neural plasticity had to be changed (Schweighofer, Arbib, & Kawato, 1998; Schweighofer, Spaelstra, Arbib, & Kawato, 1998):

1. LTD alone would continually weaken those synapses. We thus had to include a complementary term for strengthening synapses that would not wash out the learning. This is the obverse of the problem with Hebbian strengthening of synapses which Peter Milner (1957) was the first to address.
2. Classical conditioning studies introduced the notion of eligibility to account for the fact that the conditioned stimulus must occur shortly before the unconditioned stimulus if conditioning is to take place. We introduced a version based on the following analysis: If a Purkinje cell fires as part of the modulation of an action, it will be around 200 ms before the inferior olive can send climbing fiber signals that encode a visually observed motor error. The strength of the LTD must thus peak for those synapses that were engaged in firing a Purkinje cell around 200 ms before the climbing fiber activates that Purkinje cell.

This was a nice example of two-way reduction between the high level of motor learning and the low-level of synaptic plasticity, with the latter suggesting specific challenges for probing the dynamics of the molecular biology of the parallel fiber → Purkinje cell synapse (Schweighofer & Arbib, 1998).

Another study concerned prism adaptation in throwing at a target. It incorporated the observation by Tom Thach of Washington University in St. Louis that parallel fibers were long enough to bridge several microcomplexes, and thus adjustment of their synapses on the Purkinje cells of two microcomplexes could serve not only to tune the modulation of their corresponding MPGs, but could also help tune the coordination between them. Our model (Arbib, Schweighofer, & Thach, 1995) successfully addressed data from Thach's lab (Martin, Keating, Goodkin, Bastian, & Thach, 1996a, 1996b), both behavioral data on normal subjects, and data on the type of cerebellar lesion that would impair adaptation. Martin et al. found that only in some subjects would adaptation to over-arm throwing

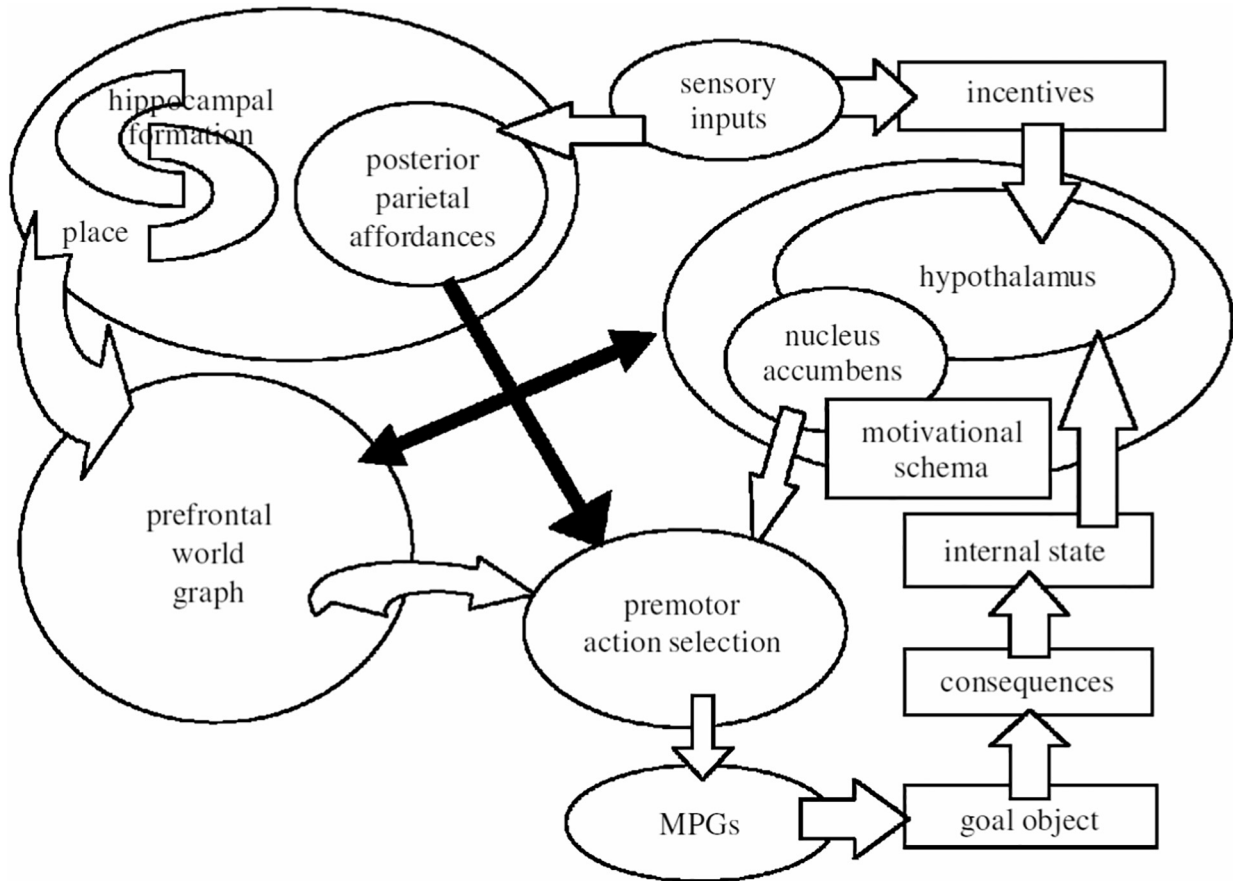
to a prism transfer to under-arm throwing. These data were a surprise – I had thought prism adaptation involved a global shift within the visual system but the data showed relatively private shifts in the sensorimotor transformations for different tasks, a result we modeled via degree of overlap of the relevant microcomplexes.

Martin et al. also showed that after a session with prisms on, a new period of adaptation was required to throw accurately when the prisms were removed. However, after hundreds of blocks of trials, each involving adaptation and readaptation to the prisms, a (very dedicated) subject eventually reached a stage at which no adaptation was required when the prisms were donned or doffed. The basic model rested on the fact that hand areas in cerebellum and cortex are richly endowed with fibers encoding eye position, whereas there is no reason for evolution to have favored fibers encoding prism on/off. Thus, to complete our model, we hypothesized that a neutral mix of fibers from cerebral cortex was available to the relevant cerebellar microcomplexes, and thus a very sparse subset could convey features that might correlate with prism on/off even though neither evolution nor experience had previously selected for them. The model worked as follows: Because many fibers encoded eye position, learning could rapidly adjust enough synapses to adaptively change cerebellar modulation of the arm-throw circuitry. However, because the prism on/off-related fibers were so sparse, the chance of their being modified adaptively was very small, and thus the number of trials for their adaptation to become effective was very large.

Schweighofer also collaborated with Dominey on “A model of the cerebellum in adaptive control of saccadic gain” (Schweighofer, Arbib, & Dominey, 1996a, 1996b), showing how the cerebellum may help ensure the accuracy of the brainstem saccade generator. As such, it complements Dominey's earlier models employing the basal ganglia in choosing the next saccade by involving the cerebellum in compensating for the nonlinearities in the brainstem MPG. More direct integration of cerebellum and basal ganglia is a topic of current research.

7.1.4. Navigation and the hippocampus

A return to the work with Liebhich on the World Graph (WG) model after a 15-year hiatus addressed the challenge of developing a model of the hippocampus within a much larger system that could integrate hippocampal “you are here” into a plan to move to a place that satisfied some goal. The TAM-WG model (Guazzelli, Corbacho, Bota, & Arbib, 1998) was based on the distinction (O'Keefe & Nadel, 1978) between navigation based on *locale* (exploiting a map) versus *taxon* (exploiting an affordance). Consider looking for a restaurant in a strange city. You can either follow a street looking for signs that you recognize as being for restaurants (affordances for eating), or you can follow a map supplied by the hotel concierge that shows the location of a good restaurant and which enables you to plan a route there without being able to see it for much of the way. Our Taxon Affordances Model (TAM)



gave a neural net model of the former; our WG model (based on the work with Liebllich) gave a schema-level model of the latter. Moreover, the ascription of subsystems to specific brain regions allowed us to explain data (O'Keefe, 1983) on the effects of lesioning the fornix, effectively decoupling the hippocampus from the ongoing computations.

The overall structure of the models is shown in Fig. 6. On the right is the basic motivational system, centered on the hypothalamus, which holds neural representations for basic drives such as fear, hunger and sex. The activity level can depend both on internal state signals (e.g., low blood sugar increases the hunger signal) and sensory cues (seeing food may provide an “incentive” signal that increases the hunger signal). As the animal acts, it may or may not be successful in reaching a goal object (e.g., food) and thus (in the case of food) change its internal state by becoming less or more hungry, respectively. The nucleus accumbens is modeled as the locus of reinforcement learning, yielding an adaptive bias signal for action selection dependent on the current internal state.

The middle of the figure completes the TAM model. Sensory inputs determine parietally encoded affordances for the rat’s locomotion; these provide the “menu” for premotor cortex to select (on the basis of a variety of cues, tuned by the motivated learning) appropriate commands

for the MPGs that control locomotion. This frog-inspired model might involve midbrain circuitry including superior colliculus, the homologue of tectum, so it would be useful to return to the data of [Dean, Redgrave, and Westby \(1989\)](#) implicating rat superior colliculus in both orienting and in avoidance.

Finally, on the left of the figure, we see the *locale system* (WG) for map-based navigation. Here the place (“you are here”) system of the hippocampus is augmented by a cognitive map, the world graph (WG) posited to be in prefrontal cortex. Mechanisms for updating and using WG are taken from [Lieblich and Arbib \(1982\)](#); their implementation in a neural model, and the testing of that model against neurophysiological data, remains a target for future research. The model includes circuitry that supports the ability of place cell firing to be updated by the rat’s own locomotion even in the dark ([Guazzelli, Bota, & Arbib, 2001](#)), akin to the earlier results on dynamic remapping in the saccadic system. This allows the animal to update its place-cell encoding on the basis of its recent movements even when landmarks for the new place are not currently visible.

A later paper (Arbib & Bonaiuto, 2012) placed the TAM-WG model firmly in the framework of temporal difference learning, but now viewed as *spatial* difference learning because the motivational data associated with a node

reflects not only the positive or negative reinforcement with respect to specific drives received at the place corresponding to the node, but also expected drive-related reinforcement for paths through that node, an idea sketched in the early work with Lieblisch but unpolished because this work preceded the development of TD learning.

This modeling ignored data on theta rhythms, including that on the phase shift in activity of a place cell as a rat approaches the corresponding “place.” This is not to deny the importance of rhythms; it is just that I have not made rhythms (other than locomotory ones) my concern. Grid cells (Moser, Kropff, & Moser, 2008) were not necessary as an addition to place cells in our models. More generally, of course, if one addresses a broad array of topics one cannot go as deeply as those who specialize in any one of them. On the plus side, though, by tackling a broader systems view, one may provide insights that complement those of the specialist. The contribution of the TAM-WG model is to assess the role of hippocampus in a system that offers a large scale cognitive map that complements the current “chart” in the hippocampus. Unfortunately, this scheme has not been adequately dissected by other researchers.

Going even more broadly, these models provide a backdrop for my involvement (primarily essayistic and editorial) in the study of emotion. The link is to view the basic drives considered in the WG model (such as hunger, thirst, sex and fear) as “primordial emotions” (Denton, McKinley, Farrell, & Egan, 2008) – which of course raises the question of how “real emotions” differ from these drives. What sort of brain is required to experience *Schadenfreude*? While working on his Ph.D. with me (A Neural Code for Face Representation from V1 Receptive Fields to IT Face Cells, completed in 1995), Jean-Marc Fellous pursued a parallel interest in the neural correlates of emotion. This interest took the form of offering a graduate course at USC, under my responsibility and with my minor involvement. Some years later he came back to me with a plan for an edited book on the topic and was insistent that I join him as co-editor. The book was eventually published as *Who Needs Emotions: The Brain Meets the Robot* (Fellous & Arbib, 2005), with papers by experts in the neuroscience of emotion (Jean-Marc co-authored a chapter with Joseph Ledoux) as well as by experts in AI and robotics seeking analogues of emotion to complement more conventional forms of computation.

My own chapter, “Beware the passionate robot” (Arbib, 2005), offered a synthesis of the book’s diverse contributions, but cautioned that while all the other chapters stressed the positive role of emotion, and while indeed emotion and cognition usually were positively intertwined, emotion could at time overwhelm good judgement (as illustrated by an example of my own behavior, with an analysis of the emotional states involved). Jean-Marc and I also published a paper on emotion in biology and robotics (Arbib & Fellous, 2004), stressing the interaction between the internal role of emotion (biasing decision making) and the external role (communication of emotional state,

as in facial expressions affecting social interactions, Darwin, 1872; Ekman, 1992). The basic (still unresolved) question is: “What is the possible value (or inevitability) of future robots not only simulating emotional expression but ‘having emotions’?” Various authors in our book sought to generalize concepts from the neurobiology of emotion to cover robots as well as animals, but the chemical basis and evolutionary history of animal function differs greatly from the mechanics and computations of current machines ... so not all emotions need be like human emotions. Our 2004 paper argued that the old RETIC model (Kilmer et al., 1969) could offer a framework for a view of emotion for systems that did not share our biological heritage. For more on the subject of (human) emotion, see the section on “Music and the Brain.”

7.1.5. Modeling mirror neurons in the macaque

The Mirror Neuron System model (MNS; Oztop & Arbib, 2002) offers a developmental (Devo) view of mirror neurons. Rather than positing an innate repertoire, it suggests how mirror neurons for manual actions might emerge during observation of one’s own actions. In Fig. 7, the external diagonals correspond to the dorsal path of the FARS model for converting an affordance into a grasp and a complementary path for controlling the arm to bring the hand to the desired position. Since we emphasize learning, we distinguish “potential” mirror neurons (before learning) from actual mirror neurons (after their properties are defined by the learning process). These receive both (i) efferent copy of the code for some grasps and (ii) input from circuitry that monitors the trajectory of the hand in a reference frame centered on the chosen affordance. The efferent copy acts as a “training signal” for the neurons it activates – the learning process strengthens synapses that encode trajectories like those for the current grasp. As learning progresses, the synaptic drive from (ii) will eventually be enough to activate the emerging mirror neurons relevant to that grasp even if input (i) is absent. Since observation of another individual’s action may evoke the same affordance-centered input pattern (ii) as for self-execution, these neurons thus become mirror neurons. MNS demonstrated how, as learning progresses, recognition of the grasp may occur earlier and earlier in the trajectory – though such anticipation will be a function of how precisely the trajectory is represented in the brain, which in turn is a function of attention as well as neural encoding.

A crucial aspect of the model, then, was to suggest that mirror neurons may have evolved first to monitor self-actions (see the ACQ model next) – matching intended action to observed trajectory – with their role in the observation of others (which is most emphasized in the literature) being an exaptation of this capability. Our 2002 hypothesis, that F5 mirror neurons of the macaque are sensitive to the sight of the monkey’s own hand during object grasping, was confirmed by Maranesi, Livi, and Bonini (2015). But there are also mirror neurons for ingestive and communicative

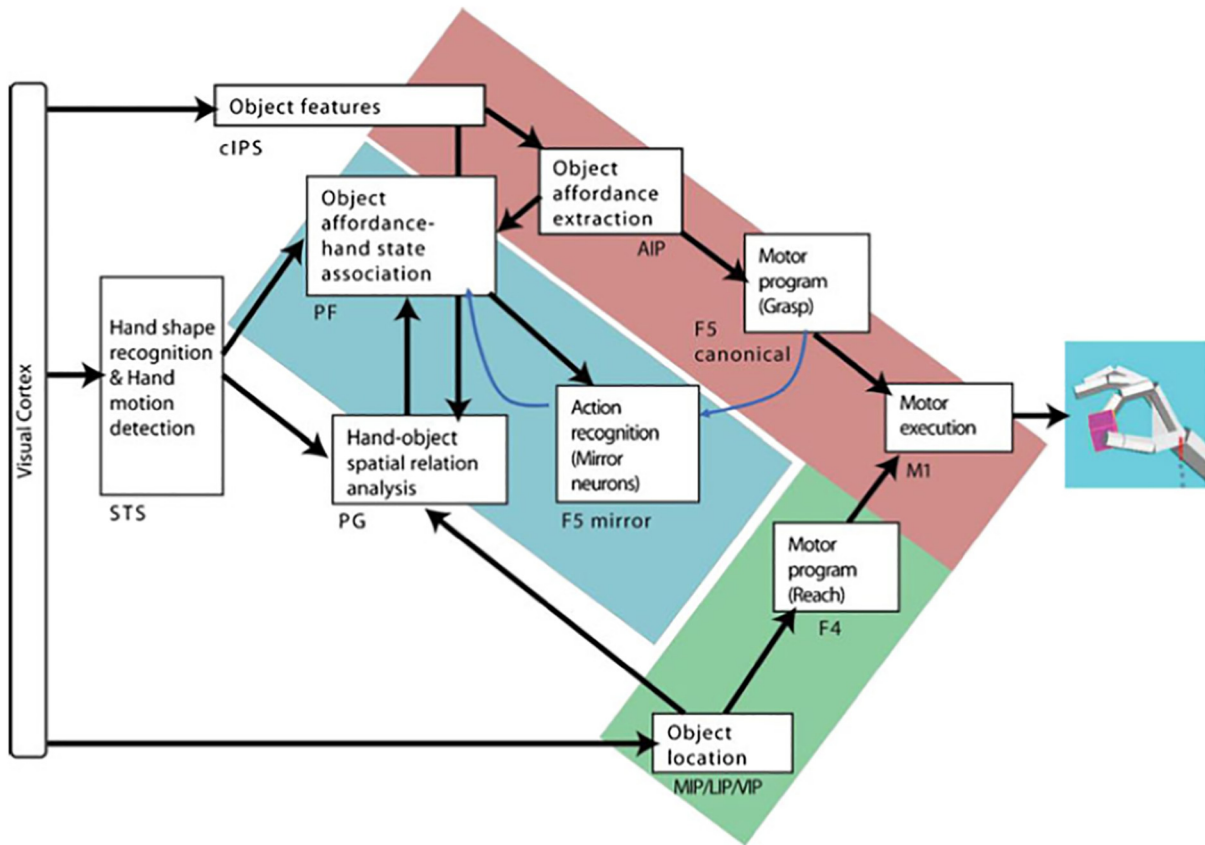


Fig. 7. The MNS Model of Learning in the Mirror Neuron System (Oztop & Arbib, 2002). Note that, whatever properties mirror neurons have, they only have by dint of being part of a larger system “beyond the mirror.”

oro-facial actions in the macaque (Ferrari, Gallese, Rizzolatti, & Fogassi, 2003) and here a new model not based on self-observation is required. Intriguingly, a recent study (Ferrari, Gerbella, Coudé, & Rozzi, 2017) shows that whereas the manual system is related to parietal-premotor circuits, the oro-facial system connects with limbic structures. Exploring the linkage between these two systems could lead to new modeling charting evolution of the linkage between action and emotion.

The MNS2 model (Bonaiuto, Rosta, & Arbib, 2007) extends the MNS model to explain *audiovisual mirror neurons* (which, for actions with distinctive sounds [like breaking a peanut], respond whether the monkey sees or hears the action, or both) and the fact that mirror neurons for grasping will not respond to a pantomimed grasp (no associated object) yet will respond to grasp of a recently visible object if the view of the object, and thus the completion of the reach-to-grasp, is subsequently obscured by a screen.

Alstermark, Lundberg, Norrsell, and Sybirska (1981) demonstrated lesions of axons leaving the spinal cord in cats that impair grasping but not reaching. They taught cats to reach into a glass tube projecting horizontally from the wall and grasp a piece of food, which the cat then brought to its mouth. After just a few trials, a lesioned cat would not try to grasp but would simply bat the food from the tube and then grasp it from the floor with its jaws.

How could this new skill emerge so quickly? The Augmented Competitive Queueing Model (ACQ; Bonaiuto & Arbib, 2010) explains such phenomena (and we argue that it applies to the monkey and LCA-m, even though it was inspired by cat data) by linking mirror neurons to *self-actions*, as emphasized above.

Mirror neurons can be activated both by efference copy of a motor command or by observing a hand-to-object trajectory associated with the action. The key to ACQ is that when an *intended action* is unsuccessful, it may appear similar to an unintended action – and then the mirror neurons for the *apparent action* can serve a “what did I just do?” function. Thus, when the lesioned cat tries to grasp the food and inadvertently knocks it out of the tube, the mirror system can recognize that this looks like a “batting” action already in the cat’s repertoire. ACQ makes two evaluations for each action:

Desirability depends on the current task or goal, and is a measure of “expected reinforcement” that will be positive if the action leads “soon enough” to achievement of the goal, but will be greater the shorter the time required to reach that goal.

Executability depends on the availability of affordances (can the action be carried out now?) and the probability of the action’s success.

At each time step, the priority of available actions is set by combining executability and desirability – the highest

priority action will then be executed (or, at least, attempted). Each time an action is performed *successfully*, its desirability is updated while executability may be left as is or increased. However, when the action is unsuccessful, *executability* of the *intended* action is reduced while *desirability* of the *apparent* action is adjusted by TD-learning.

This explains the rapid change of behavior in Alstermark's lesioned cat. Since the grasp keeps failing its executability is decreased (but its desirability is unchanged), whereas the desirability of batting increases each time it is used. Consequently, in only a few trials the priority of batting comes to exceed that of grasping, and the cat has a new plan of behavior implicit in altered desirability and executability of its actions. The model assumes that cats have mirror neurons for brachio-manual actions. This has yet to be tested. However, the suggestion is again that mirror neurons arose first for monitoring of self-actions and that this functionality is widespread.

7.2. How the brain got language

My approach to (neuro)linguistics via schema theory (Arbib & Caplan, 1979; Arbib et al., 1987) was brought back to “active status” after a 12-year inter-regnum by data on macaque mirror neurons that suggested a novel approach to the evolution of the human brain's ability to support language. This led to research that continues to the present day.

7.2.1. The mirror system hypothesis

Macaque F5 (with its mirror system for grasping) is homologous to Brodmann's area 44 in human Broca's area; and imaging studies show activation for both grasping and observation of grasping in or near Broca's area. But Broca's area in the human had been implicated in speech production. However, I had learned from Ursula Bellugi that lesions of Broca's area are equally implicated in aphasia of sign language as of spoken language. This led Rizzolatti and me to think about the role of mirror neurons in language evolution from LCA-m (the last common ancestor of macaque and human) to modern humans, with manual gesture playing an important bridging role (Arbib & Rizzolatti, 1997; Rizzolatti & Arbib, 1998). Our *Mirror System Hypothesis* (MSH) posited that the evolutionary basis for *language parity* (the hearer is generally able to “get,” more or less, the meaning the speaker intends to convey) is provided by the mirror system for grasping, rooting speech in communication based on practical tasks involving the hands. The posited path from “praxis” to communication provided a neural basis for a gestural-origins view of the evolution of brain mechanisms, unique to humans, that could support language. Even in speaking, humans gesture with their hands (these cospeech gestures provide strong evidence for the linkage of hands to these brain mechanisms), while deaf children if raised in a community with a sign language can learn it as readily as hearing children can learn a spoken language.

By 2005, I had elaborated the evolutionary path from LCA-m (around 25 million years ago) via LCA-c (the last common ancestor with chimpanzees, around 5–7 million years ago) as spanning seven stages, constituting a “road map” for further research to fill in, or modify, the details:

1. LCA-m: An ability for dexterous grasping, coupled with a *mirror system*, matching action observation and execution for manual actions. A small innate repertoire of vocal signals mediated communication.
2. LCA-c: A *simple imitation system* (based more on trying to replicate achievement of a goal rather than the means used by another to achieve it) for grasping. A small innate repertoire of manual gestures could be augmented by novel gestures developed within a group to mediate communication.

Post LCA-c (stages 3–6) biological and cultural evolution (EvoSocio) yield the following:

3. (a) An ability for *complex action recognition*, combining the ability to recognize the movement details of actions within an observed behavior (with limitations based on the familiarity of the component actions and the complexity of the assemblage); and (b) the ability to exploit this to support *complex imitation* of observed skills and the means employed.
4. “*Ad hoc*” *Pantomime*: Performing manual movements in the absence of objects to convey the need for the object or some associated behavior to others.
5. *Protosign*: A manual-based communication system, in which pantomimes are conventionalized within a community to allow meanings to be conveyed more economically and less ambiguously. Between them, 4 & 5 break through the fixed repertoire of primate vocalizations to yield an open semantics
6. *Protospeech* rests on the “invasion” of the vocal apparatus by collaterals from the communication system based on F5/Broca's area. (Other primates do not have vocal learning.)
7. The above stages (2) through (6) involve biological evolution of brain and body (Evo), though cultural evolution (Socio) is needed for a group to learn to exploit these abilities. MSH posits that, with the emergence of *Homo sapiens*, cultural rather than biological evolution is dominant as language emerges from protolanguage, with not only a widening of the lexicon but also the emergence of complex grammars to support the combination of words to express and comprehend an unlimited range of novel meanings.

Further refinements to the theory, together with the relevant data, and a careful rebuttal of alternative theories (or, in some cases, refining MSH by attention to changes suggested by these alternatives) were presented in the book *How the Brain Got Language: The Mirror System Hypothesis* (Arbib, 2012b). A sequel (Arbib, 2016a) further devel-

oped the argument, bolstering the case for (*Computational*) *Comparative Neuropsychology*, based on comparative analysis of brain, behavior and communication in humans in comparison to diverse species of monkey and great ape. It also reported on modeling developments in my group conducted after the book was published. The paper was published with 18 commentaries from experts in diverse disciplines, together with my response, and offers perhaps the best entry point for understanding this effort (but see “How the Brain Got Language, Revisited” on the development of a new road map).

7.2.2. Dyadic brain modeling for the emergence of novel gestures in apes

Studies of ape communication anchor hypotheses about the neural, behavioral and communicative repertoire of LCA-c. It is a current topic of debate among primatologists as to whether and to what extent apes acquire particular gestural forms through social interaction rather than learning to draw their expression from some innate “gestural space” (Call & Tomasello, 2007; Hobaiter & Byrne, 2011; Perlman, Tanner, & King, 2012), though I believe the evidence favors a combination of the two. In any case, both require a treatment of the socially constrained learning processes involved in the competent production and comprehension of the manual and vocal gestures (Gasser & Arbib, 2018). How are their expressions contextualized; how are they organized neurally; why are different patterns of use seen in the same community, and thus how would interaction in a physical and social world tune their communicative behaviors?

The notion of *ontogenetic ritualization* (Call & Tomasello, 2007) provides a conceptual model of how dyads of apes may generate a gestural form whereby one may influence the behavior of the other, as mutual interaction yields a truncated version of a larger instrumental action sequence originally intended to *physically* exert that influence (Gasser, Cartmill, & Arbib, 2013). Our computational version (Arbib, Ghanesh, & Gasser, 2014) instantiates *dyadic brain modeling*, simulating how interactions between agents with similar brains may differentially adapt those brains over time. This allows us to assess which basic sensorimotor processes are needed to learn communicative signals from interaction in physical and social worlds. Importantly, we seek to delineate brain mechanisms distinguishing what apes and monkeys can learn, going beyond “primate-general” circuitry building on models that may be relevant to LCA-m by augmenting the role of proprioceptive data in determining the goal of an action. This supported a transition from the role of the hands primarily in transitive actions dependent on visual and tactile information about a manipulated object to the ability to support intransitive actions (i.e., not directed at objects) such as gestures. This sets the stage for future modeling contrasting ape and human brains to crystallize debate about what supports the language-readiness of the human brain.

7.2.3. Modeling how the brain may support language

The above dyadic brain modeling starts from models of macaque brain to offer a detailed hypothesis on the brain of LCA-m as a base for working “forward,” and offers a first step toward assessing how brains support, and are changed by, social interaction. However, we also need to understand what it is about the human brain that supports language, and this in turn requires some sense of how language is processed in the brain. There have been various attempts to offer primarily conceptual (rather than computational) models of how syntax and semantics might be distributed across the human brain (as suggested by ERP, fMRI and lesion data) but although masterly reviews of data are available, no consensus model has emerged. Linguistics offers diverse models of grammar, but there is no consensus as to which model might best serve as a first approximation to what is represented in a brain processing language. Moreover, these models fail to link language to “what language is about.” This led me to develop the notion of description of a visual scene as a suitable challenge. The result was the definition of Template Construction Grammar (TCG) as a schema-based grammar system. There are, of course, other relevant modeling efforts (e.g., Brouwer & Hoeks, 2013; Chang, 2015; MacWhinney, 2014).

TCG is but one of many different computational attempts seeking to exemplify the general framework of *construction grammar* (Croft, 2001; Goldberg, 2013) in which grammar is defined by a large number of more or less language-specific constructions, each of which combines form (how to put words and/or phrases together) and meaning (how to assemble the meanings of those pieces); this is to be distinguished from those approaches in which syntactic processing is separated from semantics. Perhaps the other version of construction grammar most relevant here is the Fluid Construction grammar developed by Luc Steels. His group has employed AI systems for learning in robot dyads to model language change using, e.g., evolutionary language games repeatedly played within a community of embodied robotic agents, building on the Talking Heads experiment (Steels, 1999). Parity of meaning is achieved as an emergent property of embodied language, resulting in alignment of cognitive content. Note, however, that such studies provide at best weak constraints on our quest to understand what it is that enables a brain to support language, let alone the evolutionary forces that made the brain “language-ready.” Dyadic brain modeling provides a crucial tool to address these questions, although we may hope to learn from dialogue with colleagues approaching cultural evolution from the AI side. Other approaches (Bergen & Chang, 2005; Hinaut, Petit, Pointeau, & Dominey, 2014) are certainly relevant as well, and the quest for a unified computational construction grammar suitable for neurolinguistics remains a current research challenge.

In production, describing a visual scene, TCG is implemented by “lifting” the schema-interactions defined for

VISIONS as it links linguistic processing to “what the utterance is about;” a complementary and overlapping system handles comprehension of utterances. More on the model can be found in several papers (Arbib, 2016a; Arbib & Lee, 2008; Barres, 2017; Barrès & Lee, 2014). Here is a brief overview.

In a visual scene description task, a subject simultaneously gathers information from the image, fixating elements that become salient based on both bottom-up features and top-down hypotheses, and starts generating linguistic output based on relevant visual information. Scene description is not the be-all and end-all of language but it provides a basis for understanding how world and language are linked. The same methodology could and should be extended to look at the interpretation of questions and commands within a given context. A further challenge (which we have not yet addressed) is to extend the methodology beyond the here-and-now (Corballis, 2018).

The SemRep/TCG model of scene description (Fig. 8) incorporates Visual WM and Long Term Memory (LTM) from VISIONS but adds a SemRep (Arbib & Lee, 2008) as a hierarchical graph-like “semantic representation” of the visual scene, abstracting away from visual details that will not enter into the description. This bridges between visual scene recognition and the language system. The latter incorporates a Linguistic WM, and a Linguistic LTM encoding grammatical knowledge as a set of constructions. The Linguistic WM builds a hierarchical covering of the current SemRep by iterated application of construction instances. There are two kinds of constructions: *Lexical constructions* are applied directly to the

SemRep, associating nodes or subgraphs of the SemRep with words or idioms. *Phrasal constructions* are applied to an already partially completed construction assemblage. As in VISIONS, construction instances compete and cooperate until an assemblage reaches threshold for an utterance to be produced. Just as VISIONS allows Visual WM to request more data from low-level visual processes, TCG allows the SemRep to be updated by requesting information from the vision system when completion of an utterance requires further attention to the visual scene.

Barrès and Lee (2014) proposed a conceptual extension of TCG to a model of language comprehension structured to explain data on agrammatic aphasics in *sentence-picture matching tasks* during which the patient is asked to decide whether a sentence he hears matches a visual scene. Caramazza and Zurif (1976) showed that agrammatic aphasics could be impaired not only in production but also in their capacity to make use of syntactic cues during language comprehension. They could successfully match the correct picture with canonical active sentences such as “the lion is chasing the fat tiger”, but were no better than chance for center-embedded object relatives such as “the tiger that the lion is chasing is fat.” However, their performance was restored when world knowledge cues were available to constrain the sentence interpretation, as in “The apple that the boy is eating is red”.

In the TCG-based model of language comprehension (Fig. 9), the utterance input is fed in parallel to two interacting routes. The grammatical route (G) updates the SemRep indirectly through the creation of a construction schema assemblage in grammatical working memory; the

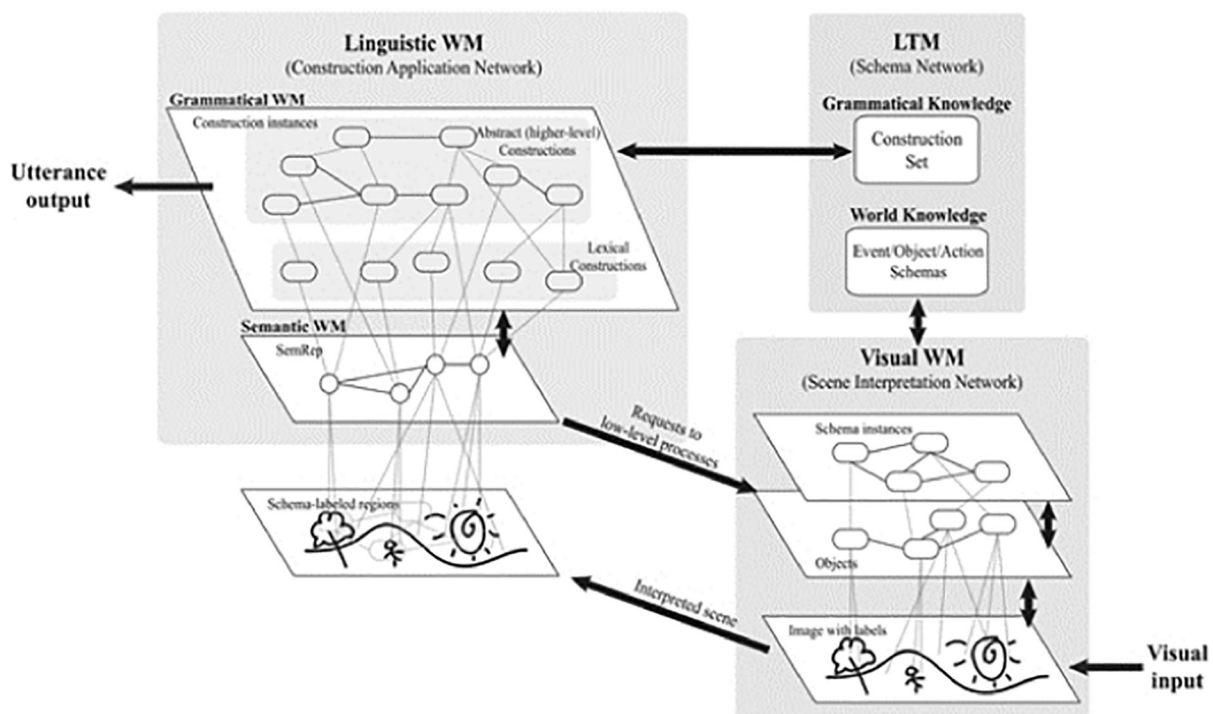


Fig. 8. The structure of the SemRep/TCG model of scene description (Arbib, 2016a).

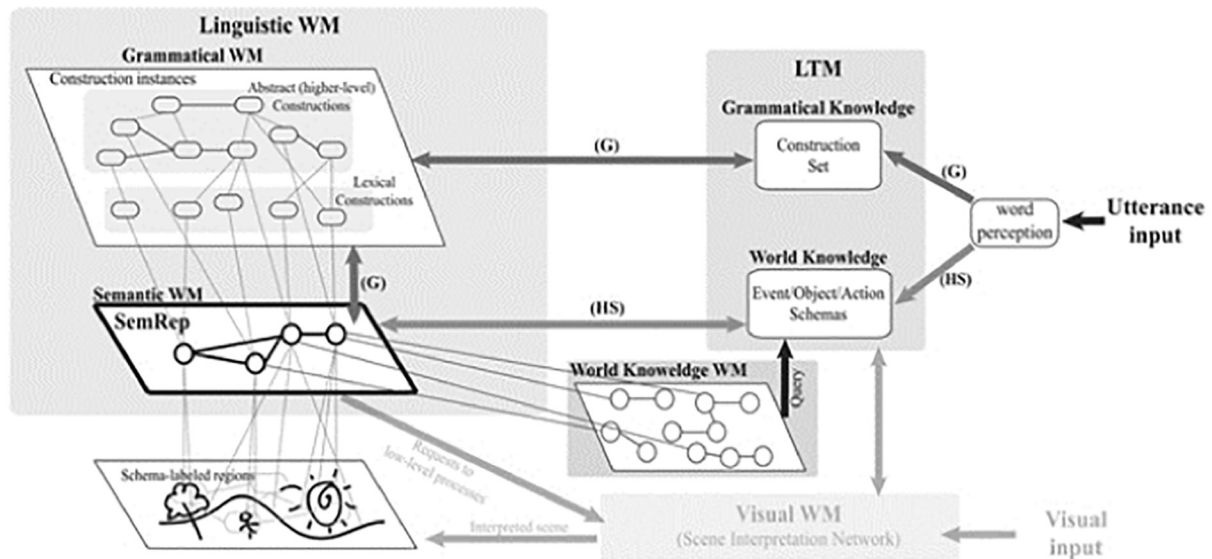


Fig. 9. TCG as employed in a two-route model of language comprehension. The grammatical route (G) and the heavy semantics route (HS) compete and cooperate to produce a SemRep as interpretation of the input utterance (Arbib, 2016a).

heavy semantic route (HS) can query world knowledge to generate SemRep nodes directly for content words but is not sensitive to grammatical cues. The currently relevant hypotheses are kept active in the World Knowledge WM where they cooperate and compete to update the SemRep. The construction assemblage on the one hand and the world knowledge hypotheses on the other contribute to growing SemRep hypotheses which compete and cooperate until the readout of the winning version. Visual WM remains a source of input for the SemRep as in the production model. Lesioning of G in the model can explain the data on agrammatic comprehension. (See Arbib, 2016a, Section 4.3, for further details.)

Some conclude from comparative studies such as that of Rilling et al. (2008) that a bigger arcuate fasciculus evolved to support language, but this sidesteps the computational questions of what the distinctively human arcuate adds beyond bandwidth. My current assessment is that the arcuate is present in monkeys and apes as part of the dorsal path for visuomanual control but has expanded in humans to support the vocal control lacking in other primates. Conversely, the ventral pathway for planning manual action on the basis of goals and the current state of the action-perception cycle expands to support the planning of communicative actions, with the details of articulation of selected actions (whether gestured, spoken, or signed) again delegated to the dorsal pathway.

The reader may have noticed a grave shortcoming of our macaque models. Apart from modeling audiovisual mirror neurons, no attention has been paid to auditory processing, and no attention at all has been paid to vocal production. Note, however, that the TCG model is neutral as to whether language is spoken or signed. In either case, visual input serves to provide data about the environment as the basis for developing a modality-neutral SemRep. In production

mode, TCG readout provides a string of words – it does not model the process whereby these words are articulated. It may thus be construed as a model associated with the ventral pathway. Other than that, it has not been linked to functional analysis of brain regions. Similarly, as we turn to the TCG model of comprehension (Fig. 9), the words are already abstracted from the sensory domain (whether visual for signed language or auditory for spoken). Elsewhere (Arbib, 2017), I described how one might integrate the TCG model with a conceptual model of sentence comprehension (Bornkessel-Schlesewsky & Schlewsky, 2013) which focuses on the auditory ventral and dorsal pathways. Further work is needed to resolve debates over the distribution of effort between the dorsal and ventral pathways in each modality.

7.3. Music and the brain

As already noted, when we were in the Vatican Wolf Singer invited me to prepare a proposal for a Strüngmann Forum (the successor to the Dahlem Conferences, under the continued direction of Julia Lupp). Each Forum has 32 members, divided into four groups each addressing a sub-theme of the overall topic. Before the meeting, some of the participants prepare background papers which everyone is expected to read before the weeklong meeting in Frankfurt (and some actually do). Then, in Frankfurt, each group meets for 4 days, sometimes singly and sometimes with another group or two (in which case, one group is “on stage,” and the others provide the audience). On Thursday night, the heroic rapporteur for each group turns 4 days of notes for his or her group into a coherent draft report. The Friday is then devoted to presentation and discussion of the reports. A challenging format with the potential to yield new insights from the interaction of people with diverse expertise.

In pondering Wolf's invitation, I sought a topic that would link to areas I knew but stretch them by confronting them with a different domain of human experience. In the end, I chose music as my focus, not only for its intrinsic interest but also because its linkage to the world of sound complemented my prior interest in how the visuo-manual system linked to audio-vocal processes in the study of language. And thus I chose the topic *Language, Music and the Brain: A Mysterious Relationship*. A planning meeting, under Julia Lupp's direction, with Tecumseh Fitch, Peter Hagoort, Larry Parsons, Uwe Seifert and Paul Verschure, set the list of invitees and possible themes.

The Forum complemented the focus on language and music as aural performances by taking seriously their relation to action (as in speech and gesture, or music and dance) without insisting on "embodiment all the way up." We addressed four themes: Semantics of Internal and External Worlds; Multiple Levels of Structure in Language and Music; The Neurobiology of Language, Speech, and Music; and Culture and Evolution. In particular "Semantics of Internal and External Worlds" explored the hypothesis that the semantics of music is primarily emotional, whereas that of language is propositional. The discussion was informed by background on aesthetic emotions (Scherer, 2013) and on the role of music in film (Cohen, 2013), among others. The subsequent book (Arbib, 2013d) gathered improved versions of the background papers, as well as one overview chapter for each theme evolved from the overnight draft of its rapporteur. I contributed an integrative first chapter, "Five terms in search of a synthesis" (Arbib, 2013b), the five terms being Action-Perception Cycle, Emotion, Language, Music, and Brain. My one regret is that some participants wanted the published version of the theme chapters to contain only established results and edited out some of their more creative remarks that had enlivened the discussion both in the formal sessions and over meals or with a glass of beer or wine.

7.4. Neuroinformatics

In 1993 (more or less) I was delighted to celebrate with a group of USC colleagues in both computer science and neuroscience the receipt of a Program Project grant from the Human Brain Project (HBP; a US initiative to advance neuroinformatics; not to be confused with the far more ambitious current European Human Brain Project). With this, I thought that we would finally achieve the full promise enshrined in the name NIBS: Neural, Informational, and Behavioral Sciences. Our USC Brain Project (USCBP) produced some excellent Ph.D. students who went on to advance the field and we did lay the basis for some good results (e.g., the brain architecture management system, BAMS: Bota, Dong, & Swanson, 2005, 2012) and a book reporting on our progress, *Computing the Brain: A Guide to Neuroinformatics* (Arbib & Grethe, 2001). Nonetheless, I count this effort among my failures. Although many of

the professors and students invested a worthy effort, most of the computer science professors failed to learn enough neuroscience and vice versa, two of my colleagues "let the side down," and I failed to devote enough effort to USCBP to solve these problems. But I think we failed to get Project Funding renewed for a different reason, and here I fault the funding process as much as myself. We sought to develop a broad integrative framework for neuroinformatics which embraced computational neuroscience as well database construction, but HBP apparently preferred neuroinformatics efforts focused primarily on developing databases for narrow areas of neuroscience (perhaps the Yale group offered the strongest exception). A certain whiff of sour grapes here. Some valuable efforts were indeed initiated with this early HBP funding.

A note on the rapid pace of computerization. When we gave our first course on neuroinformatics in 1994, we had to purchase an LCD projector since until then NIBS had only used projectors for transparencies; half the students did not have email accounts; and all had to be introduced to the use of a Web browser (we used Netscape in those pre-Google days).

The one legacy of USCBP to which my group has devoted major continuing effort is the Brain Operation Database (BODB; bodb.usc.edu/bodb), the working out of ideas for a system originally called Brain Models on the Web (BMW). To see the inspiration for this, consider my group's models of cerebellum. These are just part of a range of models from many research groups – some of which advance the field, others do not. The problem is that modeling is often piecemeal. Previous models and relevant data may be sketched in the first section of a paper, but we have lacked a support system for "going cumulative." BODB offers a first pass on providing this. The key ideas were set forth by Arbib, Plangprasopchok, Bonaiuto, and Schuler (2014) as part of a special issue of the journal *Neuroinformatics* based on presentations at the Workshop on Action, Language & Neuroinformatics that I organized in Los Angeles in 2011:

We present principles for an integrated neuroinformatics framework which makes explicit how models are grounded on empirical evidence, explain (or not) existing empirical results and make testable predictions. The new ontological framework makes explicit how models bring together structural, functional, and related empirical observations. We emphasize schematics of the model's operation linked to summaries of empirical data (SEDs) used in both the design and testing of the model, with tests comparing SEDs to summaries of simulation results (SSRs) from the model. We stress the importance of protocols for models as well as experiments. We complement the structural ontology of nested brain structures with a functional ontology of Brain Operating Principles (BOPs) for observed neural function and an ontological framework for grounding models in empirical data. We present an implementation of this onto-

logical framework in the Brain Operation Database (BODB), an environment in which modelers and experimentalists can work together by making use of their shared empirical data, models and expertise.

Although BODB has received moderate attention, it has not “caught fire.” It has a classic start-up problem. Unless it already contains many models, there is little incentive for others to use it to enter their own models and use BODB’s tools to compare it with other models in terms of data explained and the schemas and neural circuitry invoked to explain them. In an attempt to address this, we edited a major textbook, *From Neuron to Cognition via Computational Neuroscience* (Arbib & Bonaiuto, 2016), with its own website to complement it, in the hope that students will learn from the BODB chapter (Bonaiuto & Arbib, 2016, chap. 5) and the models in the book already documented in BODB to engage in projects that support a rapid expansion of the BODB entries. Here’s hoping.

Chapter 8. Constructions (2016–)

Although living for some extended periods of time in Los Angeles, Prue and I maintained our house in La Jolla throughout my time at USC. Thus, when I retired from USC in August 2016, La Jolla was already our home, and USC was too far away for active engagement. Instead, with great help from Vic Ferreira, the Chairman of Psychology, I surmounted the committee obstacles to become an Adjunct Professor of Psychology at the University of California at San Diego (UCSD), which is located in La Jolla. I am also a Contributing Faculty Member in Architecture at the NewSchool of Architecture and Design in San Diego.

The title of this Chapter is almost a pun, first published in “Tool use and constructions” (Arbib, 2012c). This was a commentary on Krist Vaesen’s “The cognitive bases of human tool use” (Vaesen, 2012), responding in part to his comments on the Mirror System Hypothesis. I noted that, inspired by Stout’s (2011) essay on stone toolmaking and the evolution of human culture and cognition, I had developed a rudimentary scenario in which complex imitation underwrites the co-evolution of language and toolmaking, with neither required to reach a critical complexity to initiate the evolution of the other (Arbib, 2011). The not-quite-pun was to seek a parallel between the role of tools in making objects (construction of things) and the approach to language offered by Construction Grammar (construction of utterances). It is thus appropriate, perhaps, that the two main themes that have occupied me since leaving USC are continuing study of the theme “How the Brain Got Language” and a relatively new exploration of the link between neuroscience and architecture (specifying the construction of the built environment and experiencing the result).

8.1. How the brain got language, revisited

The field of language evolution has ballooned in recent years. One of the final products of my time at USC was a manifesto, “Towards a Computational Comparative Neuropriatology: Framing the Language-Ready Brain” (Arbib, 2016a), which argued for a concerted effort based on the comparative study of brain, behavior and communication in monkeys, apes and humans to develop an integrated (EvoDevoSocio) approach to the biological and cultural bases of language evolution. I also advocated greater attention to the computational modeling of how the brain supports behavior and communication in monkeys, apes and humans, illustrating this with the work on macaque modeling, dyadic brain modeling, and Template Construction Grammar outlined above.

In addition to working with my group on these modeling efforts and BODB, I have engaged in extensive “missionary work” to encourage researchers to collaborate within this framework. To this end, I organized a series of workshops under the banner of the ABLE (Action, Brain, Language and Evolution) Project. These were held in Los Angeles, Bielefeld (hosted by Pia Knoeferle), Rome (hosted by Gianluca Baldassare), Atlanta (hosted by Erin Hecht and Dietrich Stout), Chicago, and – post-USC – in La Jolla.

Trying to get people in different disciplines to understand each other is hard. If somebody is an expert on ape behavior, then to get them to think about the monkey brain is hard enough, but to get them to think about what they might learn from a computational model of what the monkey brain is doing to ground their understanding of what’s going on in ape behavior is *incredibly* hard. For the La Jolla ABLE Workshop in August 2018, the organizing theme was to develop a *new* road map (replacing that provided by the 2012 book) for developing and coordinating comparative neuropriatology approaches to studying how the brain got language. I had contributors prepare drafts of their papers and make them available before the meeting, encouraging them to complement material on their own research with ideas for the new road map. Then, at the meeting itself, each session involved 2 or 3 short talks presenting key ideas from the papers followed by at least as much time for *integrative* discussion. Each day concluded with the meeting breaking into smaller groups to assess the day’s sessions in terms of implications for the new road map.

The Proceedings of this last Workshop are being published as a special double issue (volume 19, numbers 1 and 2) of *Interaction Studies*. 21 papers pave the way for a concluding paper, “The Comparative Neuropriatology 2018 (CNP-2018) Road Map for Research on How the Brain Got Language” (Arbib et al., 2018). The Mirror System Hypothesis (MSH) provided the old road map. However, MSH is not a fixed dogma but, rather, an evolving

system to be updated as new data and theory become available, and so an explicit charge to Workshop participants was that they should not privilege MSH. Consequently, a paper in this issue may show how to develop some aspect of MSH, offer an alternative, or ignore MSH entirely in exploring aspects of biological or cultural evolution missing from MSH. Rather than present the new road map here, let me simply present the aspects of language that the road map addressed, including several that had not been tackled in developing MSH.

A language provides a framework for sharing meaning in a community by combining *words* (we use the term to include, e.g., the signs of a signed language), perhaps modifying the words in the process, to express both familiar and novel meanings and to understand (more or less) the novel utterances of others (*parity of comprehension and production*). It combines an open-ended *lexicon* with a rich *grammar* that supports a *compositional semantics*. As such, a human language is a mechanism to support sharing of meaning in a community about physical and mental worlds. Further components of this ability include:

Here-and-Now: A commonly shared assumption is that the primary drive in the evolutionary path to language was the value of being able to coordinate current behavior, with joint attention supporting the sharing of perception of the current environment and plans for acting within that environment (Common Ground).

Theory of mind: The ability to talk about the mental (including emotional) states of others; this may rest on an ability, possibly shared to some extent with other species, to infer the mental states of others, and use these to predict behavior.

Displacement: Moving beyond the co-situated context, language builds on capacities for episodic memory, planning and imagination to support the ability to talk about distant events as well as about the past and the possible future, as well as counterfactuals.

Abstraction: Moving from embodied grounding to disembodied abstractions.

Building upon the new road map is indeed a worthy challenge.

8.2. Linking neuroscience and architecture

Prue and I have always had a keen interest in architecture, and our son is an architect. But my *professional* interest in architecture starts with the interactive space Ada, built as a temporary exhibit, visited by 550,000 people from May to October of 2002, at the Swiss Expo in Lausanne. Ada was designed by a team led by the computational neuroscientists Paul Verschure and Rodney Douglas (Eng, Douglas, & Verschure, 2003, 2005). Ada was designed as a perceiving, acting, adapting entity. “She” had a “brain” which was built in part of artificial neural networks and she had “emotions.” She wanted to play with her visitors. Ada’s sensors included not only cameras and microphones but also, most distinctively, floor tiles which were

pressure-sensitive. They could sense where a person was and, via a simple neural network based on where footsteps fell in succession, figure out how people were moving. By being able to change color, the tiles provided possibilities for a natural progression in visitor interaction:

Sleep: One tile color for all visitors

Wake: Visitors given different colored floor tiles

Explore: Probe for “interesting” visitors; deploy gazers (cameras which could be directed to attend to particular visitors)

Group: Try to direct visitors to a certain location in space, e.g., by floor patterns or by deploying light fingers (beams of light for pointing at individual visitors or indicating different locations in the space)

Play: Play a game selected on the basis of number of visitors grouped together.

Leave: Show a path for visitors to exit the space.

Not only did Ada’s “skin” (the floor) serve for visual communication via its patterning of lights, but Ada also used sound and music composed in real-time on the basis of her internal states and sensory input – using a computer system called Roboser to compose the music which could then be directed at different groups of people to change their level of interaction and excitement (Manzolli & Verschure, 2005).

As mentioned earlier, Fellous and I (2005) gathered experts to report on both the neuroscience of emotion and the current state of providing robots with at least the appearance of emotions. Some of these ideas were anticipated in the design of Ada. Ada continually evaluated the results of her actions and expressed “emotional states” accordingly, as part of the effort of regulating the distribution and flow of visitors. Ada’s level of overall “happiness” was translated into the soundscape and the visual environment in which the visitor was immersed, establishing a closed loop between environment and visitor. Ada “wants” to interact with people. When people participate, she is “happy”. When they do not, she is “frustrated.” The details are provided by Wassermann, Eng, Verschure, and Manzolli (2003).

Almost every year from 1986 through 2015, I gave a course on “Brain Theory and Artificial Intelligence” at USC. In 2003 and 2004, inspired by the work on Ada, instead of asking students to implement models of brain regions for their term projects, I asked them to develop “Brains for Intelligent Rooms.” We turned neuroethology inside out – instead of an animal in its environment, we considered “intelligent” rooms that surrounded their environment. Such a room’s sensors are directed inwards. In today’s surveillance society, such an idea is common; the issue then was to investigate what the study of neuroscience might add to the “information infrastructure” of a room or building so that it could better serve its inhabitants.

Nothing further happened in this direction until November of 2009 when Prue alerted me to an event put on in San

Diego by the Academy of Neuroscience for Architecture (ANFA). We attended the event, with the architect, Gordon Chong, speaking on “Design informed ... transforming evidence into innovation,” with commentary by the neuroscientist Fred Gage. They addressed ANFA’s central theme, namely *the neuroscience of the experience of architecture* – seeking to understand what is going through somebody’s head as they occupy a room, or behave in the context of different buildings. This is tightly connected with a second theme, *the neuroscience of the design process*, assessing what is going on in the head of the architect as he or she tries to envision the experience of inhabitants and design the building according to some criteria for what constitutes a good experience, both aesthetically and functionally. The “manifesto” for ANFA was provided in the book *Brain Landscape: The Coexistence of Neuroscience and Architecture* (Eberhard, 2008) by the architect John Paul Eberhard, the founding president of ANFA. He stressed that buildings where people spend much time can influence the fundamental structure of the brain and thus affect people’s thoughts and behaviors. His goal was to foster development of a “database” of neuroscience results that can support understanding how different populations will experience and benefit from schools, homes for Alzheimer’s patients, diverse workplaces, sacred spaces, and more. In the discussion period following the Chong and Gage presentations, and recalling the work on ADA and in 564, I argued that ANFA should add *neuromorphic architecture* as a third facet, assessing what it would mean for a building to “have a brain.” In due course this led to my being invited to join the ANFA Board by then-President Eduardo Macagno, a neuroscientist at UCSD, and somewhat later giving the ANFA-sponsored lecture that became the basis for the paper “Brains, Machines, and Buildings” (Arbib, 2012a).

Juhani Pallasmaa, the dean of Finnish architects and the author of many excellent books that stress the multisensory nature of the experience of architecture (Pallasmaa, 2009, 2012), emphasizes phenomenology: understanding architecture in terms of the way we experience it. I argued that understanding how different brain damage or just age-varying experience affects perception and action will enrich the phenomenology of knowing how different people experience different buildings – though one cannot jump directly from the reality of lived experience to the dynamics of neural networks without the consideration of mediating levels of representation (Arbib, 2013c). The action-perception cycle is a learning cycle, too, because as we interact with the world our new experiences, especially those that are unexpected or otherwise memorable, change our brains and thus alter the dynamics of the cycle as we continue to confront new situations. Famously, the pre-Socratic Greek philosopher Heraclitus (535–475 BC) said “*you can never step in the same river twice*.” As a neuroscientist, I would say “*you can never think with the same brain twice*.”

To close this glimpse of architecture in relation to neuroscience, consider what the noted Swiss architect Peter

Zumthor had to say in his essay “A Way of Looking At Things” (Zumthor, 2012):

When I think about architecture, images coming into my mind.

When I design, I frequently find myself sinking into old, half-forgotten memories. ... Yet at the same time, I know that all is new and that there is no direct reference to a former work of architecture.

First note that the “images” here need not be purely visual. As in episodic memory, they may combine action and a range of associated perceptual experience. But in design the aim is *not* to recall past episodes accurately, but rather to recall fragments of such experiences in the service of imagination, of creating new experiences. Thus, factoring what we know about episodic memory into developing the neuroscience of imagination is one of the current challenges that will help us bridge the borders between architecture and neuroscience. The essence of imagination is the *construction* of something new, and I am exploring whether viewing even visual perception as a process of construction – recall VISIONS and its posited “lifting” to scene description in the work on Template Construction Grammar – may provide theoretical insight into current experimental research (Maguire, Intraub, & Mullally, 2016; Schacter et al., 2012; Zeidman & Maguire, 2016).

Zumthor further states:

The challenge of developing a whole out of innumerable details, out of various functions and forms, materials and dimensions.

Details ... lead to an understanding of the whole of which they are an intrinsic part.

Construction is the art of making a meaningful whole out of many parts. I feel respect for the art of joining, the ability of craftsmen and engineers ... the knowledge of how to make things.

Here the interplay between whole and parts is crucial. A related challenge in analyzing visual perception has been only partly addressed within neuroscience: When I look at a scene, I immediately get the *gist*, whether it be a cityscape, a beach scene, or a domestic interior (Oliva & Torralba, 2006). But to fill in the details (and to possibly correct an initial misimpression) my gaze must shift from region to region in the scene, identifying agents and objects and clarifying their relationships or interactions. In architecture, the process of constructing a design is not constrained by a pre-existing scene, but the cognitive processes are similar: a first idea of the whole may guide the search for details; the attempt to mesh the details may necessitate changed notions for the pieces which may in turn propagate to affect the design of the whole. And this interaction of bottom-up and top-down design may proceed through many iterations.

A crucial difference between visual perception and architectural design is that in a visual scene the “pieces” are already there (though what pieces we attend to, and what relations we emphasize is open to the varying goals of perception within the action-perception cycle), whereas in architecture the very building blocks may change as time proceeds – but always constrained by the architect’s understanding of available materials and the skills needed to assemble them. Whether in architectural design or in making sense of our immediate environment, the interaction between the overall gist and the filling in of details must yield a representation that can guide our action appropriately. Probing these processes – from imagination and perception to construction – poses many challenges for neuroscience whose solution will enrich, and be enriched by, conversations with architecture.

Chapter 9. Reflections on modeling the brain

In the late 1960s, while still at Stanford, I flew to Boston for a planning session of the NRP, and shared the first leg of the flight home, to Chicago, with John Eccles, the Nobel Prize-winning Australian neurophysiologist who had worked with Charles Sherrington to uncover the basic role of inhibition in complementing excitation in neural circuitry. Eccles held forth on his dislike for McCulloch’s “romantic” style of neuroscience which offered bold ideas on how the brain might, or even should, work, rather than grounding all the work as Eccles did (at least until his later writings on free will) on the results of detailed experiments. I defended McCulloch and his dictum “Don’t bite the end of my finger, look where I’m pointing.” To this day, my work goes back and forth between the “McCullochian” and the “Ecclesiastical.” Some of my models are more concept-driven, as in the development of Template Construction Grammar which is based on the hope that being computationally explicit about what processes *might* support language can help primatologists become more process-oriented in the way they gather data on brain, behavior and communication in monkeys, apes and humans. Other modeling is very much data-driven, as in our work on the cerebellum.

For data-driven models, the goals of the Brain Operation Database of Chapter 7 are highly relevant. Whether or not BODB succeeds, a major neuroinformatics effort will be required that implements the principles on which it is based. With the overwhelming amount of new data now pouring forth, it will be essential – both for the design of new experiments and for the design of models – that databases not only provide access to the detailed results of specific studies, but that agreement is reached on how to extract from these results summaries of what we know, SEDs. We also need ways to compare different models, not only to assess to what extent they use certain SEDs in their design and succeed or fail in explaining others, but also to assess the models as hierarchical structures, determining what part particular “subsystem modules,”

or small networks thereof, play in explanatory successes. Hopefully, this will speed further modeling which will exploit these successful modules, seeing what updates are necessary as new data become available or as integration of modules forces changes to ensure their successful integration. Such steps are necessary for brain modeling to go cumulative.

It is my hope that the brain theory community will take BODB seriously in future, whether to extend it, or to find a new strategy to address challenges such as those we set forth in designing BODB (Arbib, Plangprasopchok, et al., 2014). It should also be noted that in modeling complex systems, we may need to combine detailed neural models of some subsystems with a schema-based model of some other systems (either for economy of simulating the overall model, or because neural data are not available – as in certain cognitive skills for which only brain imaging data are available [but see Fig. 10 below]).

A further challenge is that different labs not only gather different data, they gather very different kinds of data. Masao Ito followed his basic research on LTD in the cerebellum with very detailed analysis of the molecular biology of the parallel fiber → Purkinje cell synapse (Ito, 2002). Had we tried to incorporate all these chemical pathways in our cerebellar models, we would never have been able to offer new insights into the role of the cerebellum in motor learning. On the other hand, we did suggest a refined view of cerebellar plasticity informed by our system-level modeling that offers new hypotheses for exploration by those who work at the molecular level.

Not only is there a problem of the flood of data at different levels. Study of the role of a neural system for a given task may offer a dataset that at first seems little related to data on its operation in another task. Here, a broader approach may be of benefit even if one wishes to emphasize just one neural system in one’s own research. One example. My work with Curt Boylls on the cerebellum (Chapter 3) focused on locomotion, while the work at USC (Chapter 7) focused on arm and hand movements. In short, we looked at the role of cerebellum in the control of the

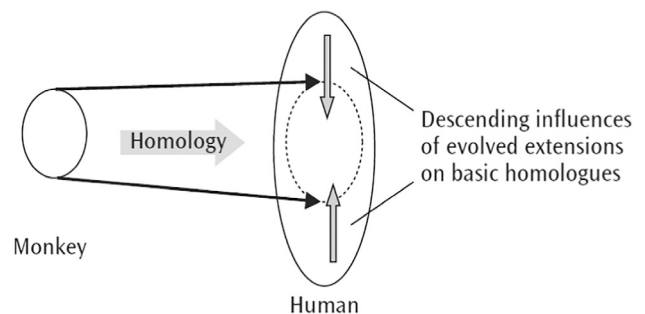


Fig. 10. A strategy for modeling the human brain: Use known circuitry in the macaque brain to suggest details of circuitry in homologous regions of the human brain; extend these models using hypotheses on brain evolution; then test the result against lesion data, neurological disorders, or human brain imaging studies by the use of synthetic brain imaging.

skeletomuscular system. Richard Thompson, a neuroscience professor at USC, specialized in the study of learning in the cerebellum and associated circuitry, using classical conditioning of rabbits' reflex blinking to an air puff by repeated presentation of a tone as the key preparation (Thompson & Steinmetz, 2009). I suggested that we collaborate on constructing an integrative theoretical framework for the motor control and classical conditioning roles of cerebellum by viewing the failure to blink before the air puff as a form of motor error. Thompson was not interested. Here the conflict was not between more concept-driven and data-driven modeling so much as whether one's motivation is to study a focal subproblem in more and more detail or to step back to bring several datasets and several modeling methodologies into view and see if a more unified perspective is possible.

I find something of the same issue when I edit articles, or review articles submitted to journals. Some authors respond well to suggestions for situating their work in a broader context; others are committed to a narrow focus and will not budge. Of course, if the work is of high merit, it deserves to be published no matter what scope I may see for establishing wider connections; but undue specialization can impede the progress of science. When I headed the curriculum committee for the USC Neuroscience Graduate Program, I introduced a four-course core, with courses in molecular, cellular, cognitive and computational neuroscience, only to find that many faculty insisted that their students take only two of the courses. They preferred that students stay in the lab to work toward renewal of research grants rather than get a comprehensive (but still very partial) overview of neuroscience. The catch of a narrow focus (as a world-wide phenomenon, in no sense restricted to USC) is that when I review a neuroscience paper, I often find that an excellent piece of experimental work is followed by a discussion that is amateurish and naïve because the authors have been “protected” from computational neuroscience or theory more generally. (Of course, the counter-criticism is that my own work has at times slighted subtleties of experimental design.)

In Chapter 1 of *How the Brain Got Language* (Arbib, 2012b), I recounted the story of the drunk looking under a lamp post not because that was where he had lost his keys but because that was where the light was brightest. I suggested that for truly complex problems (like language evolution, or large-scale brain modeling), we might view the scientific effort in terms of (sober) scientists each looking under their own lamp post to find pieces of a jigsaw puzzle. A complementary effort is needed to assemble the pieces found to date and assess to what extent they support a coherent theory. In any case, whatever the state of data gathering and modeling, the Modeling-Experimentation cycle must continue and, at times, concept-driven models, even “romantic” models, may be more beneficial than data-driven models. Copernicus pursued his heliocentric theory of the solar system long before further research could bring it to the point where it could

predict planetary observations better than could Ptolemy's scheme. “Don't bite the end of my finger. Look where I am pointing.”

Another challenge for brain modeling that goes beyond the issue of levels or choice of tasks is that much data relevant to our understanding of the human brain comes from animal data on physiology, molecular biology and genetics. We still have much to do to understand when and how this “injection” of animal data into the human brain is justified. In developing a comparative neuroprimatology, seeking an integrative understanding of the similarities and differences of brain mechanisms in monkeys, apes and humans (but not forgetting what we had learned from frogs and rats), we had three foci:

- (i) Monkeys: Here we have neurophysiological details on activity of single neurons and circuits linked to detailed neuroanatomy. For us, the key data were related to the role of vision in the generation and recognition of manual actions, with lesser attention to auditory processing of sounds linked to such actions or to monkey vocalizations.
- (ii) Apes: The different data on praxic and communicative behavior gleaned from studies of captive apes versus those in the wild raise issues about the relation of adult behavior not only to the genetics of the brain but also to the “culture” in which an animal is raised (EvoDevoSocio). We focused on the relation of certain praxic actions to communicative manual gestures. However, ape neuroscience data are limited to anatomical studies and to brain imaging addressing connectivity in anesthetized (i.e., non-behaving) chimpanzees (Hecht et al., 2013), and the latter data are now being limited or prohibited.
- (iii) Data on the human brain come primarily from neuropsychology (effects of brain damage due to lesions), the study of brain diseases (not only changes in behavior and cognition due to the disease itself, but also the effects of drugs on such changes), as well as varied brain imaging techniques, including ERPs (with relatively precise timing but poor localization) and fMRI (with poor timing but relatively precise localization – but at far lower resolution than single cell recording). Neurolinguistics is the attempt to relate such data to language.

Computational comparative neuroprimatology is the attempt to knit together these very diverse datasets using computational models of neural and schema networks. In many cases – such as the reach-to-grasp and the control of eye movements – we may assume that the underlying circuitry as charted in the monkey is relevant to filling in the details obtained from brain imaging and other human studies. Mihail Bota and I addressed some of the neuroinformatics challenges for comparing macaque and human circuitry by developing a prototype for the Neurohomology Database, long defunct, but our papers (Arbib &

Bota, 2003; Bota & Arbib, 2004) offer analyses of continuing relevance, but in need of updating in light of new data.

We developed two tools for related modeling. Synthetic Brain Imaging (SBI) offers algorithms for averaging over the synaptic activity revealed by simulations of models of neural circuitry to predict region-by-region activity of the kind measured by brain imaging. SBI can bridge between neuron-based modeling of animal brains and studies of the human brain for which scant information about the underlying circuitry is available (Arbib, Fagg, & Grafton, 2002; Bonaiuto & Arbib, 2014; Griego, Cortes, Winder, & Tagamets, 2016). The methodology is to develop neural network models of human brain mechanisms for which one believes the relevant circuitry is similar to that revealed by animal (e.g., monkey) single cell neurophysiology and then process simulation results at the neural network level to infer predictions that can be tested against human fMRI or other non-invasive measures. One can also apply variations of this methodology to schema networks. Related strategies also apply to the development of Synthetic ERPs to allow further testing against human data (Barrès, Simons, & Arbib, 2013). Fig. 10 illustrates an extension of the approach for mechanisms – such as those subserving language – for which nonhuman neural circuitry does not suffice: We use hypotheses about evolution of brain mechanisms to suggest how macaque circuitry is modified and expanded upon in the architecture of the human brain, then use the resultant model to make predictions for human brain imaging or for lesions.

All this leads to the photograph (Fig. 11) taken, at my request, by Jean-Arcady Meyer in 2005. I was in Paris for a Summer School on “Mathematics and Brain,” and visiting speakers were put up at Hotel Jack’s (an atypical name for a French hotel). But I noticed, as nobody else at the conference did, that the little alley at the side of the hotel was called “Impasse du Petit Modèle” – *the dead end of the little model*. I have adopted the name of this alley as the slogan for my research. If we really want to under-

stand the brains of creatures interacting with the world around them, then no little model, or even a few simple but basic principles, will suffice. We must understand how to orchestrate many different models of different parts of the brain and at many different levels of detail, while seeking the many core principles they exemplify in different combinations, to chart more and more of a vast landscape spanning from diverse facets of cybernetics to brain theory and cognitive science, with insights from AI and robotics thrown in for good measure.

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Fig. 11. Impasse du Petit Modèle. Photo by Jean-Arcady Meyer.

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